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COUNTRY
**GROWTH&JOBS**
REPORT

# Philippines

## Running Uphill: Growth, Jobs, and the Quest for Productivity



COUNTRY
**GROWTH&JOBS**
REPORT

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# ABBREVIATIONS AND ACRONYMS

| | |
|---|---|
| AI | Artificial Intelligence |
| BARMM | Bangsamoro Autonomous Region in Muslim Mindanao |
| BCP | Business Continuation Plan |
| CAB | Civil Aeronautics Board |
| CFO | Commission on Filipinos Overseas |
| CMCI | Cities and Municipalities Competitiveness Index |
| DBM | Department of Budget and Management |
| DEPDev | Department of Economy, Economy, Planning, and Development |
| DILG | Department of the Interior and Local Government |
| DOLE | Department of Labor and Employment |
| DOST | Department of Science and Technology |
| DOT | Department of Transport |
| DTI | Department of Trade and Industry |
| EBET | Enterprise Based Education and Training |
| EPIRA | Electric Power Industry Reform Act |
| ERP | Enterprise Resource Planning |
| EUDR | European Union Deforestation Regulation |
| EWE | Extreme Weather Event |
| FDI | Foreign Direct Investment |
| FIRB | Fiscal Incentives Review Board |
| FLFP | Female Labor Force Participation |
| FTA | Free Trade Agreement |
| GBFs | General Business Functions |
| GDP | Gross National Product |
| GFC | Global Financial Crisis |
| GNI | Gross National Income |
| IFP | Infrastructure Flagship Project |
| IRT | Internal Revenue Allotment |
| IT-BPO | Information Technology Business Process Outsourcing |
| LGU | Local Government Unit |
| LIR | Low-Income Region |
| MARINA | Maritime Industry Authority |
| MIC | Middle-Income Country |
| MIR | Middle-Income Region |
| MNC | Multi-National Corporation |
| NCR | National Capital Region |
| NTA | National Tax Allotment |
| NTC | National Telecommunications Commission |
| PDP | Philippine Development Plan |
| POEA | Philippine Overseas Employment Administration |
| PPP | Public-Private Partnership |
| PSA | Public Service Act |
| REER | Real Effective Exchange Rate |
| RORO | Roll On/Roll Off |
| RTWPB | Regional Tripartite Wages and Productivity Board |
| SEC | Securities and Exchanges Commission |
| SMEs | Small and Medium Enterprises |
| STEM | Science, Technology, Engineering and Mathematics |
| TESDA | Technical Education and Skills Development Authority |
| TFP | Total Factor Productivity |
| WAP | Working-Age Population |

# FOREWORD

The global economy is undergoing profound and rapid transformation. Geopolitical tensions are reshaping supply chains and weakening multilateralism. Protectionist policies are gaining ground, while accelerated advances in artificial intelligence, demographic transitions, and the escalating impacts of climate change are altering how economies grow and labor markets function. These forces compound global uncertainty, threatening to displace workers, destabilize economies, and exacerbate inequality. The Philippines, like many countries, must navigate these evolving dynamics. Without deliberate and forward-looking policies, the country risks moving toward a future marked by economic stagnation, widening disparities, and missed opportunities.

Despite these global challenges, the Philippines remains committed to advancing its development agenda. While the COVID-19 pandemic temporarily disrupted growth and poverty reduction efforts, the country has made notable progress. Post-pandemic, the country's economic growth has become among the highest in Asia and the world's emerging economies. Poverty incidence declined to 15.5 percent in 2023, down from 18.1 percent in 2021 and even lower than the pre-pandemic level of 16.7 percent in 2018. The labor market has similarly improved, with the unemployment rate falling to 4.3 percent in 2024—surpassing the Government's target range of 4.4 to 4.7 percent.

Nevertheless, significant structural challenges persist. Capital deepening and labor force expansion have primarily driven economic growth, while gains in total factor productivity remain limited. Although some regional convergence has occurred, stark welfare disparities endure. The National Capital Region's gross domestic product per capita is nearly nine times that of the Bangsamoro Autonomous Region in Muslim Mindanao. Moreover, while job quantity has improved, job quality remains a concern. Informality continues to dominate the labor market, with the informal workforce averaging around 16 million from 2012 to 2024.

The country's demographic window of opportunity heightens the urgency of addressing these challenges. As the working-age population grows more rapidly than the dependent population, there is potential to realize a demographic dividend. Yet, we can unlock this potential only through timely and strategic reforms and investments. Without them, we lose the opportunity.

Historically, the Philippine economy has relied heavily on consumption and factor accumulation. The country must anchor growth on broader, more resilient foundations to secure long-term, inclusive prosperity: investment-led, innovation-driven, and export-oriented strategies. This anchor requires attracting high-quality investments aligned with the report's three-pillar framework: (1) foundational investments in infrastructure and human capital to enhance productivity and generate quality jobs; (2) a regulatory and governance environment that reduces risks and costs for investors; and (3) targeted public interventions to catalyze private sector participation in areas where market failures persist.

While job creation has increased, the next imperative is to raise labor productivity, particularly in underperforming sectors such as agriculture and small and medium-sized enterprises. This demands sustained investment in infrastructure, broader adoption of technology, and strengthened institutional capacity to foster innovation and efficiency. A more conducive business environment is equally critical. Streamlined regulations, improved public service delivery, and digital transformation can reduce transaction costs and spur private sector expansion. Enhancing competition in key sectors—such as energy, agriculture, and digital connectivity—will also help eliminate structural bottlenecks that constrain growth. Furthermore, leveraging well-designed free trade agreements can expand market access, promote reform, and boost the productivity of domestic firms through global competition.

Ultimately, inclusive development depends on investing in people and the systems that support them. A healthy, well-educated workforce with labor market-relevant skills is central to innovation and sustainable growth. Yet, meaningful progress is not measured by national averages alone—it must also reflect the equitable distribution of opportunities across all regions. Physical and digital connectivity infrastructure has been instrumental in fostering spatial convergence. At the same time, enhancing regulatory efficiency and transparency at the local government level is essential for stimulating quality job creation and sustained development.

Public investment can drive growth only if it is fiscally sustainable. The fiscal space remains constrained, mainly due to elevated government borrowing in the wake of the pandemic. This constraint underscores the importance of prioritizing public spending that delivers the highest returns. We must direct limited resources toward quality healthcare and education, food security, and infrastructure that enhances productivity and inclusion. Beyond budget allocations, improving spending efficiency and ensuring that programs are grounded in evidence and measurable outcomes are equally vital.

Amid global uncertainties, this is a pivotal moment for the Philippines. The country can accelerate its development momentum with sound policies and strategic investments and emerge as a regional leader. The decisions made today will shape the nation's future for generations. This report offers in-depth analysis and actionable recommendations on how addressing productivity constraints can pave the way for sustained, inclusive economic growth. This report will contribute to an informed policy dialogue, support rigorous research, and inspire collective action toward a future where all Filipinos enjoy a firmly rooted, comfortable, and secure life.

**Arsenio M. Balisacan**
*Secretary, Department of Economy, Planning and Development (DEPDev)*

Over the past 15 years, the Philippines ranked among the top quartile of the fastest-growing middle-income countries. Growth was widespread and pro-poor. Low- and middle-income regions grew twice as fast as high-income ones and real incomes of the bottom 40 percent rose faster than those of the wealthiest 20 percent. Nearly 12 million jobs were created, driving unemployment and underemployment to record lows of 3.8 and 11.9 percent, respectively.

But the global financial environment has grown more complex. The world economy is slowing. Disruptive technologies are radically changing industries. And uncertainty is rising. Against this backdrop, maintaining the Philippines' positive trajectory will demand a refreshed focus to boost the private sector and create more and better jobs. This can be achieved by strengthening infrastructure foundations, enabling business through better policies and regulations, and mobilizing private capital at scale.

This *Growth and Jobs Report* for the Philippines is the first of a new generation of World Bank reports that places jobs at the center of growth analysis. It offers a rigorous, evidence-based assessment of how growth has occurred, where constraints remain, and what reforms are needed to shift to productivity-driven development and build a dynamic private sector that unlocks opportunities for people.

The Philippines' recent experience is characterized by both momentum and risk. Investment-to-GDP rose from 17.1 percent in 2000–2009 to 22.5 percent in 2010–2023, driven by rising public infrastructure investments and private capital mobilization. But structural challenges persist. Three out of four new jobs are in low-productivity activities, innovation remains limited, and 91 percent of ten-year-olds cannot read or comprehend a simple text. These factors threaten to constrain long-term growth and job creation. Addressing these gaps will require business-enabling policies and greater private capital mobilization—through streamlined regulations, improved access to finance, and risk-sharing instruments that help firms grow and create jobs at scale.

This report goes beyond diagnosis to offer concrete, implementable solutions that are aligned with the Philippines' development goals and grounded in evidence. If fully enacted, the proposed reforms could raise GDP growth to 6.8 percent annually, generate over 5 million new jobs, and increase real wages by nearly 13 percent by 2040. The World Bank stands ready to support this agenda—whether by financing foundational investments, assisting regulatory reform, or helping design smart interventions to mobilize private capital. The report makes clear the path forward: Achieving the Philippines' long-term development goals will depend not just on how fast the economy grows, but on the quality of the jobs it creates

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# EXECUTIVE SUMMARY

**Since 2010, the Philippines has forged ahead, achieving record low unemployment and doubling its GDP.** Rapid growth put the country in the top quartile of fastest growing middle-income countries (MICs), while the 11.7 million jobs created led to a record low 3.8 percent unemployment rate in 2024. Employment grew faster than the working-age population, by 0.4 percentage points annually, shifting to wage-earning jobs in slightly higher productivity sectors. Economic growth was fueled by a spatial catch-up, while incomes of the poor grew relatively rapidly.

**Growth was fueled by pro-investment reforms, macroeconomic stability, and surging public and private investment.** Foundational infrastructure spending, structural reforms and private capital mobilization led to the growth surge. Capital accumulation accounted for over 90 percent of growth, reflecting high investment returns. Lagging regions contributed to growth and job creation, with most new jobs in non-tradables, although IT and IT-enabled services boomed.

**A commitment to deeper reforms is needed to achieve the national ambition of becoming *a middle-class* economy by 2040.** This will require faster-for-longer and *different* growth—similar to that of the Republic of Korea at similar development levels, sustaining 6 to 10 percent annually for decades. This contrasts with the Philippines' 5.2 percent average growth over the past 14 years, driven by investment growth, mainly in non-tradables. This report diagnoses the challenges to growth and job creation, and proposes reforms to shift the country onto a higher development path.

## Foundational infrastructure gaps remain a key challenge:

- **Connectivity infrastructure, energy, and resilience:** Despite investments in connectivity infrastructure, transport costs remain high. Key enabling services—energy, telecoms, logistics—continue to increase in cost, while climate-related events threaten the level and stability of growth.
- **Skills scarcity:** Low human capital, and gaps in STEM and digital skills hinder technology adoption. Talent misallocation, particularly with low female labor force participation, increases these challenges.

## Complex regulations that limit private capital mobilization and market contestability:

- **Competition deficits:** Concentrated market power in key sectors, regulatory capture, and limited competition have affected growth and job creation of top-tier firms. Tradable sectors, critical for productivity, are stunted by an appreciated exchange rate and high input costs.
- **Regulatory inefficiencies:** Complex permitting processes in enabling services and distortions in agriculture erode gains from past reforms and serve to misallocate resources.

## Priority outcomes for reform focus:

Sustaining spatial convergence, enhancing productivity, allocating resources more effectively, and improving regional integration will boost real annual GDP growth to 6.8 percent, raise employment by 7.0 percent, and lift wages by 12.9 percent by 2040. Reforms can be classified along three pillars:

1. **Foundational infrastructure and human capital investment:** Invest in connectivity and in resilience to protect jobs and boost growth in lagging regions. Incentivize firms' upskilling of workers to prepare the workforce to leverage AI advancements.
2. **Better regulations and governance:** Enhance competition by streamlining business regulations, improving local governance, and reducing the cost of moving goods, services, ideas, capital and people.
3. **Private capital mobilization:** Strengthen the innovation ecosystem and remove barriers to technology adoption, and foster domestic-foreign linkages for increased productivity.

# PREFACE

## The Report

**In 2024, the Philippines achieved record-low unemployment and more than doubled its GDP compared with 2010.** At the historical growth rate from 1990 to 2010, the economy would have taken 19 years to double. Instead, GDP growth since 2010 has been consistent with the economy doubling every 13.5 years. Faster GDP growth came with employment outpacing working-age population (WAP) growth by 0.4 percentage points per year (and with 11.7 million additional jobs). Unemployment fell to 3.8 percent in 2024.[1] New jobs were better jobs: increased waged jobs rather than self-employment, and in slightly more productive sectors. This report addresses two linked questions: why this happened, and how this process can be accelerated in a complex global context.

**Faster growth was driven by pro-investment reforms implemented in a period of high returns to investment and facilitated by public foundational infrastructure investment.** Macroeconomic and structural reforms enhanced stability and lowered investment costs. Public investment, increasing from 2.5 to 5.0 percent of GDP on average in the past decade, contributed to providing some of the public goods needed. Given the remaining gaps in investment relative to global standards, these reforms led to high returns to private investment, creating the right conditions to attract more investment from home and abroad, particularly to lagging regions. Spatial convergence became another engine of growth.

**A more complex external environment and ambitious national targets call for accelerating growth and improving labor outcomes.** Potential growth must rise to create more productive jobs. This requires improved connectivity infrastructure, better local governance, and innovation policies to enhance productivity, together with regulatory and trade reforms to optimize resource allocation and tap into the global economy. It also requires stronger skills to prepare for the jobs of the future.

**In the following six chapters, this report looks at these issues in detail.**

**The first two chapters look at macro and micro features of growth and job creation.** They centrally position better labor outcomes as an objective of faster growth. They examine how policies supported or inhibited job creation and growth, and provide long-term growth projections.

**The subsequent three chapters look at specific development challenges.** Chapter 3 looks at spatial growth and job creation dynamics; Chapter 4 looks at technology adoption and productivity

1 Underemployment also reached a record low of 11.9 percent.

dynamics; and Chapter 5 looks at how climate events affect firms' performance and decisions on investment and jobs.[2]

**Spatial convergence is essential for growth and job creation.** Disparities in income, productivity, and infrastructure across the Philippines highlight the need to address these gaps. However, spatial disparities also provide an avenue for a more attainable way to accelerate growth: returns to investment remain relatively high in low- and middle-income regions. The middle-income trap regularity that makes growth of upper middle-income regions (MIRs), such as Metro Manila, more challenging is less of a binding constraint to accelerate growth in low-income regions or lower MIRs, such as the Bangsamoro Autonomous Region in Muslim Mindanao (BARMM), Palawan, or the Visayas. Chapter 3 examines these disparities and provides recommendations to boost job creation and GDP growth across regions.

**Boosting productivity and innovation will also be required to sustain growth, and to create more and better jobs.** For firms operating in the Philippines, technology adoption is critical for productivity, growth and quality job creation. Disruptive digital technologies that permeate the economy may erode or enhance the comparative advantage of the IT-enabled and business services sectors, which have driven growth and job creation over recent decades. A well-functioning innovation ecosystem for more sophisticated firms is also crucial. Chapter 4 examines firms' technology adoption, identifies the main obstacles that firms face, and proposes interventions to ensure that disruptive technologies and innovation enhance growth and jobs prospects.

**Private sector resilience to climate shocks is key.** Extreme and infrequent natural events, as well as gradual and recurrent ones, affect livelihoods by influencing the level and volatility of growth, and labor outcomes. These events disrupt supply chains, lead to workers absenteeism and damage productive assets. Chapter 5 examines the impact of climate events on firms' outcomes and decisions. It proposes interventions to preserve growth and jobs with more frequent climate shocks.

**A changing global landscape requires a new playbook of reforms to accelerate growth and improve labor outcomes.** The Philippines has demonstrated remarkable growth since 2010. As it enters a new global context marked by trade policy uncertainty, fast-evolving, labor-saving technologies, and increasingly frequent climate shocks, it may need a new playbook to move closer to its national ambition. This report provides one. The first five chapters of this report conclude with highlights of policy recommendations. Then, Chapter 6 presents a modeling framework to estimate how the recommendations, if fully implemented, impact growth, job creation and welfare. It provides a reality check on attainable GDP, job and wage growth for the Philippines moving forward.

2 The choice of focus chapters is based on three facts. First, that spatial disparities suggest convergence is crucial to sustained growth moving forward. Second, that global technological disruptions present important opportunities and challenges to some of the most dynamic sectors of the economy that underpinned growth and job creation, aside from being a force for increased productivity growth. Third, that climate change will continue to shape firms' outcomes and decisions, with key implications on the Philippine private sector.

# The Framework

**Over the past 15 years, the Philippines has achieved rapid and pro-poor growth, marked by more productive and better-paying job opportunities.** Yet significant gaps remain across regions and relative to global peers. Frontier firms struggle to drive innovation and generate employment at scale, while the economy's gradual inward orientation poses a constraint on future growth and on quality job creation. To address these challenges, this report builds on the idea that development is a dynamic process fueled by the virtuous interaction between creating opportunities for the population, and building their capabilities, with job creation serving as a central engine of inclusive and sustainable progress. Employment not only provides income, but also instills dignity, purpose, and resilience in individuals and communities. To realize its transformative potential, public policy needs to ensure adequate investment in the foundations: physical and human capital; a business-enabling policy environment; and smart, targeted interventions to mobilize private capital. This approach aligns with the World Bank Group's strategic framework for private sector development and job creation (World Bank, 2025).

## Pillar I: Foundational investments in infrastructure and human capital

**Inclusive development relies on investing in people and in the systems that support them.** Building human capital—through improved education, healthcare, and skills training—ensures that individuals can seize opportunities and contribute meaningfully to economic growth. Investment in essential infrastructure—such as transport and energy networks, or digital connectivity—is crucial for linking people to markets, services, and to each other.

**Foundational investments must be calibrated to address regional disparities.** In relatively less developed areas, such as the BARMM region or rural Luzon, the highest-return investments focus on connecting farms to markets, electrification, or basic services. These interventions are critical to enabling catch-up growth, as they enhance agricultural productivity, improve labor mobility, and connect isolated communities to better economic opportunities. Complementing these efforts with strengthened early childhood education, health systems, and nutrition programs can lay a strong foundation for long-term growth and job creation.

**In more advanced regions, foundational investments of high returns focus on upgrading skills and building infrastructure to support the shift toward higher-value economic activities.** For example, expanding tertiary education capacity, such as scaling up the Technological University of the Philippines at Batangas, can help meet the rising demand for engineers and technical professionals, particularly as firms establish new research and development hubs in these regions.[3]

3 In 2024, Dyson announced the set-up of a technology center in Batangas, representing an investment of about US$200 million, and aiming to hire an additional 400 engineers and more than 50 graduate engineers.

### Pillar II: A business-enabling policy environment

**A stable and competitive policy environment is key to increasing the returns of foundational investments and to facilitate the crowding-in of private investment.** A business-enabling policy environment encourages the private sector to invest, expand, and innovate. Such an environment is characterized by low regulatory uncertainty, efficient administrative procedures, and strong institutions that build investor confidence and foster competition.

**Strong institutions matter for foundational investments to be cost effective.** Regional disparities demand institutional strengthening at the local level. In lagging regions, the administrative capacity of local governments is key for investment projects and service delivery to be executed on time and within budget. Strengthened governance and improved coordination across national and subnational agencies are vital to making development efforts more effective and responsive.

### Pillar III: Smart interventions to mobilize the private sector

**Smart interventions to correct market failures are key to mobilizing private capital.** Foundational investment in infrastructure and human capital, coupled with business-friendly regulations, is key to crowding in private investment. In some cases, however, where market failures are pervasive, targeted smart interventions are also needed. These interventions are designed to address, for example, high uncertainty, coordination failures, slow diffusion of technologies, or limited access to finance, especially among small and medium enterprises (SMEs).

- In high-income, innovation-ready regions such as Metro Manila, interventions that nurture an innovation ecosystem, with strong protection for intellectual property, access to venture and risk capital and smart risk-sharing tools such as credit guarantees, and a deep pool of highly skilled workers, can support the development of globally competitive firms capable of generating high-quality jobs. For large infrastructure investments, foreign exchange risk hedging facilities can also help.
- In emerging growth centers such as Cebu or Davao, where firms are moving up the value chain, programs that support SMEs in connecting to larger multinational companies (MNCs) can promote technological upgrading, and reduce trade and investment costs.
- In lower-income regions, targeted support for smallholder farmers, informal enterprises, and early-stage entrepreneurs can help unleash the potential of the agribusiness sector. The interventions include credit guarantees, blended finance mechanisms, business development services, and market access platforms to help firms grow and formalize.

**The central question of this report is how the Philippines can move into a path of higher economic growth and quality job creation.** The report examines what has worked and what requires fixing in the development path observed over the past decades, and the experience of global and regional peers, as well as the current global economic landscape. It provides a set of recommendations that are grounded on the three pillars: (i) building the foundations; (ii) building a

business-friendly regulatory and institutional ecosystem; and (iii) building the conditions for increased private capital mobilization. The recommended interventions are in turn designed to be adaptive, data-driven, and phased to align with evolving regionally heterogeneous needs.

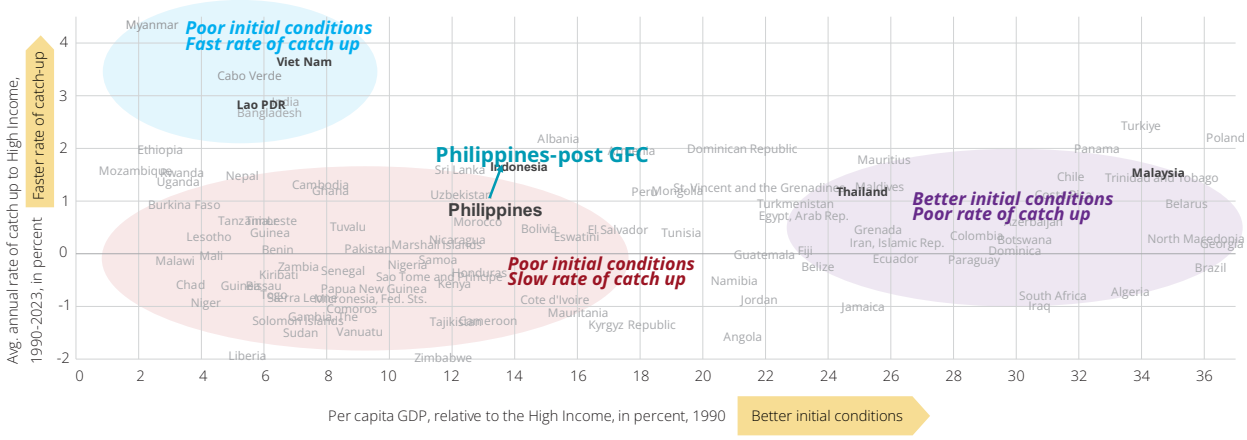
# OVERVIEW

## Introduction: Growth, jobs, and the path ahead

**Over the past 15 years, the Philippines has doubled its GDP, increased employment rates and improved job quality.** Despite this progress, growth remains low to approach the national ambition of a middle-class, poverty-free economy by 2040. At 5.2 percent per year, GDP growth lags regional peers such as the Republic of Korea or Malaysia, which sustained 7–10 percent GDP growth at similar income levels to those of the Philippines. Without decisive reforms, the Philippines risks stagnating in a cycle of slow expansion and limited job quality. This report examines the forces driving and constraining progress. It outlines urgent reforms to boost growth and jobs in the context of a more complex global economy. The reforms are organized along three pillars: foundational investments in infrastructure and human capital; better regulations and governance; and smart interventions to mobilize private capital. They aim to achieve four outcomes: (i) spatial convergence; (ii) productivity growth; (iii) resource allocation; and (iv) regional and global integration. If fully implemented, they could boost annual GDP growth by 1.4 percentage points (increasing GDP by one-quarter by 2040), real wage growth by 12.9 percent, and create new jobs for 5.1 million people by 2040.

### Figure 1

**Initial conditions relative to high-income countries and the pace of catching up**
(GDP per capita, PPP (constant 2021 US$), 1990–2023)




Source: World Development Indicators (WDI); World Bank staff calculations.

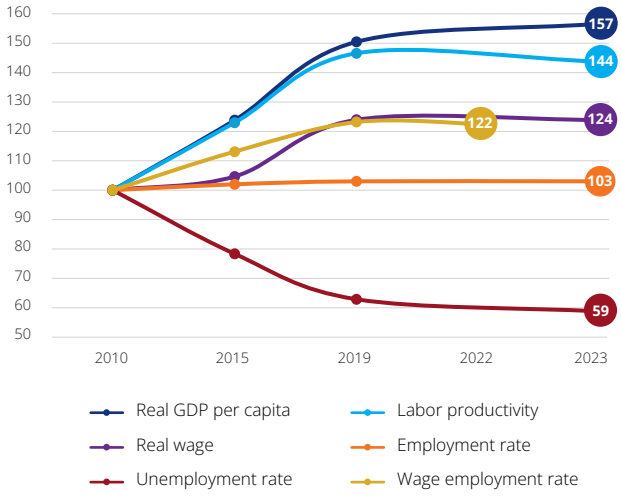
**The Philippines has transitioned from decades of sluggish growth to become one of East Asia's faster growers** (Figure 1). Progress has brought the country closer to upper middle-income status, but it has been uneven. From 1990 to 2023, the Philippines' convergence rate to high-income economies averaged just 0.8 percent, well behind Indonesia (1.8 percent), Thailand (1.3 percent), and Malaysia (1.6 percent). Growth was particularly weak before the global financial crisis (GFC),

earning the country the label "sick man of Asia". However, post-GFC, the convergence rate surged to 1.8 percent despite pandemic disruptions. Real GDP growth accelerated from 3.7 percent (1990–2010) to 5.2 percent (2010–2023), with per capita growth rising from 1.5 to 3.5 percent. At this pace, the economy will double in size every 13.5 years, faster than would have been the case at its pre-GFC growth rates.

**Job creation matched the good growth performance.** Since 2010, unemployment rates declined, reaching a record low of 3.8 percent in 2024, with 11.7 million jobs created. This was mainly driven by improvements in labor market conditions for women: while the female working-age population grew at 2.0 percent, the number of women actively participating in the labor market grew at 2.8 percent anually. Moreover, the labor outcomes show progress in occupational types: more wage-employment rather than self-employment, and more jobs in construction and domestic services relative to agriculture, where productivity is the lowest.

## Figure 2

**Growth, productivity gains, and better labor outcomes**
(Index 2010=100)




Source: Philippine Statistics Authority (PSA); World Bank staff calculations.

**Rising labor productivity has supported real wage growth.** Labor productivity increased by 44 percent, accompanied by a 24-percent rise in real wages (Figure 2). The partial transmission from labor productivity to wages reflects that a sizable share of productivity gains occurred in non-wage-earning sectors (agriculture and the informal sectors), while another share is likely to have been appropriated by firms' profit margins.

**The post-2010 growth surge was driven by private investment, plus the right conditions and high returns.** Foundational infrastructure investments, particularly in connectivity, together with a few pro-investment reforms, accelerated private investment, underpinning growth. Nonetheless, more physical and human capita is needed, as well as a better regulatory environment, to boost productivity and innovation.

**Approaching the national ambition of becoming a middle-class country, where nobody is poor by 2040, requires faster growth and better paying jobs.** At similar levels of income per capita, the experiences of the Republic of Korea and Malaysia were based on real annual GDP growth rates of 7–10 percent, sustained over decades, on account of better integration into the global marketplace and faster productivity growth. This is significantly higher than the Philippines' 5.2 average over the past 14 years, on account of growth in non-tradable sectors and investment acceleration.

**In a more complex global economy, accelerating GDP and wage growth with employment creation at this stage of development requires deep reform commitment.** External factors make faster growth more challenging: rising global policy uncertainty, and trade and investment fragmentation, labor-saving technological disruptions such as automation and Gen-AI, and escalating climate risks that add volatility and costs for firms, farms, households, and the Government. To achieve the Philippine Development Plan's (PDP) target of 6.5–8.0 percent annual GDP growth and create jobs, bold, sustained reforms are essential, as is their full implementation.

**This overview presents the report's key messages, analyzing the drivers and barriers of growth, and job creation over the past 15 years and a path forward.** It outlines a reform agenda that, if fully implemented, can accelerate GDP growth closer to 7 percent, and sustain employment expansion while driving real wage gains. The choices made today will determine whether the Philippines remains on an average growth path or leaps toward a more dynamic, innovative, and open economy.

# Four features of growth and job creation in the Philippines

## Feature 1: Growth was broad-based, pro-poor, spatially balanced on the back of job creation

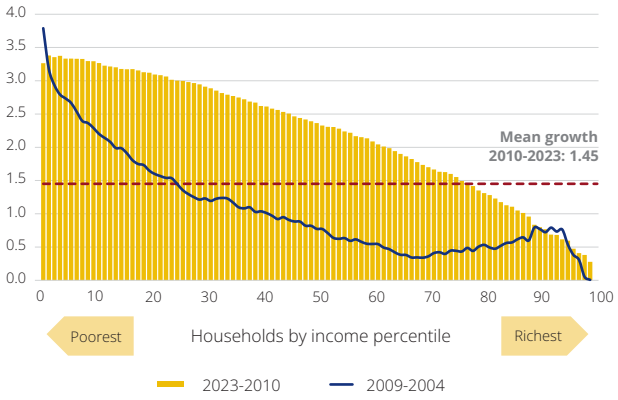
**Pro-poor growth in the post-2010 period was on the back of better labor outcomes.** During 2010–23, real incomes of the bottom 40 percent grew faster than those of the wealthiest 20 percent (Figure 3). Key to this process was a shift from self-employment (mostly in agriculture) to wage jobs (mostly in services). Urban jobs paid better than rural ones as they also revealed higher labor productivity. Employment growth responded to GDP growth, particularly in lower-productivity regions such as BARMM and Eastern Visayas. Social protection also helped. The Pantawid Pamilyang Pilipino Program (4Ps), a conditional cash transfer program, increased its coverage to support poor households, from 1 to over 4 million between 2010 and 2022.[4]

**Post-2010 growth has also become more spatially balanced, contrasting with the spatial divergence observed pre-2010** (Figure 4). Low- and medium-income regions drove aggregate GDP growth after the GFC. Between the pre-GFC and post-GFC periods, almost all the GDP growth increase came from medium-income regions (MIRs), with low-income regions (LIRs) contributing a small portion, while Metro Manila's contribution remained stable. This trend departs from past growth dynamics dominated by high-productivity regions such as Metro Manila. During 2000–2009, LIRs' value added per worker grew at an anemic annual rate of 2.3 percent, that of MIRs at 2.1 percent, while the National Capital Region (NCR) grew at 3.0 percent. This trend reversed post-2010, with LIRs and MIRs growing at 3.2 and 2.5 percent, respectively, while NCR value added per worker growth slowed to 1.5 percent annually.

4 The growth incidence curve of 2004–2009 contrasts sharply with that of 2010–2023. While it also shows pro-poor growth, it is consistent with: (i) lower average growth and (ii) lower income growth of the middle class (those between the bottom 40 and the top 10 percent).

## Figure 3

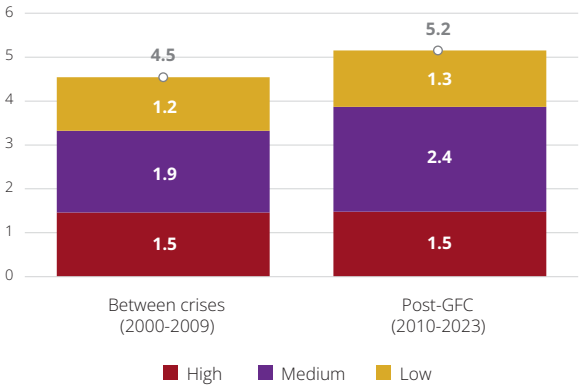
**Growth was pro-poor…**
(Incidence curves, annual real income growth, in percent)




Source: PSA; World Bank staff calculations.

## Figure 4

**…and came mostly through low- and middle-income regions' catching up**
(Contribution to GDP growth by region, CAGR, in percent)

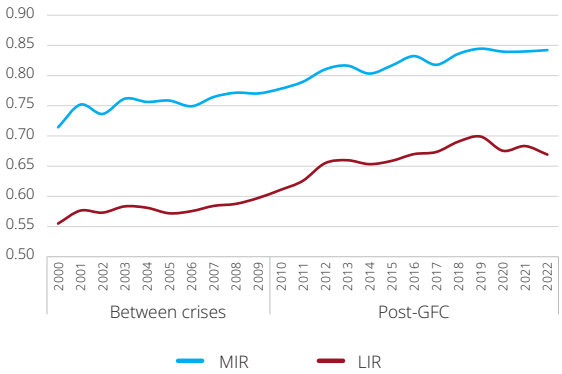



Source: PSA; World Bank staff calculations.

**Better jobs and more formal firms were key for spatial income convergence** (Figure 5, Figure 6). In both LIRs and MIRs growth came with better-quality jobs. LIRs' wage employment increased from 56 percent of the NCR's in 2000 to 69 percent in 2022. In MIRs, it rose from 71 to 84 percent. These trends in the convergence of wage employment are aligned with formal firm creation, substantially faster in LIRs and MIRs than in the NCR. They are also aligned with the increasing importance of total factor productivity (TFP) in contributing to growth in LIRs (1.2 percentage points) and MIRs (0.9 percentage points) compared with in the NCR (slightly negative).

## Figure 5

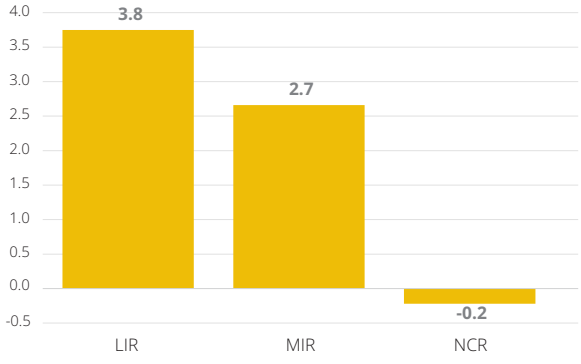
**In low- and middle-income regions, wage-employment convergence was key for growth…**
(Wage employment share in total employment, index NCR=1)




Source: PSA Labor Force Surveys (LFS); World Bank staff calculations.

## Figure 6

**…on the back of new firms emerging**
(Annual growth in number of formal firms 2012–19, in percent)




Source: PSA Annual Survey of Philippine Business and Industry (ASPBI); World Bank staff calculations.

**Sustaining spatial convergence requires Local Government Units (LGUs) to play a key role.** The 1991 Local Government Code ensured that LGUs manage over 80 percent of the country's road network, directly influencing trade and investment attraction at the regional level, as well as the provision of basic services including health and education. Their role expanded further with the 2019 Mandanas-Garcia ruling, implemented in 2022, which increased the share of national tax revenues allocated to LGUs from 2.9 to 4.4 percent of GDP. Challenges such as limited administrative capacity, inefficient budgeting practices, and low capital outlay execution rates (just over 50 percent) constrain LGUs' effectiveness in driving regional growth and job creation, emphasizing the need for improved local governance and capacity-building moving forward.

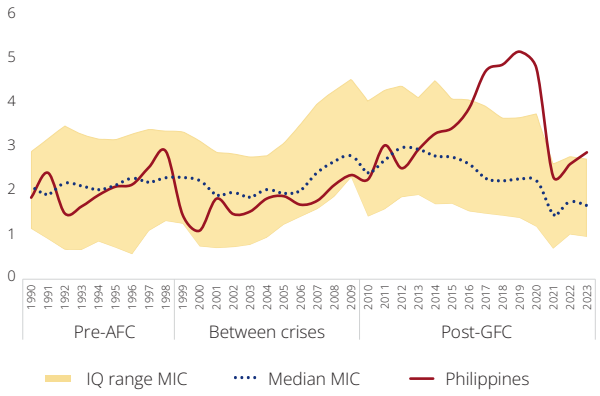
## Feature 2: Growth was driven by capital accumulation rather than by productivity growth

**Capital accumulation accounted for over 90 percent of the Philippines' growth post-GFC.** Capital accumulation contributed over 5 percentage points during the peak of the investment acceleration period (2017–2019), placing the Philippines among the middle-income countries (MICs) that most relied on investment for growth (Figure 7). Labor and TFP contributions remained considerably lower. Still, as a share of GDP, investment remained 10 and 11 percentage points below that observed in Viet Nam and Indonesia, respectively, with space for capital deepening (Figure 8). While investment is needed to keep growing, so is productivity growth, which has longer-lasting effects. In fact, evidence shows that productivity accounts for most of the difference in standards of living across countries.[5]

### Figure 7

**Growth came with a large contribution of capital accumulation...**

(Capital contribution to growth, in percentage points)

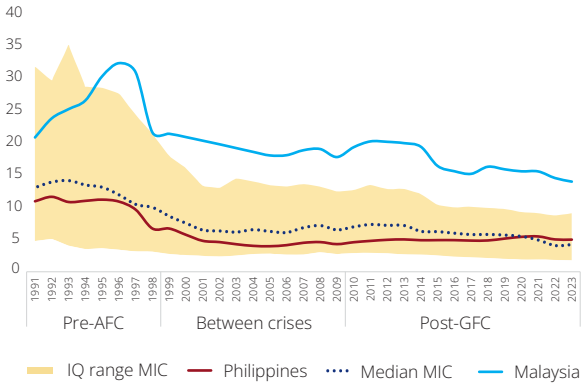



Source: Macro Poverty Outlook (MPO); World Bank staff calculations.

### Figure 8

**...although there is still space for capital deepening**

(Capital stock per worker, in percent of US capital/ worker)




Source: MPO; World Bank staff calculations

5 Investment is subject to diminishing returns, while productivity is not. Krugman (1997) argues that "productivity isn't everything, but in the long run, it is almost everything. A country's ability to improve its standard of living over time depends almost entirely on its ability to raise its output per worker".

**Public investment focused on connectivity soared.** Public investment, stagnant at around 3 percent of GDP during the period 2000–2015, increased significantly under the Build, Build, Build (2017) and Build Better More (2022) initiatives, committing to maintain infrastructure spending at a floor of 5 percent of GDP, and with a focus on connectivity. To meet infrastructure needs within fiscal constraints, implementing reforms to the public-private partnership (PPP) code aimed at streamlining approvals, improving transparency, ensuring equitable risk-sharing, and enhancing private sector participation all remain critical.

**With higher public investment, pro-investment reforms, and increased macroeconomic stability the profitability of private investment increased.** This led to a private investment acceleration. Private investment increased from 14.3 percent of GDP in 1999–2009, to 18.3 percent in 2010–2023. The increase was on the back of increased macro stability (the Philippines first achieved investment grade in 2013), and pro-investment reforms (e.g., PEZA framework, retail trade liberalization, fiscal incentives and trade agreements), coupled with public investment that reduced costs of inter-island transport, e.g., Roll-on/Roll-off (RORO) ports.[6]

## Feature 3: Sectoral re-allocation of jobs helped growth, but stagnant top tier firms hampered it

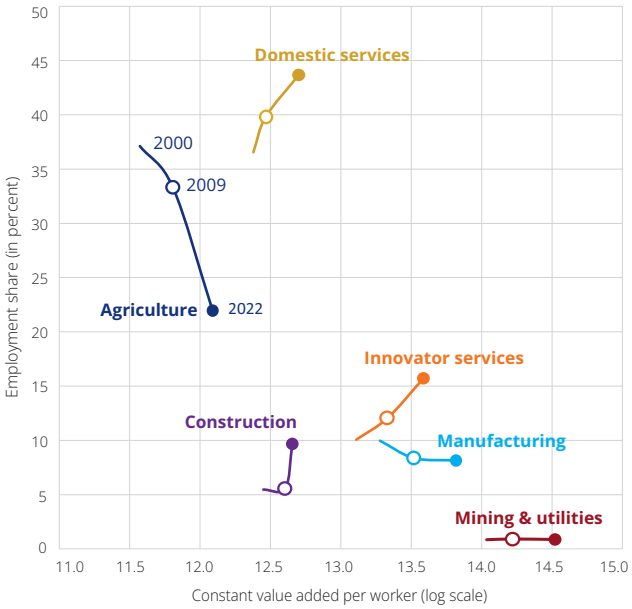
**Growth in the Philippines has shifted toward higher productivity sectors, improving allocative efficiency and helping workers climb the economic ladder** (Figure 9). Wage employment rose from 55 percent in 2010 to 64 percent in 2022, making the Philippines among the MICs with the fastest wage-employment growth (Figure 10). It was driven by strong gains in construction, commerce, and modern services such as finance and business services. While opportunities in services have reduced self-employment in low-productivity agriculture, 22 percent of jobs remain in this sector, highlighting the need for further job creation in higher productivity sectors.

**While job creation accelerated, the most productive firms struggled to expand, limiting the creation of high-wage employment and slowing overall wage growth** (Figure 11). In 2012, the Philippines' top manufacturing firms (quintile 5) paid three times the median wage paid by the least productive firms (quintile 1) and employed seven times more workers; in services, these gaps were five- and threefold, respectively. Yet, over the past decade, job growth occurred in mid-productivity firms (quintiles 2–4), with top manufacturing firms shedding workers and top services sector firms expanding modestly. In well-functioning economies, instead, top firms drive growth and absorb workers from less productive ones. This striking result deserves policy attention and action.

6 Roll-on/Roll-off (RORO) terminals were often developed under public-private partnerships, or build-operate-transfer schemes in which the private sector finances, constructs and operates infrastructure projects for a fixed period before it transfers ownership to the government.

## Figure 9

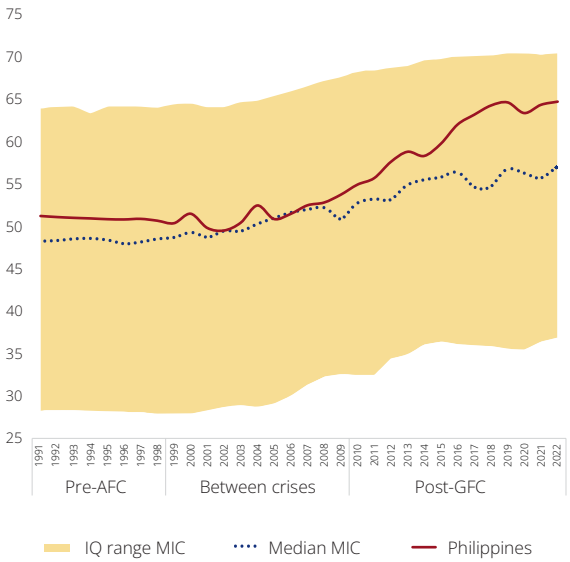
**Growth was enhanced by reallocation of workers to high(er) productivity sectors…**
(Aggregate employment and labor productivity)




Source: PSA; World Bank staff calculations.

## Figure 10

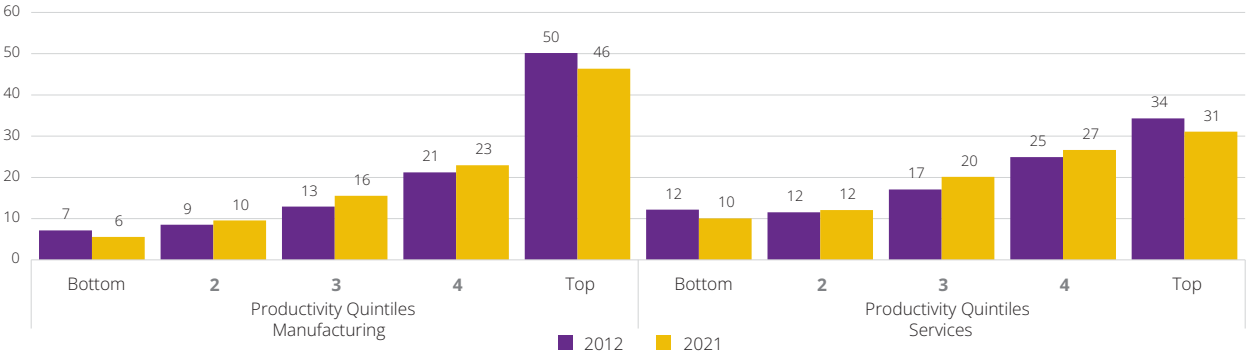
**…and by better types of jobs, as wage employment soared**
(Share of wage employment in total employment, in percent)




Source: Darvas (2012) and Darvas (2021); World Bank staff calculations.

## Figure 11

**However, while productivity and employment shares are positively linked, allocative efficiency at the firm level stagnated as top productivity firms failed to scale up**
(Employment shares by productivity and sector, in percent of sector employment)




Source: PSA ASPBI; World Bank staff calculations.

**Stagnant job creation among frontier firms in the Philippines reflects limited competition and persistent regulatory and capability constraints.** Unlike the Republic of Korea's chaebols, which drove industrial expansion, Philippine conglomerates largely remain in rent-seeking, low-competition industries.[7] Strengthening antitrust enforcement and opening key sectors to foreign

7 See Mendoza (2024).

competition can unlock productivity growth. Recent de jure reforms opening the transport, telecoms, and renewable energy sectors to foreign entrants could enhance competition, but complex business regulations remain de facto barriers. Registering a foreign company takes 106 days, placing the Philippines in the bottom quintile for business entry among 50 countries surveyed and in quintile 4 for market competition.[8] Legal complexities of terminating permanent employees discourage employers from opening these positions. Innovation struggles also stem from capability constraints: 90 percent of 10-year-olds are deficient in reading comprehension, undermining future workforce productivity.

## Feature 4: Increased inward orientation of growth and jobs

**Three in four new jobs created since 2010 have been in non-tradable sectors** (Figure 12). As the economy doubled, its tradable sector shrank. Assets of non-tradable firms grew faster than those of firms in tradables, as more credit and foreign direct investment (FDI) flowed into them. The export-to-GDP ratio dropped from one-third to slightly more than one-quarter, while the number of goods exporters declined by 23 percent. Exports of semiconductors, the Philippines' main manufacturing export, declined from 11.4 to 7.7 percent of GDP during the period. Services exports, instead, rose from 8 to 10 percent of GDP, driven by IT and business services, now contributing 27 percent of total exports, and 1.4 million jobs for Filipinos. Yet, jobs in these sectors are unlikely to experience rapid productivity growth as jobs in tradables do, since tasks in non-tradable jobs are less scalable than those in tradable sector jobs.[9]

**The appreciation of the real effective exchange rate (REER), a complex business climate, and relatively high energy and logistics costs, led to a relative contraction of the tradable sector.** The REER has appreciated since the early 2000s (Figure 13) on the back of high remittance inflows, and increasing tourism and business process outsourcing (BPO) exports.[10] A stronger peso made exports more expensive, reducing profits and discouraging investment in tradable industries. Analysis from this report shows that a 1-percent real appreciation leads to a 0.59- percent decline in investment by tradable firms.[11] In addition, a complex business environment, high energy costs, and competition challenges in key logistics sectors put pressure on the cost structure of tradables (particularly of manufacturing).

8 See World Bank (2024a).

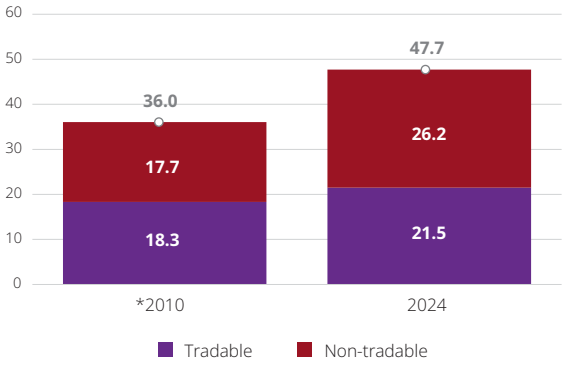
9 One software programmer can write code for a million users, or a semiconductor worker can test and pack thousands of semiconductors per week with the right technology, while for non-tradables scaling is more difficult: one line cook cannot make food for a million guests, nor can one hairdresser give a million haircuts in a week.

10 The link between remittances and the appreciation of the REER in the Philippines has been explored by Tuaño-Amador et al. (2007), Bayangos and Jansen (2011). Williamson and de Dios (2014) also link remittances inflows with REER appreciations in Philippines (p. 57). Hien et al. (2020) show the link between remittances and REER appreciations for all developing economies in Asia.

11 The results emerge from a regression analysis relying on a panel of 389 publicly listed firms in the non-financial economic sectors, that classifies sectors into tradables and non-tradables and spans the period 2006–2023. The analysis relies on a difference-in-differences approach to examine the impact of REER changes on firms' asset growth. Firms in the tradable sector, which are more exposed to exchange rate fluctuations, serve as the treated group, while firms in the non-tradable sector serve as the control group. The objective is to identify whether REER changes differentially affect investment in tradable versus non-tradable firms, isolating the sensitivity of investment decisions to exchange rate dynamics.

## Figure 12

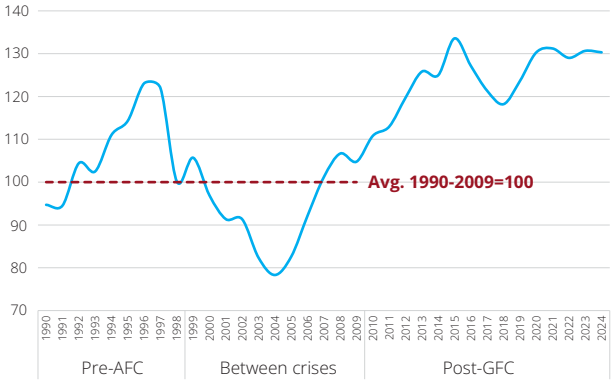
**Non-tradables accounted for 8.5 out of 11.7 million jobs created...**
(Employment by type of sector, in millions)




Source: PSA; World Bank staff calculations.

## Figure 13

**...as the appreciation of the real exchange rate made non-tradables more profitable**
(Multilateral, avg. 1990-2009=100)




Source: Darvas (2012) and Darvas (2021); World Bank staff calculations.

**The relative decline of tradables puts long-term growth and quality job creation at risk.** Philippine firms in globally connected sectors outperform those in domestic sectors in productivity, wages paid, and the way they use key inputs such as energy. Their productivity also grows after they start exporting, or after being acquired by a multinational corporation (MNC). A decrease in importance of the tradable sector reduces the scope for these productivity and wage gains, limiting potential output and quality job growth. Reforms to increase competition in logistics, such as the full implementation of the PSA amendment and Konektadong Pinoy, simplification of business regulations, investments in connectivity, and reductions in trade and investment costs through regional trade agreements are critical for creating jobs in the tradable sector. Equally relevant is ensuring that firms invest in the design of reskilling programs for workers to fit the needs of modern manufacturing and digitally-enabled services with export potential. The Enterprise Based Education and Training (EBET) initiative is a step in that direction.

## Box 1

### Looking back: What worked and what didn't

**The remarkable transformation of the Philippines over the past 15 years provides a solid foundation to keep growing.** But achieving the national ambition of a middle-class society requires yet faster GDP growth, and the creation of higher productivity jobs. The way to achieve that is to double down on reforms. Looking back allows us to identify what worked and needs sustaining, and what didn't and needs fixing.

**Effective Catalysts of Faster Growth and Job Creation:**

- Pro-investment policies and stable macroeconomic management attracted capital inflows.
- Strategic infrastructure investments substantially improved regional connectivity, directly boosting job creation, and crowding in private investment.
- Economic Zones helped jump-start the information technology business process outsourcing (IT-BPO) industry.
- Social protection policies and targeted interventions effectively reduced poverty and improved income distribution.

**Persistent Barriers to Faster Growth and Job Creation:**

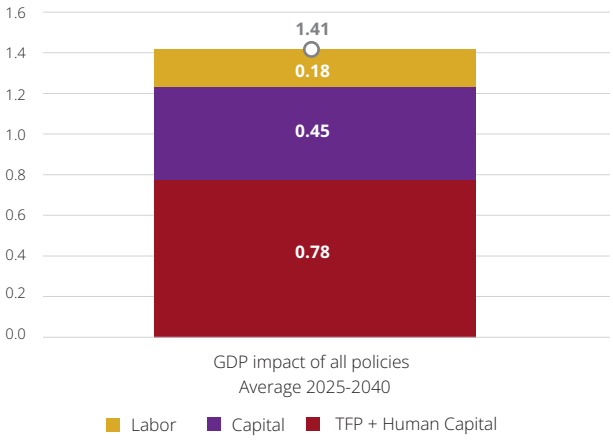
- Weak productivity growth due to low tech adoption constrained high-quality job creation.
- Low capacity of LGUs that struggle to deliver quality services linked to human capital.
- Limited market contestability discouraged frontier firms' dynamism; land market distortions created limited destruction in agriculture; labor market distortions led to some talent misallocation.
- An appreciation of the REER, coupled with high costs for key manufacturing sectors reduced competitiveness of tradable sectors and impeded the creation of higher productivity employment.
- Insufficient FDI spillovers limited technological and managerial advancements within local firms.
- Limited adaptation investment, and misaligned incentives increased exposure to climate shocks.

# Charting the path forward: Targeting 7 percent growth and quality job creation

**The Philippines stands at a pivotal juncture in a new global context.** Middle-income countries face new challenges. Trade policy uncertainty combined with slow global growth suggest that global demand will not act as a locomotive for developing economies such as the Philippines, all else equal. The scope for technology transfers from trade and investment may be reduced, highlighting the need for domestic technology adoption. Technological disruptions such as GenAI may erode some of the comparative advantage of the Philippines' star sector, IT-BPO, weakening the link between growth and jobs. Climate change will further affect the growth and jobs landscape.

## Figure 14

**The proposed reforms will accelerate TFP and human capital, while also increasing investment and labor contributions to GDP growth**
(Contribution of proposed reforms to additional GDP growth relative to the baseline, in p.p.)




Source: World Bank staff elaboration based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024).

**Accelerating GDP growth from 5.4 to 6.8 percent by 2040, and increasing employment and wage growth, are feasible.** The roadmap laid out in this report shows how: reforms to boost productivity and human capital are estimated to add 0.78 percentage points to growth annually, those to boost capital deepening a further 0.45 percentage points, and those leading to increased labor force participation another 0.18 percentage points, altogether delivering a 1.41-percentage-point lift to annual GDP growth (Figure 14). This shifts the growth trajectory from 5.4 to 6.8 percent. The reforms are grouped along three pillars: foundational infrastructure and human capital investment, better regulations and governance, and private capital mobilization. They are designed to work together to overcome the country's longstanding constraints. Their full impact hinges on determined implementation.

**The 12 priority reforms are presented below.** They are grouped into the three pillars described previously: foundational investments in infrastructure and human capital; better regulations and governance; and private capital mobilization. The full set of reforms is presented in Chapter 6.[12]

## 1. Foundational investments in infrastructure and human capital

**Fast growth has been on the back of investment growth.** However, the infrastructure gap remains high, and more marked in the human capital space. Three actions will help narrow it.

**First, sustain public investment with a focus on connectivity infrastructure.** The increases in public investment rates have been key for growth. Sustaining them to keep providing the needed public goods, particularly for connectivity, will help accelerate spatial convergence and catalyze growth and job creation. This requires commitment to domestic revenue mobilization (public investment will be growth-enhancing if it is consistent with fiscal sustainability), and improved public investment management, for maximum efficiency of expenditure.

**Second, support private climate adaptation by removing bottlenecks and prioritizing resilience.** Each additional day of heavy rain reduces SME productivity by 1 percent and wages by 1.4 percent, highlighting the economic cost of climate impacts. LGUs can reduce these losses by prioritizing flood control investments in high-risk economic hubs and streamlining building permits

12 The 12 priority recommendations are identified considering economic impact and feasibility, given the capacity of implementers.

to cut reconstruction time and aftershock costs. At the national level, the Department of Budget and Management (DBM) can operationalize the "value-for-money" principle in procurement to account for climate resilience, ensuring that durable materials, not just the lowest price, guide public investments.

**Third, boost human capital by fast-tracking implementation of the Enterprise Based Education and Training (EBET) Act, and scaling up STEM and digital skills to better address the AI challenge.** The EBET Act is key to supporting flexible, industry-aligned worker upskilling. Technology adoption in the Philippines remains low—just 3 percent of firms are fully digital, compared with 11 percent in Viet Nam. A major constraint is the lack of workforce skills, especially among innovative firms. The Technical Education and Skills Development Authority (TESDA), as lead implementor, can operationalize the EBET Act as a cost-sharing scheme that ensures training programs are co-designed with industry, continuously updated, and responsive to evolving needs. Ensuring active public-private dialogue is key to making training relevant, adaptive, and scalable.[13] Also, STEM and digital skills are needed to move up the business services value chain and stay ahead of AI disruption. The IT-BPO sector powers 27 percent of services exports and employs 1.4 million workers—but many roles are vulnerable to automation. To preserve jobs, the Philippines must shift to higher-value segments such as telemedicine, finance, and professional services. This requires a bold push: expanding in STEM education, investing in digital skills at scale, and using mutual recognition agreements to attract global and diaspora talent. Without this shift, the country risks losing ground to AI; with such a shift, it can lead to the next wave of service-driven growth.

## 2. Better regulations and governance

**First, ensure that regulations catch up with infrastructure progress to maximize investment impact.** Public infrastructure has been key for spatial convergence, evidenced by the RORO port system expansion. While the resulting lower transport cost have led to more firms and jobs, outdated regulations limit the gains. Cargo transport still requires separate permits for land, sea, and air, while cabotage restrictions prevent foreign operators from competing in domestic shipping. By streamlining permits and fully liberalizing shipping, the Department of Transport (DOT) can help cut connectivity costs and accelerate economic integration.

**Second, remove de facto barriers to market entry to make openness reforms work.** Legal reforms have opened markets de jure, but failure to implement still blocks entry. It takes 106 days for a foreign company to register in the Philippines, 30 percent longer than for a domestic firm, and 91 days more than in Singapore, due largely to procedural delays. Telecom newcomers must secure a congressional franchise, a process that takes an average of five years. Setting up an onshore wind project involves 373 government interactions, with 173 duplications caused by poor agency coordination. Securities and Exchanges Commission (SEC) action on registration, LGU streamlining of permitting, and coordinated whole-of-government reform will reduce red tape and enable real market contestability.

13 Because firms cannot appropriate all the benefits from training its workers (as workers could leave after the firm incurs in the expense, and thus other firms may benefit), on-the-job training is typically under supplied. This market failure justifies the public intervention.

**Third, facilitate land consolidation to enable productivity-enhancing reallocation in agriculture.** In dynamic sectors, underperforming producers exit and resources shift to more productive ones, raising overall output. In Philippine agriculture, this process is stalled by land retention limits of 5 hectares per owner. These restrictions are designed to prevent the concentration of land ownership and preserve the source of income of rural communities. Cooperatives and joint ventures offer legal pathways for clustering and committing to those objectives. However, these arrangements remain rare due to high complexity and costs. The Department of Agriculture and LGUs can unlock scale by streamlining support for cooperative formation, simplifying joint venture processes, and actively promoting models that lower entry barriers for commercial partnerships.[14]

**Fourth, adopt a proportional contribution system for part-time work to boost female labor force participation (FLFP) and correct talent misallocation.** High fixed employer contributions make part-time hiring costly, discouraging flexible work options despite legal allowances. This disproportionately affects women, who dominate higher education—55 percent of college graduates—but remain underrepresented in the workforce. The Department of Labor and Employment (DOLE) can address this by aligning contributions with hours worked, making part-time jobs more viable and unlocking underused talent at scale.

**Fifth, negotiate and implement deep trade agreements that drive domestic reform and global integration.** The Philippines' trade network covers only one-third of global GDP, limiting diversification and constraining access to higher-value markets. While existing agreements focus primarily on tariffs, deep trade agreements go further, tackling domestic barriers such as restrictive services regulations, burdensome customs procedures, and uncompetitive procurement rules. These are essential to reduce trade and investment costs, and enable firms to join global value chains. The Department of Trade and Industry (DTI) can lead by advancing the EU-Philippines Deep Trade Agreement and forging new partnerships in emerging markets, embedding domestic reforms in negotiations. In a volatile global environment, open and rules-based trade—anchored in transparency, low barriers, and broad participation—is a proven driver of resilient and inclusive growth.

**Sixth, implement competition-enhancing reforms in energy, logistics, and telecoms to cut costs for tradable sectors.** The Philippines faces some of the highest electricity, internet, and transport costs in the region, raising input costs for firms competing globally. Revisions to the Electric Power Industry Reform Act (EPIRA) of 2001 can increase competition in power generation and reduce electricity prices. Full implementation of the PSA amendment and Konektadong Pinoy will improve efficiency and lower costs in telecom and logistics. LGUs play a key role in streamlining permitting processes for renewable energy, which is a major bottleneck. Better design of regulatory agencies such as the Civil Aeronautics Board (CAB), the Maritime Industry Authority (MARINA), and the National Telecommunications Commission (NTC) will help reduce costs and improve quality in logistics and telecom services.

14 In Peru, for example, the government has supported land consolidation by providing legal services on a cost-shared basis to small holders. This has facilitated the consolidation to supply at scale. Today, Peru is a key player in various global agriculture supply chains.

**Seventh, strengthen local service delivery by building LGU capacity and aligning incentives.** The DBM and Department of the Interior and Local Government (DILG) can play a central role in enabling LGUs to deliver better services—essential for human capital gains and quality job creation. While LGUs now have a larger role in service delivery, performance depends on financial management, resource availability, and spending quality. First, the National Tax Allotment (NTA) formula can be refined to better match local needs while minimizing disincentives for revenue collection. Second, performance-based transfers—building on metrics such as the Seal of Good Local Governance—offer a foundation for scaling operational improvements. Third, sector-specific, results-linked grants tied to enforceable service standards create levers to address persistent delivery gaps.

## 3. Private capital mobilization

**The Philippines' investment-grade status and sustained macroeconomic stability have lowered economy-wide risks and helped attract growing volumes of private investment.** However, sector-level constraints, including fragmented supply chains, regulatory opacity, and capability gaps among firms, continue to suppress capital flows into tradables, innovation-intensive industries, and emerging growth regions. While reforms to infrastructure and the regulatory environment lay the groundwork, they may not be sufficient to mobilize private capital at the scale needed to accelerate productivity and job creation. Public action is needed to reduce investment risk through instruments such as credit guarantees, blended finance, and mechanisms to support early-stage innovation. These tools are especially relevant in climate-aligned sectors, digitally-enabled value chains, and SME-MNC linkages where perceived risks remain high. Even in an investment-grade economy, such interventions remain critical to crowd in capital, provided that they are fiscally prudent, transparent, and avoid contingent liabilities. Two interventions are suggested.

**First, introduce supplier development programs to link SMEs with MNCs and large firms, and close the productivity spillover gap.** FDI has increased, but Philippine SMEs are not benefiting. Evidence shows weak capability on the SME side is a binding constraint. The DTI can address this by designing targeted supplier development programs that build SME readiness and facilitate MNC linkages. These programs should be piloted with built-in impact evaluations to ensure that they are adaptive, cost-effective, grounded in good international practices.[15]

**Second, focus innovation support where conditions are ripe, consolidate programs for scale, and create room for deeper venture capital (VC) markets.** Despite a wide range of programs, R&D investment remains low, constrained by fragmentation and limited scale of innovation policies. The Department of Economy, Economy, Planning, and Development (DEPDev) and the Department of Science and Technology (DOST) can boost impact by merging duplicative efforts and concentrating resources in high-readiness areas such as Metro Manila, where firms' innovation

15 Increased FDI in Philippines helped create better paying jobs and increase efficiency (including in the use of energy). However, there is limited evidence on productivity spillovers on domestic firms. The exception is in energy efficiency where FDI, particularly from EU MNCs shows positive spillover effects along the value chain, suggesting MNCs transfer knowledge to local suppliers if they face incentives to do it (in this case related to scope 3 reporting of ESG standards).

capabilities are taking root. To deepen VC markets, amend regulations to explicitly allow pension funds to invest in VC, subject to their mandates and risk tolerance, unlocking a vital source of long-term capital.

## Projected outcomes and the roadmap

**The implementation of the set of reforms recommended in this report is estimated to increase annual GDP growth to 6.8 percent, create over 5.1 million additional jobs, and boost real wages by 12.9 percent by 2040.**[16] The reforms lead to increases in labor force participation while also increasing demand for labor, through productivity gains. This combination results in increased employment with 12.9 percent higher real wages by 2040, driven by wage increases in industry and services.[17] Results show more balanced growth across regions and diversified exports, leading to reduced vulnerability. The estimated increase in GDP growth rates brings it closer to the rates at which the Rep. of Korea was growing at the same level of income per capita of the Philippines today, and to the target range envisioned in the PDP.

**For the Philippines, transitioning out of middle-income status is within reach.** It requires foundational investments, better regulations and smart interventions to mobilize private capital so that spatial convergence is sustained, innovation and productivity growth nurtured, distortions preventing a better allocation of resources and talent removed, and regional global opportunities capitalized upon. This report provides a concrete roadmap. Building on the momentum of the past 15 years, decisive action can put the Philippines on higher path of economic development.

16 In the baseline scenario, real wages are expected to grow slightly below 2.7 percent per year, while under the reform scenario, these are expected to grow at 3.4 percent per year. The cumulative difference over the 2025–2040 period is of 12.9 percent higher wages.

17 In a baseline scenario without reforms, the labor force would have reached 62.8 million people by 2040, from 48.8 million in 2025. Under the reform scenario, this report estimates the labor force to increase to 67.7 million by 2040. The reforms encourage participation in the labor force, bringing in additional 4.9 million Filipinos into it. Similarly, the reforms boost employment levels by 5.05 million, from a baseline of 64.9 million by 2040 under no reforms, to an estimated 69.9 million under the reform scenario. Thus, the jobs created by this reform roadmap are the result of more Filipinos participating in the labor force, and a smaller pool of unemployed.

# CHAPTER 1
# MACROECONOMIC FEATURES OF GROWTH AND JOBS

*This chapter examines the macroeconomic trends of growth and job creation in the Philippines over past decades. Since the GFC, GDP has grown rapidly and employment outpaced population growth. Growth had four drivers: spatial convergence, labor reallocation, investment, and the relative expansion of the non-tradable sector. Enhanced macroeconomic stability, public investment, and de jure reforms supported growth, but challenges in competition, regulation implementation, and declining tradable sector profitability have hindered productivity. Fast growth and job creation rely on implementing current and new reforms that involve foundational investment, better regulations, and private capital mobilization, which are critical in sustaining spatial convergence, boosting firms' productivity, improving resource allocation, and expanding tradable sectors.*

## Patterns of growth and jobs

**Over the past 14 years, GDP growth in the Philippines has accelerated.** Despite the disruption caused by the COVID-19 pandemic, growth has been robust, with convergence to high-income countries' (HICs) rates averaging 0.8 percent since 1990 and 1.9 percent post-GFC. To understand these dynamics, growth is analyzed across three subperiods: pre-Asian financial crisis (AFC) (1990–1998), between crises (1999–2009), and post-GFC (2010–2023). GDP per capita growth increased from 0.4 percent in the pre-AFC period to 4.6 percent with low volatility post-GFC. Growth showed four features: it was (i) convergence-driven; (ii) capital-led; (iii) aided by productivity-enhancing reallocation of jobs; and (iv) dominated by the relative expansion of non-tradables over tradables.

### Convergence-driven, broad-based growth and job creation

**Growth was broad-based favoring low-income regions, and the poorest 40 percent of Filipinos.**[18] Real income growth (2010–2023) averaged 3.1 percent for the bottom 40 percent, exceeding the median and top 20 percent (Figure 1.1), driven by job creation and social protection programs (4Ps). Low- and medium-productivity regions drove the acceleration in aggregate GDP growth (Figure 1.2), while the contribution of Metro Manila remained stable across the two periods.

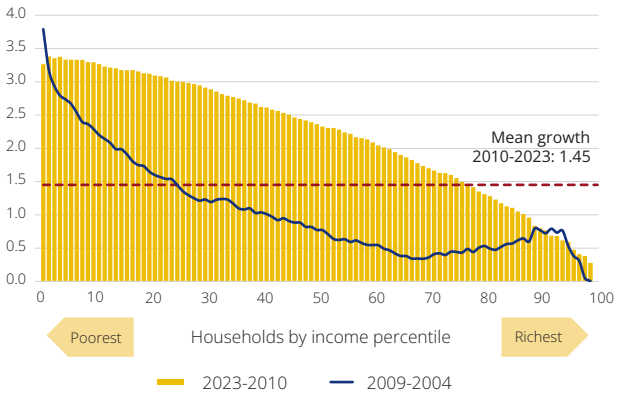
**The labor market responded well to increased growth, generating both more and better jobs.** Between 2010 and 2022, over 11.7 million jobs were created, raising the rate of those employed among the 15 to 64 working-age population from 61 to 64 percent (Figure 1.3). The average annual

18 Chapter 2 delves into firm dynamics, highlighting the role of lagging and middle firms in driving growth, while Chapter 3 explores the spatial convergence that underpinned the broad-based regional contributions to the Philippines' economic expansion.

unemployment rate also declined from 7.6 to 5.6 percent (3.8 percent in 2024). In addition, job quality improved as employment shifted from agricultural self-employment to higher-productivity wage jobs in the services sector.

## Figure 1.1

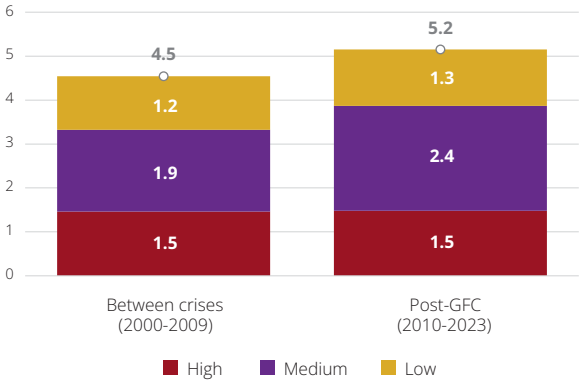
**Growth incidence curve**
(Annual growth, in percent)




Source: PSA; World Bank staff calculations.

## Figure 1.2

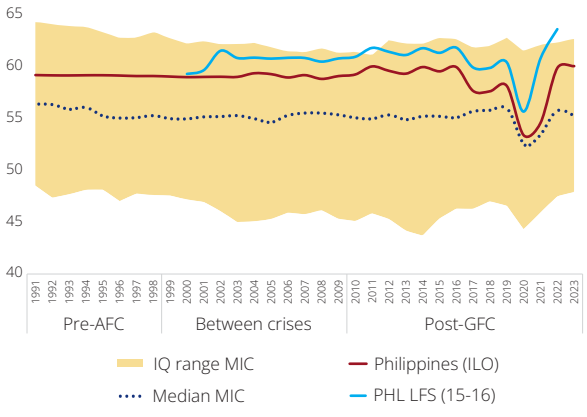
**Contribution to GDP growth by region**
(CAGR, in percent)




Source: PSA; World Bank staff calculations.

## Figure 1.3

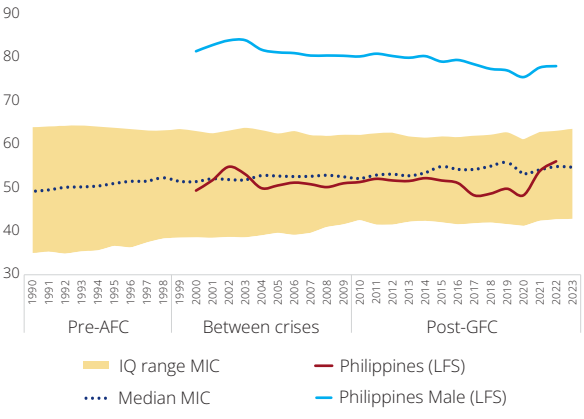
**Employment to population ratio**
(In percent of WAP aged 15+)




Source: PSA LFS; ILO; World Bank staff calculations. Note: PHL LFS (15-64) corresponds to the employment rate of ages 15 to 64 calculated based on LFS.

## Figure 1.4

**Female labor force participation rate**
(In percent of female labor force ages 15–64)




Source: PSA LFS; ILO; World Bank staff calculations.

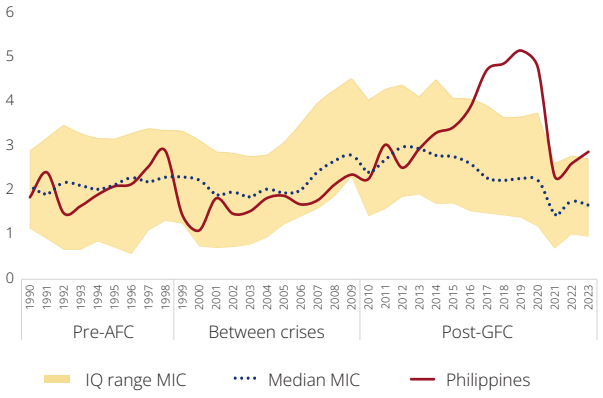
**There is still room for increased inclusion** (Figure 1.4). Despite moderate improvement, female labor force participation rate (FLFP, 56 percent) remains significantly lower than that for males (73 percent), with stark regional disparities. Because females outperform males in schooling, low FLFP results in a misallocation of talent. Labor regulations that facilitate part-time work—key for women who often need to balance work with caregiving—could help reduce this misallocation.

## Investment-driven growth

**Post-GFC growth in the Philippines was driven by capital accumulation** (Figure 1.5), **which accounted for over 90 percent of growth, while TFP contributed less than 10 percent.** The investment-to-GDP ratio has increased but still trails peers such as Viet Nam by more than 10 percentage points. Along with pro-investment reforms and increased macro stability that lowered borrowing costs, a key driver of capital accumulation has been significant public investment, particularly through programs such as "Build, Build, Build" and its successor "Build Better More," which prioritized infrastructure development. Public investment declined to 3 percent of GDP between 2000 and 2015, but has since rebounded, with infrastructure spending exceeding 5 percent of GDP. This has helped narrow infrastructure gaps, enhancing connectivity. Nonetheless, the Philippines' capital stock-per-worker ratio remains close to the median in MICs, suggesting further scope for capital deepening (Figure 1.6). Challenges remain in ensuring efficient allocation and execution of public investment projects.

**Figure 1.5**

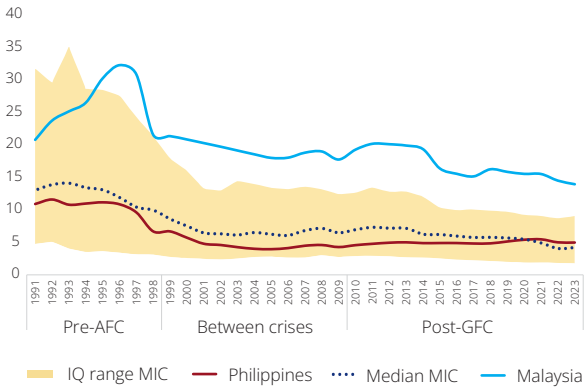
**Capital contribution to growth**
(In percent)




Source: MPO; World Bank staff calculations.

**Figure 1.6**

**Capital stock per worker**
(In percent of US capital per worker ratio)




Source: MPO; World Bank staff calculations

**Further increases in investment will require increasing saving rates.** Capital deepening is constrained by domestic saving and external financing. Overseas remittances, a crucial source of foreign exchange, accounts for 9 percent of GDP and around one-third of private saving. It has provided a critical buffer, financing both household consumption, and physical and human capital. Fiscal discipline, corporate saving, and policies to encourage saving by households, are needed to sustain investment growth. External sources will also contribute while they add to external risks.

## Productivity-enhancing reallocation of jobs, with limits

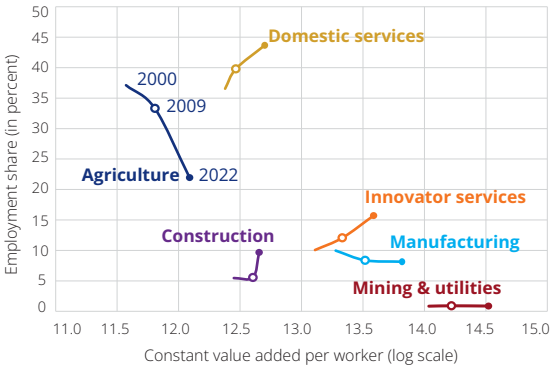
**Labor shifted from low-productivity agriculture to higher-productivity services, increasing aggregate productivity** (Figure 1.7).[19] A shift of labor from low- to (slightly) higher-productivity

19 See Standardized Tables for more details on structural change variables.

sectors (mainly from agriculture to services; retail, wholesale and construction) was matched with an improvement in the types of jobs available, thus leading to aggregate productivity growth.

## Figure 1.7

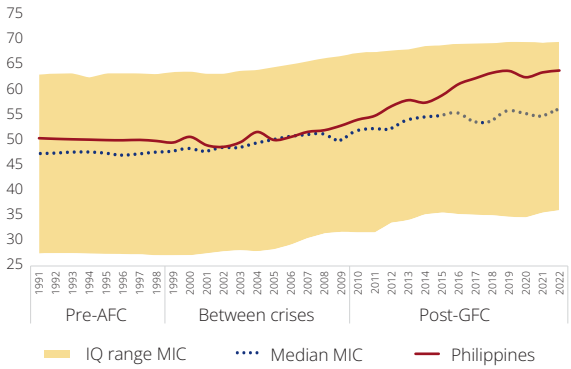
**Employment growth and labor productivity**
(By sector)




Source: PSA; World Bank staff calculations.

## Figure 1.8

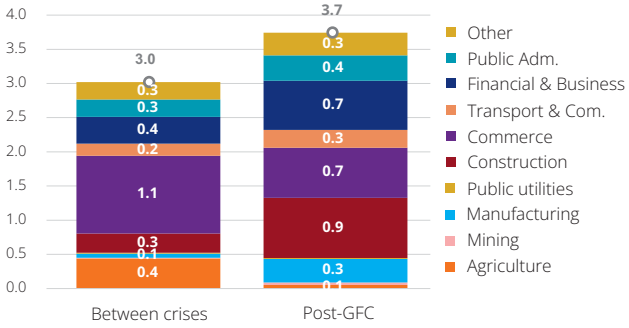
**Wage employment**
(In percent of total employment)




Source: GMID; World Bank staff calculations.

## Figure 1.9

**Annual wage employment growth by sector**
(CAGR, in percent)




Source: GMID; World Bank staff calculations.

**Wage employment rose from 55 to 64 percent in 2010–2022** (Figure 1.8). It was led by sectors such as financial and business services, commerce, and construction (Figure 1.9), which created 2.1, 2.2, and 2.6 million wage jobs, respectively. By contrast, wage job creation in manufacturing was lower (1 million) and even negative in key subsectors such as textiles and apparel. This led to a stagnant share of manufacturing employment in the post-GFC period, despite overall growth in the sector.

## Box 2

### Agribusiness as an opportunity for increased productivity and wages in rural areas

**Agribusiness represents a key pathway for transforming the Philippines' agriculture sector, enhancing productivity, and creating better-quality jobs in rural areas.** Although agriculture employs 22 percent of the workforce, it remains the least productive sector, with labor productivity

at only one-sixth that of manufacturing and one-third that of services. However, agribusiness activities (food production, animal fats, and oils) have shown potential, achieving exports worth US$6.4 billion in 2023 and growing at an annual rate of 5.2 percent since 2005. Success stories such as Malagos chocolate and Vita Coco highlight the potential of value-added agricultural products to access high-value markets and drive rural job creation.

**To unlock this potential, the agribusiness sector must address distortions that limit its potential, such as property right uncertainty, clustering challenges, misaligned incentives, and limited farm-to-market connectivity.** Land tenure security, infrastructure, and climate resilience can increase efficiency and sustainability. Reforms to facilitate clustering to achieve gains from scale while preserving the source of income of rural communities, and expanding value chains and compliance with international standards, such as the EU's Deforestation Regulation (EUDR), can further enhance export competitiveness while supporting farmers' incomes and rural development.

### Recommendations

- **Land tenure security:** Ensure clearly defined property rights to improve access to financing.
- **Land clustering:** Encourage economies of scale by facilitating clustering of small plots.
- **Incentive reform:** Redirect subsidies to promote high-value crops, price water right for efficient resource use, reduce import tariffs on corn, to reduce feed costs in livestock.
- **Infrastructure development:** Improve connectivity between farms and markets to reduce transaction costs and increase market access.
- **Climate resilience:** Support farms in adapting to climate change through crop insurance reform and sustainable practices.
- **Certification and standards compliance:** Enhance traceability and sustainability in supply chains to meet international standards such as the EU deforestation Regulation (EUDR).

## Inward-looking growth

**Since 2010, three out of four new jobs have been created in non-tradable sectors** (Figure 1.10). Non-tradable sectors outpaced tradables in dynamism and contribution to employment. Most salient of these are retail, construction and domestic services. Credit and asset growth were also concentrated in non-tradables, further reinforcing their dominance. While tradable sectors such as IT and business services experienced robust growth, other tradables, including manufacturing, stagnated due to competition, REER appreciation, and a challenging business environment.

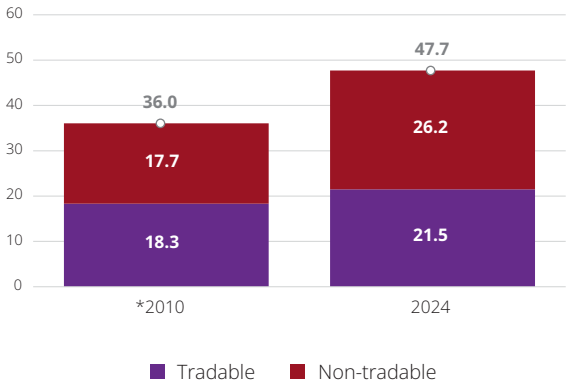
**The Philippines' inward focus shows in its declining export performance, with the export-to-GDP ratio falling from 33 to 27 percent between 2010 and 2023.** The decline is reflected in growing "missing exports", as the country moved from being an overachiever to an average performer, in contrast to Viet Nam (Figure 1.11). Non-tradable sectors have also increasingly attracted FDI, highlighting their growing role in the economy. Three factors have contributed to the relative decline

in the size of the tradable sector:(i) the appreciation of the REER; (ii) high energy prices relative to those in the region; and (iii) complex business regulations that predominantly affect the tradable sector.

**The appreciation of the REER since the early 2000s has contributed to the relative growth of non-tradables** (Figure 1.12). A real appreciation makes imports cheaper and exports relatively more expensive, reducing the relative profitability of the tradable sector, thus reducing incentives for firms to invest in technology adoption. Firm-level analysis for listed firms in the Philippines shows that following a real appreciation, an average firm reduces investment by 0.26 percentage points, while an average firm in the tradable sector reduces it by more than double that (0.59 percentage points).[20] Considering the 30-percent appreciation of the REER during the post-GFC period, this accounts for a 10-percent reduction in investment in the tradable sector relative to non-tradables.

## Figure 1.10

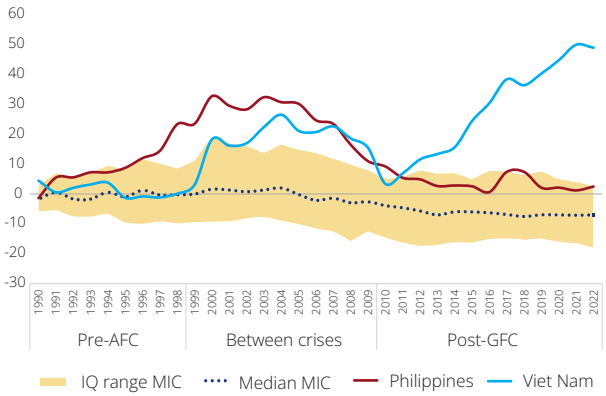
**The non-tradable sector accounted for three out of four new jobs created since 2010**
(Employment by type of sector, in million)




Source: PSA; World Bank staff calculations.

## Figure 1.11

**Export performance in the decline**
(Observed exports - predicted exports, in percent of GDP)




Source: Ganz et al. (2025); World Bank staff calculations. Note: exports are predicted using a gravity model.

**High electricity prices in the Philippines have accelerated deindustrialization.** Electricity prices in the Philippines have been consistently high relative to neighbors. In 2023–2024, Philippine rates averaged 15 cents per kWh, compared with 7.1 cents in Indonesia, 7.7 cents in Viet Nam, and 12.8 cents in Malaysia (Figure 1.13). Recent research that exploits variation in electricity prices across Philippine regions finds that regions with high power prices consistently show lower manufacturing output and employment.[21] These relatively high power prices are linked to structural weaknesses in the energy sector—limited generation capacity,

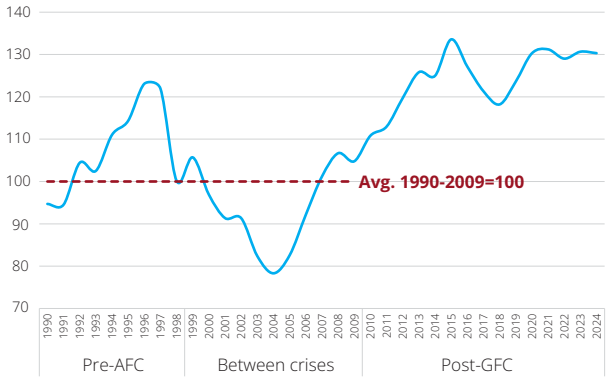
20 The results emerge from a regression analysis relying on a panel of 389 publicly listed firms in the non-financial economic sectors, that classifies sectors into tradables and non-tradables and spans the period 2006–2023. The analysis relies on a difference-in-difference approach to examine the impact of REER changes on firms' asset growth. Tradeable firms, which are more exposed to exchange rate fluctuations, serve as the treated group, non-tradable firms as the control. The objective is to identify whether REER changes differentially affect investment in tradable versus non-tradable firms, isolating the sensitivity ofinvestment decisions to exchange rate dynamics.

21 Ravago et al. (2019).

outdated contracts, competition challenges, regulatory delays—which have raised costs and reduced reliability.[22] This has pushed firms away from energy-intensive manufacturing into services, often in non-tradables.

## Figure 1.12

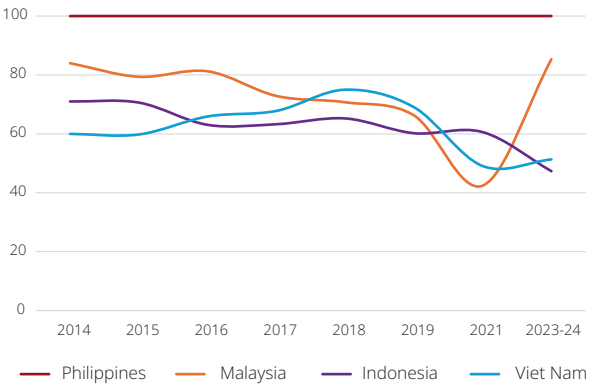
**Real effective exchange rate**
(Multilateral, avg. 1990–2009=100)




Source: Darvas (2012); Darvas (2021); World Bank staff calculations.

## Figure 1.13

**The Philippines has among the highest electricity prices in the region**
(Business rates, US$ cents per kWh, PHL=100)




Source: World Bank Doing Business database (WB DB); GlobalPetrolPrices (2025); BestBroadbandDeals (2025); World Bank staff calculations.

**A complex business environment disproportionately hurts tradables and has also led to reduced competitiveness in the tradable sector.** The World Bank's Business Ready assessment of 2024 shows a combination of complex regulations and operational efficiency challenges that distinctively affect the performance of Philippine firms in tradables. They increase operational costs, delay market entry, and create inefficiencies. For example, registering a new domestic company takes 75 days (106 days if foreign), longer than the 3-day average in more efficient economies, while transferring property can take up to 90 days. Environmental permits, often required for export-related industries, can take up to two years to obtain. Import procedures take 15 days to clear border control, compared with under a day in leading economies.[23] Delays and inefficiencies weaken competitiveness in Philippine firms in global markets, hindering investment and export growth.

**Faster growth in tradables, crucial for potential output to expand, will require three actions.** First, implementation of pro-competition reforms in key enabling services (transport and logistics) to reduce input costs and partially offset the REER appreciation. Second, increased competition in the energy sector, including through better regulations to reduce entry barriers in renewable energy (see Box 3). Third, regional trade agreements to better connect with the regional blocs.

22 Ravago (2022).
23 World Bank (2024a).

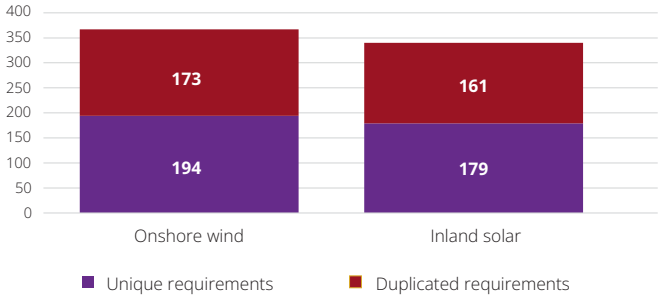
## Box 3

### More sources of energy for increased access and affordability of energy

**Additional sources of energy can help increase availability and affordability in the Philippines. This requires addressing key competition challenges.** The recent Department of Energy (DOE) decision to amend rules and regulation so of the Renewable Energy (RE) Act of 2008 allowing FDI up to 100 percent equity in renewables helps increase competition in the segment of generation. Foreign investors report the DOE decision to have led to important investment announcements. For example, the Danish Copenhagen Infrastructure Partners announced a plan to develop two wind electricity generation projects in the Bicol and Eastern Visayas regions, with an investment of US$4.1 billion, expecting to generate 330 jobs.[24]

**Figure 1.14**

**Government processes and renewable energy projects**




Source: World Bank staff. Field work mapping of processes.

**De jure restrictions lifted; de facto barriers remain.** Complex permitting processes are a barrier to entry for foreign investors, whose decision tends to be more elastic to investment climate than that of domestic investors. The permitting process in Philippines involves multiple agencies and documentation requirements (Figure 1.14), leading to delays and increased costs for developers. Addressing these inefficiencies can help not just to lower investment costs, but also to increase competition in the sector.

**The Energy Virtual One-Stop-Shop (EVOSS) Law is a necessary reform to streamline the permitting process for RE projects.** It creates a centralized digital platform for accessing information and submitting applications. This initiative, enacted into law in 2019, addresses critical bottlenecks by reducing redundancy and improving transparency, thereby simplifying the complex web of requirements and interactions developers must navigate, potentially reducing delays and administrative burdens. However, the EVOSS law needs to be complemented by efforts beyond the energy sector, such as land use planning, environmental assessments, and community consent, which involve multiple cross-sectoral regulations and agencies.

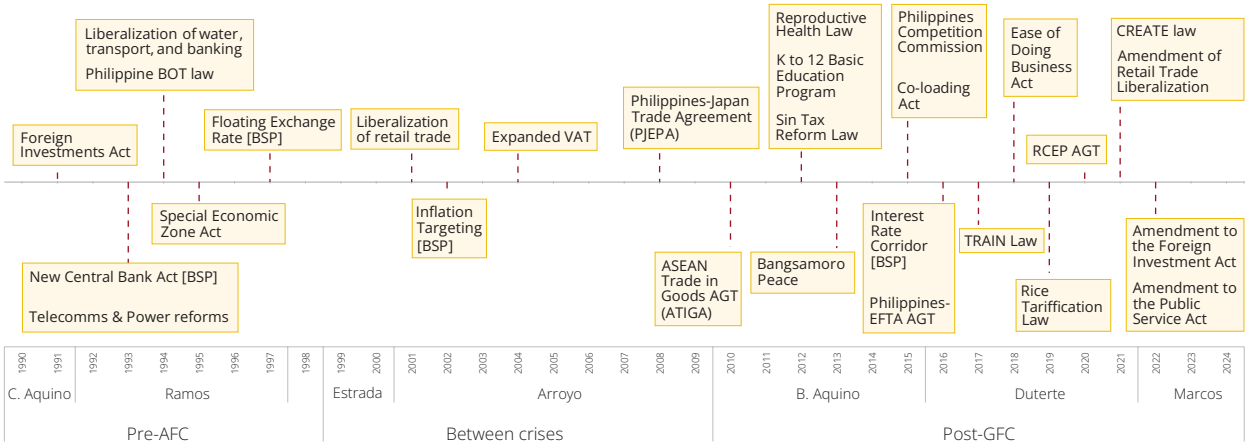
**The multi-agency nature of the permitting challenge requires a whole-of-government approach.** The effectiveness of EVOSS is limited by the lack of interoperability and data sharing among government agencies (national and local). Institutional coordination, harmonization of sectoral regulations, and comprehensive digitalization of processes across all relevant agencies are needed.

24 FDI markets database.

# Policy framework

## Figure 1.15

**Most salient policy reforms in the Philippines**




Source: Authors' elaboration based on Sicat (2002); Arellano Law Foundation (2024); Chan Robles (2024).

**Macroeconomic stability had been improving in the Philippines prior to the GFC.** Reforms such as the adoption of inflation targeting and a floating exchange rate (Figure 1.15), alongside the interest rate corridor mechanism, have improved inflation management (Figure 1.16). Relative stability in GDP growth kept unemployment low and stable, averaging 3.6 percent over the past decade, except for a spike during the COVID-19 pandemic (Figure 1.17). Fiscal policy reforms, including the 2005 VAT increase and the 2017 Comprehensive Tax Reform Program, stabilized revenues during periods of fiscal stress. The introduction of the 2022 Medium-Term Fiscal Framework helped anchor expectations, as debt ratios normalize post-COVID-19 (Figure 1.18).

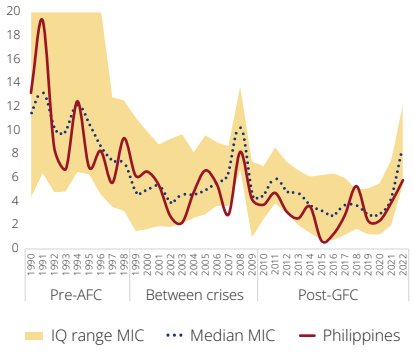
**Broadening the tax base and rationalizing fiscal incentives have bolstered revenue, though they remain relatively low.** The 2012 Sin Tax Law and updates, and the 2017 TRAIN Law increased revenue through adjustments in excises. The 2021 CREATE Law reduced corporate income tax and streamlined fiscal incentives through oversight by the Fiscal Incentives Review Board (FIRB). The 2024 CREATE MORE Law, following private sector demands, shifted incentive authority back to Investment Promotion Agencies, undermining fiscal oversight and strategic alignment.

**Prudent fiscal management enabled the country to achieve investment-grade ratings in 2013, reducing borrowing costs and enhancing investor confidence.** Post-pandemic recovery strategies have focused on maintaining public investment, fiscal discipline, and minor revenue mobilization reforms to minimize output losses while supporting growth. The reduction in borrowing costs has proved to be a key determinant of investment growth in the Philippines.[25]

25 Firm-level analysis of Philippines publicly listed firms shows that increases in the policy rate reduces firms' leverage and firms' investment. For a detailed discussion, see Background Paper – GJR Macro Chapter.

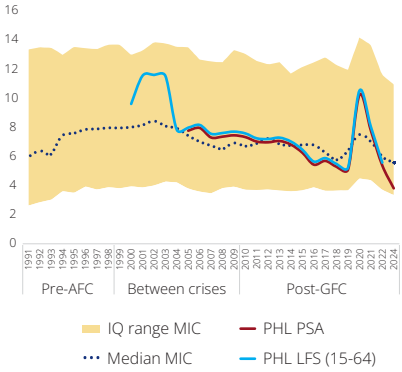
## Figure 1.16

**Inflation**
(CPI, in percent)



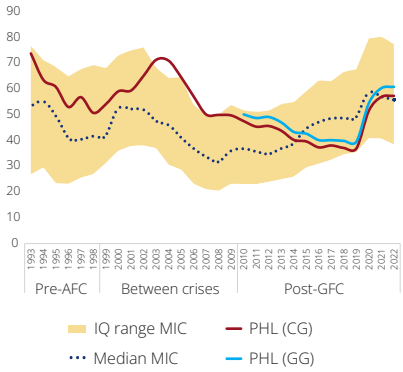

## Figure 1.17

**Unemployment**
(In percent of labor force, 15+)




## Figure 1.18

**Government debt**
(In percent of GDP)




Source: PSA, International Labour Organization (2025), International Monetary Fund (2025); World Bank staff calculations. Note: PHL PSA unemployment rate corresponds to the percent of labor force aged 15 and over, while PHL LFS (15-64) corresponds to the percent of labor force between 15 and 64. CG stands for Central Government, GG for General Government.

**While structural reforms have attempted to address market distortions that stunt potential growth, many remain.** These distortions prevent the efficient allocation of resources, the creation of opportunities and the build-up of capabilities, and include those regulating how the Philippines can interact with the world economy. For example, there are restrictions on FDI into key sectors (agriculture, media and, until recently, telecoms and logistics), and to the cross-border movement of professionals (that could help narrow skills gap, for example in health). Others are related to how domestic markets work. In agriculture, this includes limits to farm size, or complexity to set up joint ventures to cluster farms and achieve scale, and product-specific subsidies. In services the lack of an open sky policy or cabotage restrictions hurt not only the transport sector but all others through value chain effects, including labor-intensive sectors such as tourism (Figure 1.19).

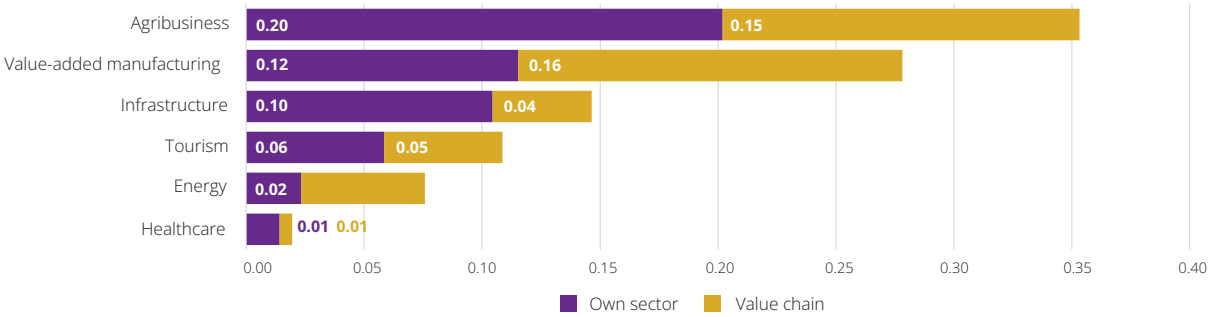
**Pro-market reforms, especially in telecommunications and air transport, served to enhance competitiveness, but implementation remains partial.** The 2015 Philippine Competition Act established a legal framework for promoting fair competition, with the Philippine Competition Commission pursuing notable cases. However, resource constraints and overlapping regulatory mandates in sectors such as telecommunications and energy hinder enforcement. Recent amendments to the Public Service Act (PSA) and liberalization of renewable energy have attracted significant FDI, and special economic zones have also helped,[26] but implementation challenges and incentive compatibility challenges in regulatory design persist (e.g., private sector is part of regulatory boards).

26 SEZs have helped boost investment and exports, particularly in manufacturing and IT-business process outsourcing (IT-BPO). The Philippine Economic Zone Authority was established in 1995 and offers firms simpler registration processes, tax incentives or exemptions. Zones have spatially balanced development, with growth in Cebu, Clark, and Davao. SEZs had existed since the 1960s, but it was the PEZA Act in 1995 that increased the role of the private sector in running them.

**In addition to fostering competition, ensuring a level playing field for all investors remains crucial.** High market concentration in key industries, coupled with regulatory capture risks related to incumbents playing crucial roles in regulatory agencies, have limited the opportunities for new entrants. Complex business regulations add to the mix. The Ease-of-Doing-Business Act aimed to reduce barriers, but uneven implementation across local governments undermines impact. Simplifying regulations, addressing national-local duplication in permitting requirements, and ensuring transparency in investment processes can unlock private sector growth.

## Figure 1.19

**Distortions that prevent the efficient allocation of resources have large impacts**
(Sectoral GDP elasticities to internal distortion reductions – selected sectors)




Source: World Bank's CEM3.0 Distortion Tool.

**Trade policy.** The Philippines has pursued Free Trade Agreements (FTAs) to reduce trade costs, but most are shallow, focusing on the border element of market access rather than on deeper reform areas.[27] The Philippines has 10 active agreements covering 33 percent of global GDP, primarily in the Asia-Pacific region. Key partners include China, Japan, the Rep. of Korea, and India, with most agreements operational since the mid-2000s. Progress in reducing non-tariff barriers, improving services trade, and simplifying customs procedures is slow.[28] Deep FTAs could help address domestic and international distortions, enhancing competitiveness. Streamlined customs processes, and smart export promotion will also help position the Philippines globally.

**Human capital.** Low human capital remains a constraint for long-term growth.[29] As of 2020, only half of children are expected to reach their productive potential by age 18, with nine out of ten 10-year-olds unable to read or understand simple texts.[30] Stunting affects one in four children, further limiting future workforce productivity and employability, particularly in high-growth sectors such as IT-BPO. Investments in foundational education, health, and skills development in the meantime are essential to address these gaps and make human capital the third engine of inclusive growth.

27 The depth of an agreement is measured by the number of policy areas it covers, with a maximum of 52 policy areas (Mattoo et al., 2020). The deepest agreement signed by the Philippines is the Regional Comprehensive Economic Partnership (RCEP), which includes 30 policy areas. All other agreements signed by the Philippines cover fewer than 25 policy areas.

28 Cascading in the context of import duty structures consists of duties on raw materials and intermediates required to produce a final good being lower than those on the final good itself. The aim of cascading is to promote domestic value addition, while the unintended consequence is to create an anti-export bias, as it increases effective protection.

29 This report acknowledges human capital as a key engine of inclusive growth. For a more detailed discussion, see World Bank (2024e).

30 For a discussion on how to increase the human capital index see World Bank (2024e).

## Box 4

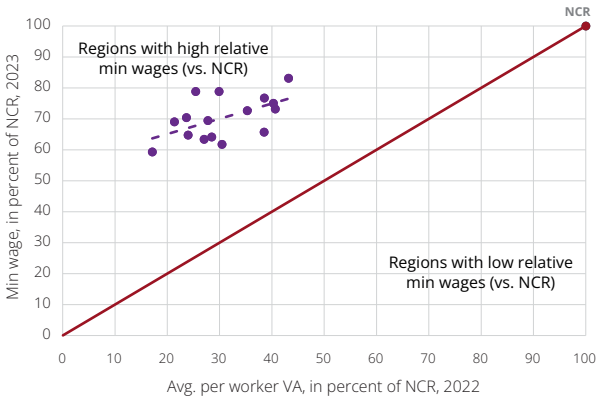
### Minimum wages in the Philippines

**Minimum wages in the Philippines are complex, varying by location and sector.** They are set at the regional level by Regional Tripartite Wages and Productivity Boards (RTWPBs). They determine wage levels based on economic conditions, cost of living, and other factors.[31] Each RTWPB publicly consults, assesses economic indicators (inflation, business competitiveness, and living costs), and issues legally binding wage orders that establish minimum wage levels. Exemptions exist for certain establishments, such as small firms with fewer than 10 employees and businesses severely affected by crises.

**This system results in minimum wages in low and middle-income regions that are high relative to labor productivity** (Figure 1.20). All low- and middle-income regions have high minimum wages with respect to the NCR given their relative productivity (140 percent higher than the NCR equivalent, on average). These differences are only partially explained by price differences (the average difference drops to 110 percent of the NCR equivalent). When mandated wages are too high relative to productivity level, firms will struggle to afford them, resulting in workers, especially lower-skilled workers, being priced out of formal employment. This can reduce the number of formal jobs for these workers, contributing to high levels of informality among lower-skilled workers (Figure 1.21).

#### Figure 1.20

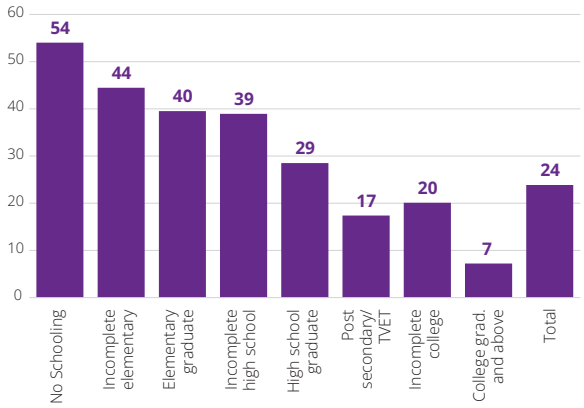
**All regions have high labor productivity-adjusted minimum wages compared to NCR...**
(Wages and productivity)




Source: PSA; World Bank staff calculations.

#### Figure 1.21

**...resulting in many working informally, with a share of the labor force (mainly the unskilled), earning below the minimum wage**
(Percent of the working population)




Source: PSA; World Bank staff calculations.

31 This system was established under Republic Act No. 6727 (Wage Rationalization Act of 1989), decentralizing wage-setting to account for regional disparities. The National Wages and Productivity Commission (NWPC), under the Department of Labor and Employment (DOLE), oversees the RTWPBs to ensure compliance with national policies.

**Recommendations for a more effective minimum wage system**

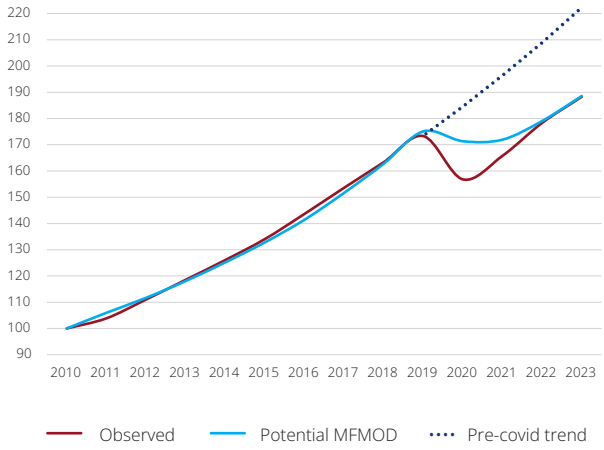
- **Improve connectivity for higher productivity:** Higher wages grounded on higher productivity can be achieved provided infrastructure and logistics bottlenecks are addressed.
- **Complement wage policies with active labor market programs to develop workers' skills:** Investments in skills development and employment facilitation programs can support higher wages by increasing productivity and employability of workers.
- **Simplify and rationalize minimum wage-setting:** The Philippines has a highly fragmented wage-setting system, with multiple wage categories and frequent adjustments. A simpler process improves compliance and reduces uncertainty for businesses and workers.[32]

A more adaptive and evidence-based minimum wage policy—integrating productivity, cost-of-living variations, and economic conditions—would support both workers and businesses, contributing to more inclusive labor market growth in the Philippines.

# Potential output

**Figure 1.22**

**Potential and observed GDP**
(Levels, index 2010=100)




Source: Macro-Fiscal Model (MFMod); World Bank staff calculations.

**The Philippines' potential output has grown but remains low to support rapid convergence with high-income economies.** At around 6 percent, potential and observed output growth have been aligned in recent years (Figure 1.22). The scarring effects from the COVID-19 pandemic highlight the need for structural reforms to expand potential output, and to facilitate faster convergence and better job creation. These reforms, to focus on creating opportunities and building capabilities to sustain the catch-up process, to turn the productivity engine of growth, allocate resources better, and to tap into the global economy for increased scale and learning, could help align potential output growth the targets that the Philippines has set itself in the PDP.

32 The recommendations draw on consultations with key counterparts and on the Philippine Jobs Report, World Bank (2023).

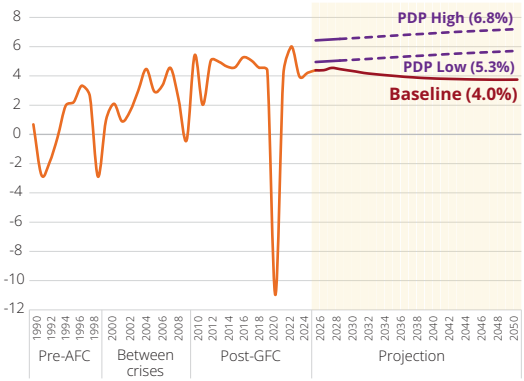
# Growth projections

**The baseline projections in this report show that under a business-as-usual scenario, aggregate GDP growth will stabilize at an average of 5.1 percent and per capita GDP at 4 percent by 2050.**[33] This contrasts with targets set up by the Government of annual GDP growth rates of 6.5 to 8 percent.[34] With average population growth projected at 1.1 percent annually, the Government's targets imply growth of GDP per capita in the range of 5.3 to 6.8 percent (Figure 1.23). Baseline projections in this report estimate average GDP growth of 5.1 percent annually (4 percent per capita) for the 2025–2050 period, assuming TFP growth at 1.3 percent, an investment-to-GDP ratio converging to 22 percent, and slower labor force expansion due to demographic trends.

**Small differences in growth rates have significant long-term effects on income levels.** Achieving the Government's upper aggregate GDP growth target of 8.0 percent (6.8 in per capita terms) would elevate gross national income (GNI) per capita to US$25,710 by 2050 at prices of 2025 (making it 37 percent above the threshold for HICs), comparable to Poland's current living standards (Figure 1.24). Instead, under the baseline scenario (status-quo reform pace), by 2050, GNI per capita is estimated at US$12,200 (or 33 percent below the threshold for HICs), similar to Brazil's today. These projections emphasize the critical nature of reforms if the Philippines is to converge to high-income status.

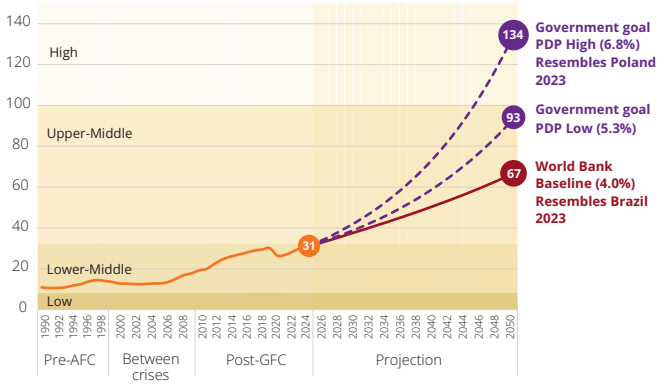
## Figure 1.23

**GDP per capita growth projections**
(In percent)




## Figure 1.24

**GNI per capita projections**
(In percent of high-income threshold)




Source: MFMod; World Bank staff calculations. Note: PDP projections for GDP until 2028. We assume constant GDP growth for the rest of the projection. HICs are assumed to grow at the post-GFC per capita CAGR (1 percent).

33 5.4 percent on aggregate and 4.1 percent per capita by 2040.

34 The baseline also contrasts with catch-up growth rates observed for countries that have sustained growth accelerations such as the Rep. of Korea which, at comparable levels of income per capita was growing at annual rates between 7 and 10 percent.

# CHAPTER 2
# MICROECONOMIC FEATURES OF GROWTH AND JOBS

*This chapter examines the Philippines' job creation and growth through a firm-level lens. It looks at firms' growth productivity dynamics. It shows evidence of productivity convergence, but also of allocative efficiency challenges and limited growth at the frontier, where wages paid are highest. Job-creating firms are mostly in the middle of the distribution rather than at the top. This is similar to other countries in the region, but in contrast to what is observed in advanced economies. Firms and jobs are also flowing into inward-oriented sectors. This poses a challenge: productivity, wages and growth tend to be higher in globally exposed sectors. The chapter discusses policies to encourage technology diffusion, boost FDI productivity spillovers to domestic firms, improve allocative efficiency through more competition and better regulations, and encourage outward integration through a reduction in trade costs.*

## Patterns of productivity growth

**Productivity growth in the Philippines has converged with that of HICs, but it remains moderate compared with regional peers.** Between 2000 and 2019, labor and TFP in the Philippines increased substantially.[35] However, compared with peers, productivity growth in manufacturing since 2006 has been moderate relative to Indonesia or Viet Nam. For services, productivity growth was faster than observed in the average Vietnamese firm (Figure 2.1).

**Productivity has increased markedly in manufacturing compared with services, a trend that was accentuated during the pandemic.** Productivity growth has been more rapid in the manufacturing sector at 53 percent (2006–2019), compared with the services sector at 26 percent (Figure 2.1). This is linked to the relatively higher competition pressures in manufacturing relative to services, and (relatedly) to the existence of relatively more regulatory barriers in the sector.[36] Manufacturing TFP has continued growing post-pandemic, while services has stagnated.

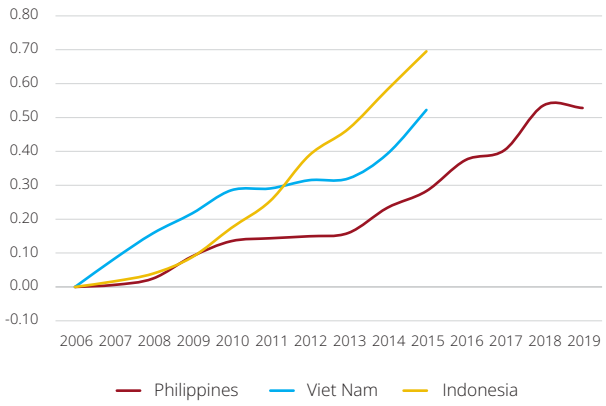
35 The increase in productivity coincided with the increasing role of capital deepening.

36 Background paper 2, Figure 3. Manufacturing productivity increased more pronouncedly, especially after 2019, while services stagnated, reflecting the differential impact of the pandemic on services and manufacturing.

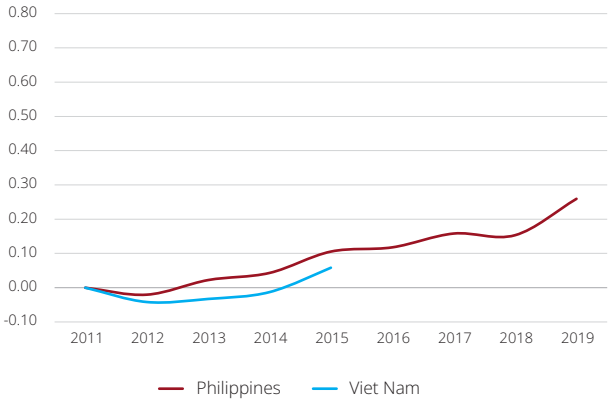
**Figure 2.1**

**Productivity growth (TFP) in selected EAP countries**

a. TFP growth of the average manufacturing firm




b. TFP growth of the average service firm




Source: PSA Census of Philippine Business and Industry (CPBI) 2018 (Philippines); National Statistical Offices of Viet Nam and Indonesia; World Bank staff calculations

# The allocative challenge

**The aggregate productivity gains mask a reallocation problem.** In well-functioning markets, the productivity of the private sector grows through three channels: (i) firms get better at what they do (within-channel); (ii) more productive firms expand absorbing workers and capital from less productive ones (between, or reallocation channel); and (iii) less productive firms exit, and more productive ones enter the market (entry-exit channel). In the Philippines, productivity growth has been driven by within-firm gains, while reallocation (between channel) has been contributing negatively. A similar pattern is found in some regional peers such as Indonesia, Viet Nam, and China. Instead, Poland and Colombia, which have shown dynamism in recent years, show productivity growth coming from both within and between channels (Table 1). Challenges to competition, distortions in land, labor, or capital markets such as land size limits, high costs of hiring and firing, entry barriers or credit subsidies can be key drivers of the lack of a productivity-enhancing reallocation of resources.

**Manufacturing firms show notable productivity growth over their lifecycles, but struggle to scale up over their lifecycle.** Philippine manufacturing firms have productivity growth paths over their lifecycles substantially better than those observed in developing economies (e.g., India, or Mexico, Figure 2.2, panel a). However, this growth has not translated into firms scaling up employment over their lifecycle (Figure 2.2, panel b), even during the period when the economy was experiencing an output expansion. An alternative source of data, balance sheet information from Philippine publicly listed firms, shows that challenge is real: over 90 percent of listed firms remained in their original asset-size class over the 2006–2023 period, reflecting barriers to grow.

## Table 1

**Decomposition of aggregate productivity growth into within- and reallocation channels**
(In percent of total growth)

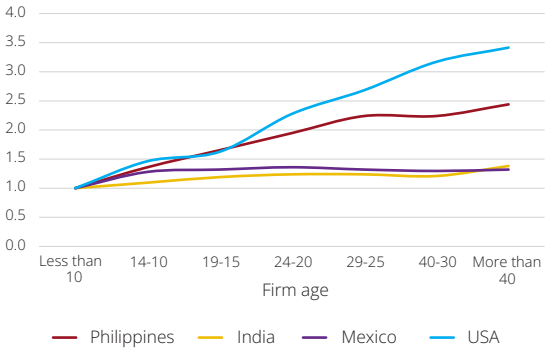
| | PHL | | IDN | | MYS | | VNM | | COL | | CHN | | POL | | USA | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2012-2018 | | 1996-2015 | | 2000-2015 | | 2011-2015 | | 2015-2019 | | 2007-2013 | | 2011-2021 | | 1997-2013 | |
| | M | S | M | S | M | S | M | S | M | S | M | S | M | S | M | S |
| **Within** | 45 | 67 | 91 | - | 67 | - | 89 | 55 | 56 | - | 105 | - | 59 | - | 58 | - |
| **Between** | -9 | 6 | -1 | - | 5 | - | -11 | -2 | 44 | - | -18 | - | -41 | - | 15 | - |
| **Entry** | 24 | 43 | 8 | - | 8 | - | 10 | 35 | - | - | 18 | - | - | - | 5 | - |
| **Exit** | 40 | -16 | 2 | - | 19 | - | 11 | 12 | - | - | -6 | - | - | - | 22 | - |

Source: Private Sector Data Dashboard (Poland and Colombia) and Firm Foundations of Growth; East Asia and Pacific Chief Economist Office (EAP comparators); World Bank staff calculations. Note: M stands for Manufacture, S for services. Data does not indicate whether a firm exited due to becoming smaller (i.e., less than 20 employees) or due to an actual exit. Additionally, the data does not account for firm entries, as some firms were experiencing "survey entry" without joining the population of firms. The terms have been normalized to add up to 100 percent for each country. Poland and Colombia report the decomposition for labor productivity rather than for TFP.

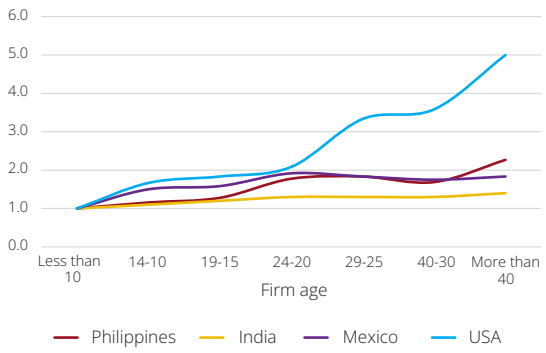
## Figure 2.2

**Firms in manufacturing do not grow as they age, while entry rates have slowed down**

a. Productivity growth by firm age
(TFP index, young firm=1)




b. Employment growth by firm age
(Employment index, young firm=1)

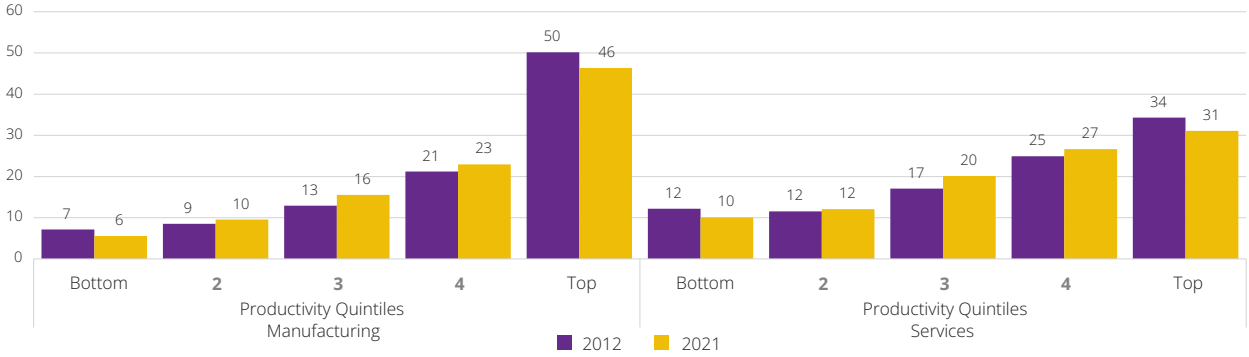



Source: PSA CPBI 2018; Hsieh and Klenow (2014); World Bank staff calculations

**Firms at the top are not scaling up.** In Philippines, more productive firms are also those employing the largest share of workers (Figure 2.3). This is good news and suggests a positive static story of allocation. However, the dynamic story is more challenging. During the past decade, top productivity firms have seen their employment shares fall rather than increase. Between 2012 and 2021, most employment growth was driven by firms that reached the middle range in productivity distribution. Top productivity firms, instead, shed employment, which is a challenge for quality job creation, as these are the ones that create better paying jobs (Figure 2.4). Complex separation processes for permanent workers may disproportionately constrain job growth in more sophisticated firms that require permanent rather than temporary employees. Also, limited competition in the sectors in which these top firms operate deters innovation and growth.

## Figure 2.3

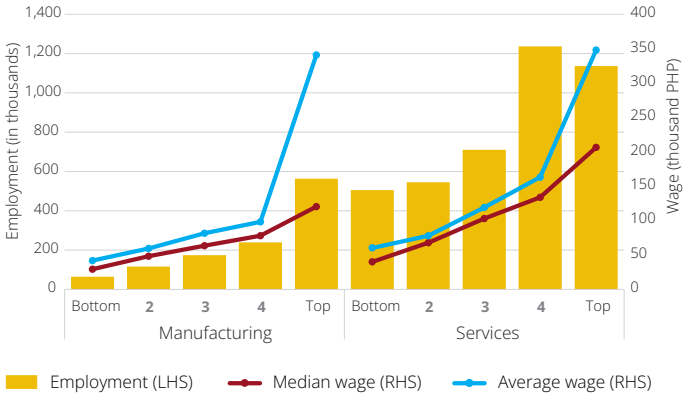
**Employment shares by productivity and sector**
(In percent of sector employment)




Source: PSA ASPBI; World Bank staff calculations.

## Figure 2.4

**Most productive firms in manufacturing and services pay higher wages and employ more people**




Source: PSA ASPBI; World Bank staff calculations.

**Top firms in the Philippines are dominated by family-owned conglomerates and heavily invested in less competition-prone sectors.** An analysis of the main investments of these conglomerates shows that they are concentrated in rent-rich, non-tradable sectors such as utilities, real estate, and transport, shielded from international competition, and offer high returns through market power rather than innovation or productivity gains. This allocation of investment reflects structural incentives shaped by a political economy in which conglomerates benefit from regulatory protection.[37]

# The inward shift

**The decline in the export-to-GDP ratio reveals a firm-level re-orientation away from exporting.** The decline in the base of exporters is visible in a 23-percent reduction of systematic exporters of merchandise, from 3,500 in 2011 to 2,700 in 2022 (Figure 2.5, panel a).

**The shift in international orientation coincided with faster growth of firms in the non-tradable sector than in the tradable sector.** The analysis of listed firms reveals the following: (i) asset growth was faster post-GFC (consistent with an overall acceleration of growth post-GFC); and (ii) post-GFC, the median firm in the non-tradable sector experienced twice as high asset growth as that in the tradable

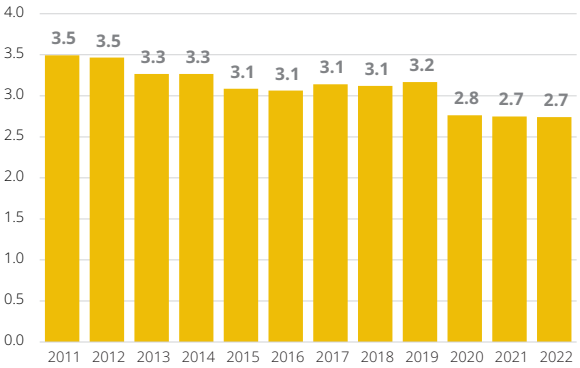
37 Mendoza and Juco (2024).

sector (Figure 2.5, panel b). This is linked to the appreciation of the REER observed since 2005, which increased the relative profitability of investing in non-tradables relative to tradables.[38]

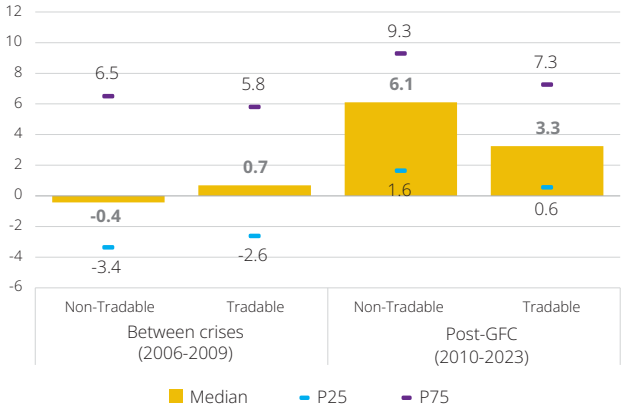
## Figure 2.5

**The Philippines' inward shift**

a. Number of merchandise exporters exporting over US$100,000, in thousands




b. Asset growth, period CAGR, in percent

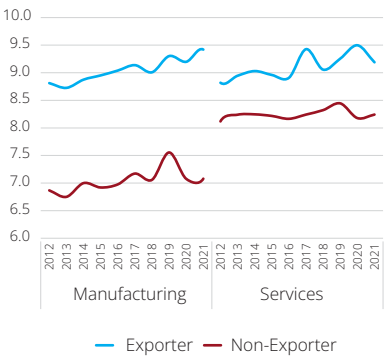



Source: panel (a): World Bank staff calculation using the PSA trade transaction database (accessible at the Data Enclave Center); panel (b): World Bank staff calculation using publicly listed firms' data from the Philippine Stock Exchange (PSE).

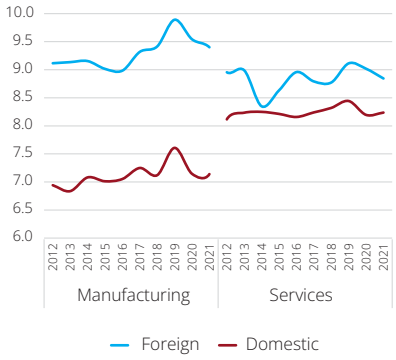
## Figure 2.6

**Outward oriented and large firms are exceptional performers in productivity**

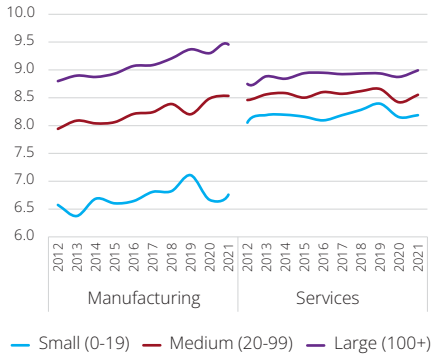
a. Exporters vs. non-exporters




b. Foreign vs. domestically owned




c. Size (number of employees)




Source: PSA ASPBI; World Bank staff calculations.

**The inward shift will take a toll on productivity moving forward, as exporters, and foreign-owned firms consistently outperformed domestically oriented ones.** Exporters achieved TFP

38 Evidence from Philippine firms show that exporters and foreign firms are not just more productive, on average, than non-exporters, or domestic owned firms. Their productivity also grows faster as they start exporting or they are acquired, suggesting some learning by exporting, and by internationalization at play, as found across countries. See, for example, Lovo and Varela (2022).

levels 1.5 to 2 times higher than domestic firms, while foreign-owned firms showed higher productivity, particularly in manufacturing (Figure 2.6, panels a and b).[39]

# Policy deep dives

## Competition challenges reduce the scope for productivity enhancing reallocation

**Market competition and business environment.** No single measure alone can provide a comprehensive picture of competition dynamics within a market.[40] The framework used for this section identifies risks of weakening competition trends in various sectors based on four characteristics of the market: concentration, profit margins, productivity dynamics, and churn rates. Based on these characteristics, the analysis identifies industries where competition trends merit an in-depth analysis of market functioning.

**Weak competition drags productivity, as it is linked to lower allocative efficiency, reduced employment, and lower wages.** The analysis shows that the allocative efficiency of industries at "high competition risk" is, on average, about half that of "low competition risk" industries, conditional on the industry's characteristics. This confirms that competition is key in the reallocation challenge of the Philippines.[41] Similarly, wages are lower in high competition risk industries than in low risk ones (controlling for industry characteristics).

**Potential risks of weakening competition dynamics in industries represent a non-negligible portion of the Philippine economy.** The analysis shows that industries experiencing "high" risks of weakening competition in 2022, representing 7 percent of total revenues, 6 percent of employment, and 9 percent of value added (Figure 2.7, panel a). Several sectors experience the presence of risks of weakening competition (Figure 2.7, panel b). Specific subsectors include financial services, electronic equipment manufacturing, telecoms, and transport.[42] This evidence reveals potential competition issues in these industries and merits further assessment.[43]

**Loss of dynamism and structural issues have heightened competition risks.** Strong competition dynamics could be expected in industries exposed to "high" risks of weakening competition in 2021, as these industries did not signal the presence of scale economies or outstanding market failures. However, the assessment noted significant increases in concentration within these industries, contributing to the overall risk outcome. In addition, a decline in dynamism,

39 This is consistent with what is observed in the literature (Lovo and Varela, 2022). Large firms also displayed scale economies. Also, SMEs showed weaker recovery from shocks, as seen during the COVID-19 pandemic.

40 While market concentration may indicate barriers to entry and weak competition, it may also result from firms differentiating themselves through investment and gaining market shares. Similarly, high markups and price-cost margins could be attributed to substantial profits, but they might also result from reduced costs, which is an efficient outcome.

41 Average allocative efficiency is measured using the within-industry covariance between firm size and productivity.

42 Lower presence of risk to declining competition in specific sectors also experience competition issues: electric power distribution, passenger air transport, accounting and electric power generation and distribution.

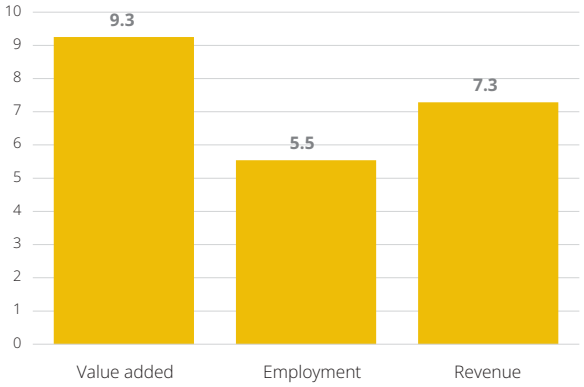
43 Almost all the high-risk industries found are deemed to be in the "competitive" category—only one wired telecoms firm was found in a natural monopoly, and two among partially contestable industries.

evidenced by a reduced rate of economic churn, was a key factor affecting the score. Firms exerted pricing power, and those with lagging productivity also contributed to the high-risk evaluation (Figure 2.7, panel b).

## Figure 2.7

**Sectors with a high presence of risks that can weaken the competition outlook represent a non-trivial part of the economy in the Philippines**

a. Economic importance of industries with the presence of risks to weakening competition, in percent, 2021




Source: PSA ASPBI; List of Establishments 2023; World Bank staff calculations.

b. Risks to weakening competition by sector, 2021

| Sector | Top 4 concentration | Profit margin | Productivity | Churn rate |
|---|---|---|---|---|
| Agriculture | High | Medium | High | High |
| Manufacturing | High | High | Low | High |
| Construction | High | High | High | High |
| Trade | High | Low | High | High |
| Transport | High | Medium | High | High |
| Information | High | High | High | Low |
| Financial | High | Medium | High | High |
| Professional | High | Medium | High | High |
| Administrative | High | Low | High | High |

Source: World Bank elaboration. Note: four factors associated with weakening competition risk are identified for each sector: the concentration of value added in the top 4 firms, the size profit margins, the growth of productivity and the churn rate in the sector. High risks (red) are associated with high concentration, high profit margins, low productivity growth, and low churn rates.

**Firms' entry rate has declined since 2015 and is relatively low compared with some regional peers, consistent with the decrease in economic churn.** The entry rate correlates with business and competition dynamics. Although an imperfect measure of the dynamic contribution to productivity growth, it provides a useful proxy for economic dynamism. In 2018, the Philippines had one of the lowest shares of young firms (< 3 y/o) at 5.5 percent among selected peers (9.5 percent in Thailand) and a declining rate between 2015 (6.5 percent) and 2019 (3.8 percent).

**In addition, firms operating in the Philippines must cope with higher costs and lower service standards in connectivity and energy compared with their peers.** Philippine firms face relatively higher operational costs for electricity and fixed broadband services (Figure 2.8).[44] The Logistics Performance Index ranks the Philippines above countries such as Morocco, on par with Indonesia, but below Thailand and Malaysia.

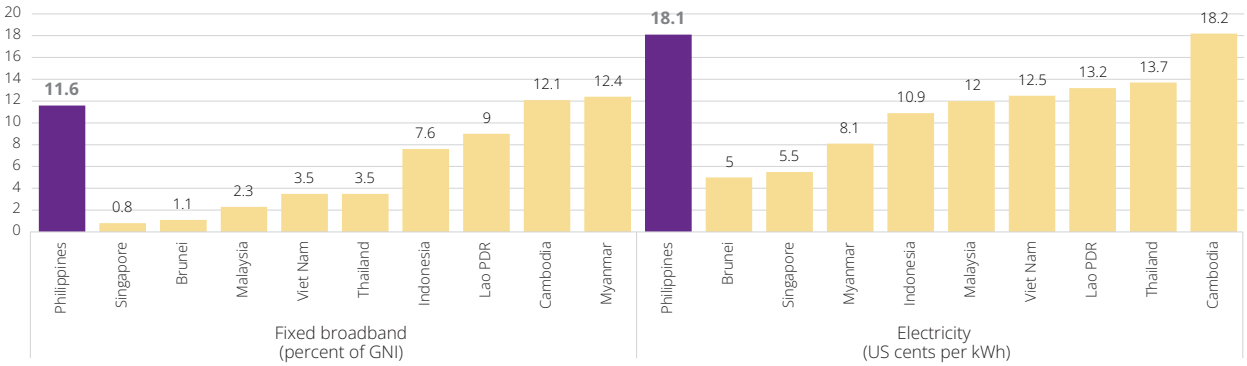
**Market restrictions remain prevalent in the transport, connectivity, and energy sectors, weakening competition and slowing innovation.** The restrictions impose barriers to entrants and weaken rivalry among existing operators. The OECD's measurement of investment restrictions in 2022 shows that the Philippines featured one of the most restrictive investment environments in transport, connectivity, and energy, albeit slightly better than Indonesia, China, and Malaysia.

44 Backbone services such as logistics, airports, roads, warehousing, telecoms., or electricity, can enhance the performance of other sectors through better, cheaper, or more varied inputs to production.

## Figure 2.8

**Operational costs of upstream broadband and electricity services**

(Cost of electricity and internet (fixed broadband) across selected economies, 2020–2021)




Source: WB DB; International Telecommunication Union (ITU); World Bank Enterprise Survey (WBES); World Bank staff calculations.

**The Government has lifted significant constraints in these sectors, but reforms must be fully implemented to eliminate restrictions.** The PSA amendment enables up to 100 percent foreign investment in transport and telecom.[45] The Konektadong Pinoy Bill aims to remove (or simplify) the legislative franchise for telecom operators.[46] The proposed amendment to the Electric Power Industry Reform Act (EPIRA) is expected to allow consumers to select electricity providers from accredited retail energy suppliers that generate power from renewable sources.[47] However, these reforms have yet to be fully implemented.

**Some reforms, such as the Amended PSA, are estimated to boost productivity, but are yet to be fully implemented.** The Amended PSA reform is expected to increase TFP by 3.2 percent on average, as it boosts competition in key enabling sectors and facilitates technology spillovers. Complementary reforms such as relaxing cabotage restrictions in maritime transport, and harmonizing logistics policy can also help boost the tradable sector and its productivity.

## Technology and innovation challenges slow down productivity growth

**Philippine firms invest less in upgrading technology and equipment than peers.** Despite the potential benefits, productive investments often require an extended period to pay themselves off. The Philippines lags other countries in technological sophistication in several business functions, both intensively—firms that use technologies do not use 'enough' of them—and extensively—only a few firms use these technologies (Figure 2.9). The share of firms that declared to invest in equipment in the past two years is below what is predicted by the Philippines' income and most of its peers.

45 Republic Act No. 11659 and its implementing rules and regulations.

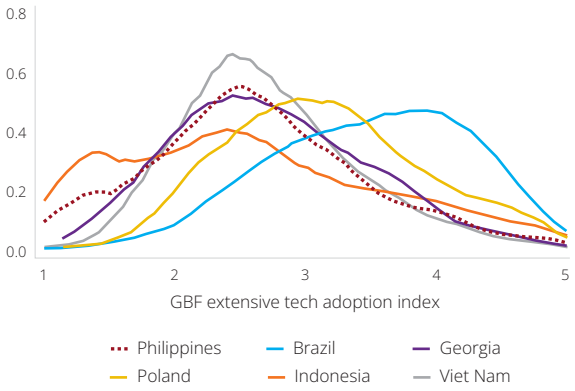
46 This is closer to the final stages of law-making in Congress: House Bill No. 6 approved by the House of Representatives and transmitted to the Senate on December 12, 2022; Senate Bill No. 2699 pending Second Reading, Special Order (August 5, 2024).

47 This is in the nascent stages of law-making with only referrals of the bills to the Committee on Energy, to name a few: House Bill Nos. 3430, 3432, 4263, 8151, 9695, 9922; Senate Bill Nos. 217, 486, 1612, 2348. The omnibus rules will enhance competition and increase consumer options and are anticipated to reduce electricity prices for participating users.

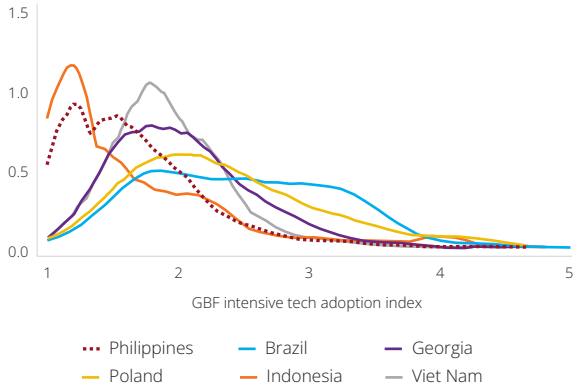
## Figure 2.9

**Filipino firms lag behind the frontier and rely little on technology for general business functions**

a. Extensive margin, density




b. Intensive margin, density




Source: Firm Adoption of Technology Survey, Philippines 2024; World Bank staff calculations. Note: The figures plot the distribution of firms across different technological sophistication. Index from 1 index score (min.- manual tech.) to 5 index score (max-advanced tech.).

**Lack of information, overconfidence, skill mismatches, and credit constraints have contributed to lower technology adoption.** Potential adopters of new technologies need to recognize the value of investing, for example by understanding the economic benefits from a technology investment that performs as expected. However, Philippine firms experience unawareness of benefits and demand uncertainty that keeps them from adopting technology (Figure 2.9). This leads to complacency and a reluctance to embrace new technologies. In addition, firms with lower technological adoption often think that they are more technologically advanced than they actually are. Credit constraints also limit technology adoption. Fewer than one in five Philippine firms finance investments through banks, non-financial institutions (NFIs), or supply chain finance, one of the lowest levels in the region.

## The lack of FDI spillovers is a missed opportunity for productivity growth

**Since 2010, FDI inflows into the Philippines have increased from 1.5 to 2.0 percent of GDP, while declining in most developing countries.** Attracting FDI has been a policy priority. Foreign firms show higher productivity than comparable domestic firms, and, according to international evidence, can lead to productivity spillovers on linked firms (such as suppliers or clients).

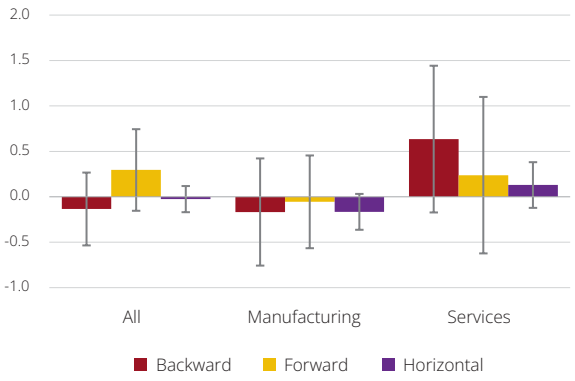
**However, evidence of spillovers from MNCs to domestic firms in the Philippines is negligible.** The extent of knowhow transfers from MNCs to domestic firms depends on the willingness to share of the MNCs (in turn, related to quality of contract enforcement or incentives) and to the ability of domestic firms to absorb that knowhow (their capabilities). In the Philippines, FDI has not led to productivity-enhancing spillover effects for domestic firms, except for energy efficiency, where there is evidence of positive spillovers from MNC clients to domestic suppliers. This indicates that incentives matter. As reporting on energy efficiency of the whole value chain is increasingly

prevalent for MNCs, knowhow transfers around energy efficiency are prevalent, while those on overall technical efficiency are not, indicating potential missed opportunities (Figure 2.10, Figure 2.11).

## Figure 2.10

**FDI in the Philippines has generated negligible productivity spillovers to domestic firms...**

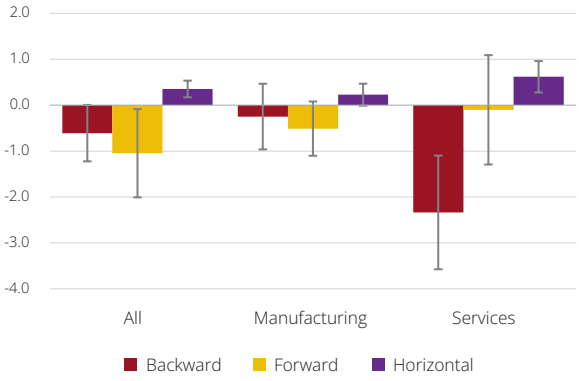
Technical efficiency (TFP) spillovers from increased FDI (In log points)




## Figure 2.11

**...but it has generated spillovers on energy efficiency along the value chain**

Energy intensity spillovers from increased FDI (In log points)




Source: PSA ASPBI/CPBI; ICA 2.0 FDI Linkages Tool; World Bank staff calculations. Note: 2012–2018 (balanced panel) and 2018 Input-Output.

**Strengthening domestic firms' capabilities will help boost spillovers.** Only domestic firms with either high productive capacity or access to financial and human resources, enabling the absorption of technology, are positioned to capture knowledge spillovers from their foreign counterparts.[48] Boosting capabilities through suppliers' development programs has proven effective to better integrate domestic SMEs with MNCs and maximize their spillovers on the local economy.[49]

48 See Crespo and Fontoura (2007) and Blalock and Gertler (2009).

49 Arráiz et al. (2013) evaluate these programs for the case of Chile. Modi and Mabert (2007) for the United States. For Asia, Lauridsen (2004) evaluated supplier development programs in Thailand in the 1990s pointing to the importance of state governance, while Abdul Hamid (2017) did so with Malaysian programs for SMES stressing the importance of skills and financing of suppliers for them to benefit.

# CHAPTER 3
# SPATIAL PATTERNS OF GROWTH AND JOBS

*This chapter examines spatial patterns of growth and job creation in the Philippines. It shows that despite stark differences—productivity in the NCR remains six times higher than in the BARMM—low- and mid-income regions have narrowed the gap over the past decade. Improvements in wage employment and productivity have driven this catch-up. Gains stem from non-tradable sector growth and rising formal employment. Infrastructure development has played a key role. Yet key barriers remain. Complex geography is exacerbated by complex regulations that increase high transport costs, including restrictive cabotage laws that limit competition in inter-island transport, limiting integration. Local governance challenges hinder better resource allocation and efficient infrastructure investment. This chapter suggests reforms to improve labor mobility, connectivity, and decentralization, and to sustain regional quality job creation and support convergence.*

## Subnational jobs dynamics and growth in the Philippines

**Stark regional disparities in per capita GDP and labor productivity in the Philippines are driven by differences in access to good jobs.** The NCR contributes over 30 percent of GDP and has income levels comparable to Mexico. In contrast, the BARMM has income levels comparable to Guinea (Figure 3.1). Disparities are closely tied to differences in labor productivity (Figure 3.2). Workers are most productive where they have access to jobs in which they can combine skills with good management, machinery, equipment, and technology, facilitating learning by doing, and benefiting from scale and scope in production.[50] Most of these jobs occur in firms. Hence firms play a crucial role in expanding the opportunities for workers to increase their productivity. Indeed, across regions in the Philippines, the share of employed workers with waged jobs is strongly correlated with labor productivity. In the NCR, 81 percent of total employment is waged. Outside the NCR, wage employment shares are much lower, ranging from 22 percent in the BARMM to 74 percent in Calabarzon (Figure 3.3).

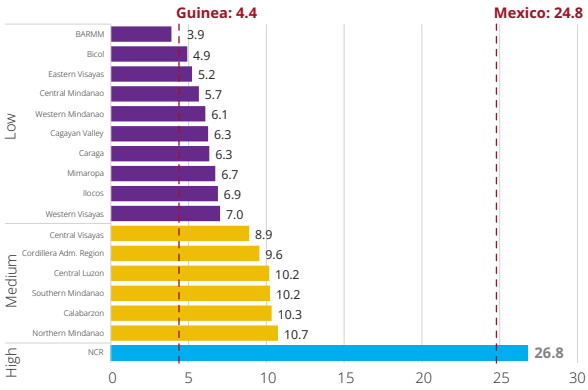
**Over the past 15 years, regions outside the NCR have shown signs of catch-up, with low- and medium-income regions experiencing faster growth in labor productivity.** In LIRs, defined as those with regional per capita GDP below US$8,000 in international PPP terms, annual growth in value added per worker rose from a 2.3 percent during 2000–2009 to 3.2 percent between 2009 and 2022. Similarly, MIRs, defined as those with per capita GDP between US$8,000 and US$11,000, saw an increase from 2.1 to 2.5 percent. By contrast, the NCR experienced a decline in productivity growth, from 3.0 to 1.5 percent annually (Figure 3.4).

50 This has been documented internationally. For the Philippines, Hasan and Jandoc (2010) reveal a substantial positive wage and productivity premium for wage workers over those comparable but self-employed workers.

## Figure 3.1

**Economic activity is highly unequal across the Philippine's 17 regions**

(Regional per capita GDP, in thousand PPP adj. intl. U$S)

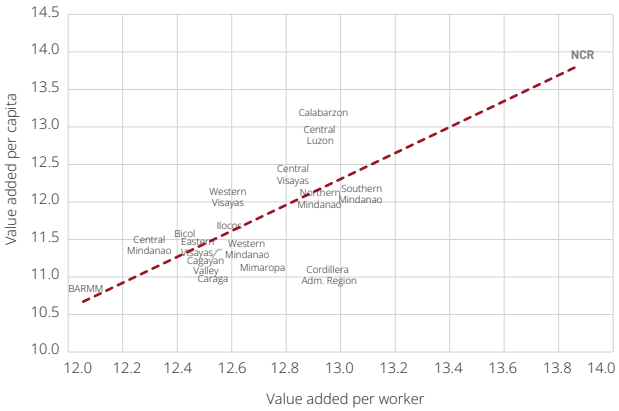



Source: PSA regional accounts and LFS; World Bank staff calculations.

## Figure 3.2

**Differences in per capita GDP correlate strongly with differences in labor productivity**

(Ln per worker/capita value added, current prices, 2022)

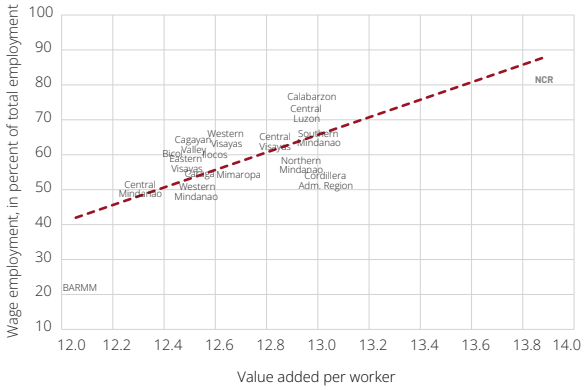



Source: PSA regional accounts and LFS; World Bank staff calculations.

## Figure 3.3

**Regional wage employment and value added per worker**

(Ln per worker value added and wage employment, 2022)

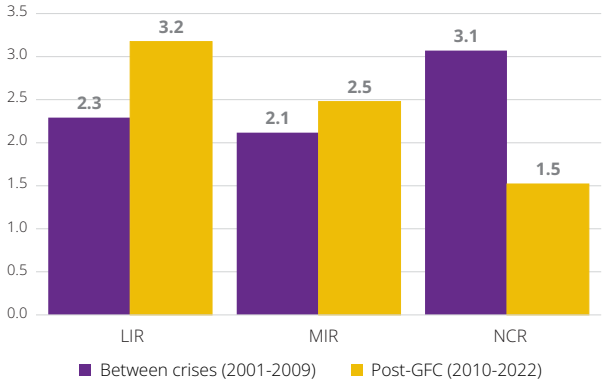



Source: PSA regional accounts and LFS; World Bank staff calculations.

## Figure 3.4

**Growth in value added per worker, by region income group**

(In percent)



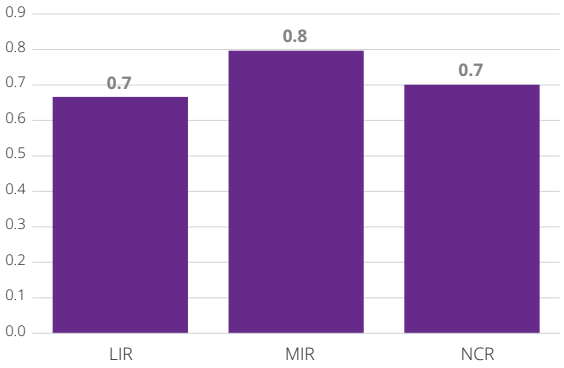

Source: PSA regional accounts and LFS; World Bank staff calculations.

**The productivity catch-up of LIRs and MIRs with respect to the NCR was driven by the expansion of formal firms.** While average TFP within formal firms followed a similar trend across the three regional groupings (Figure 3.5), LIRs and MIRs experienced much faster growth in the number of formal firms than in the NCR (Figure 3.6). As a result, aggregate TFP growth of formal

firms was substantially higher in LIRs and MIRs compared with the NCR (Figure 3.7), with a substantial impact on value added per worker (Figure 3.8).[51]

## Figure 3.5

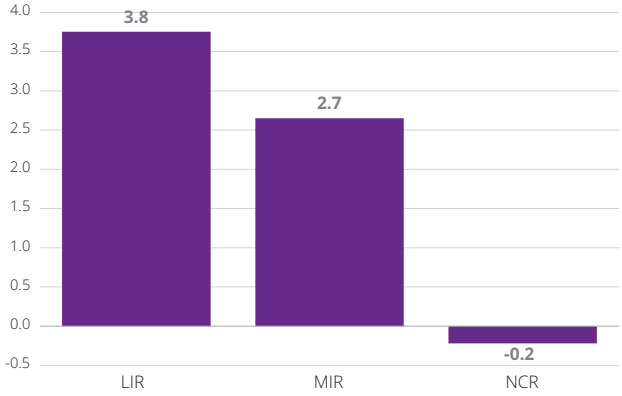
**Average TFP growth of formal firms**
(By region income group, 2012–2019 annualized, in percent)




Source: PSA ASPBI; World Bank staff calculations.

## Figure 3.6

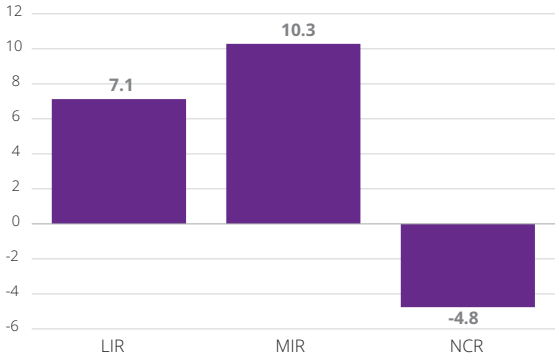
**Growth in the total number of formal firms**
(By region income group, 2012–2019 annualized, in percent)




Source: PSA ASPBI; World Bank staff calculations.

## Figure 3.7

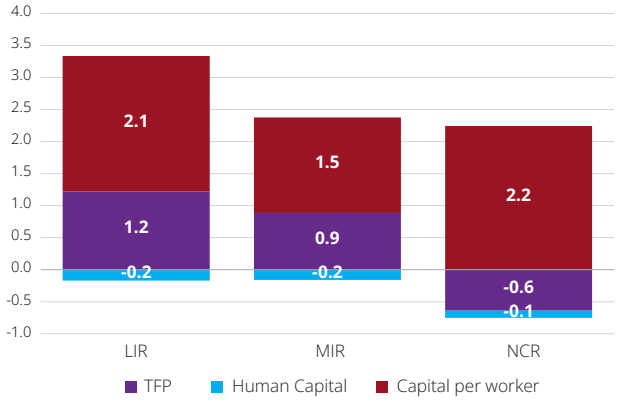
**Growth in aggregate formal firm sector TFP**
(By region income group, 2012–2019 annualized, in percent)




Source: PSA ASPBI; World Bank staff calculations

## Figure 3.8

**Growth in value added per worker by factor**
(By region income group, 2009–2022 annualized, in percent)




Source: PSA regional accounts; World Bank staff calculations.

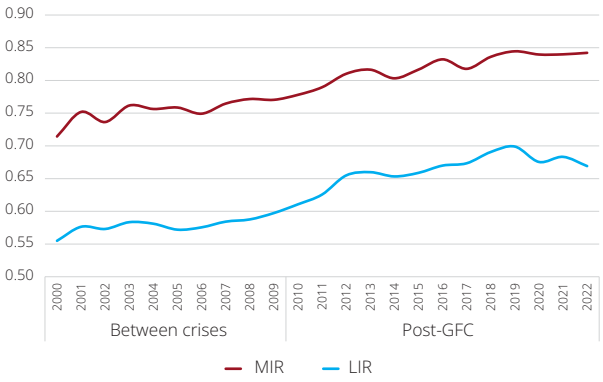
**The strong increase in formal firms in LIRs and MIRs contributed to growth in wage employment.** Between 2009 and 2022, the share of employed workers with waged jobs increased from 60 to 67 percent in LIRs and from 77 to 84 percent in MIRs, converging with NCR levels (Figure

51 While the positive average TFP growth in LIRs and MIRs might suggest that new firms in these regions were more productive than existing ones, this trend could also be explained by significant TFP gains among existing firms, which may have offset the entry of less productive new firms. Without panel data identifying new and old firms, it is therefore not possible to establish whether newly established formal firms were, on average, more productive than existing firms.

3.9). The increase in the number of formal firms was a significant contributor to this trend (Figure 3.10), while within the NCR employment was driven entirely by increases in firms' average employment levels (intensive margin), and in LIRs and MIRs employment gains were also strongly influenced by increases in the number of formal firms (extensive margin).

**Figure 3.9**

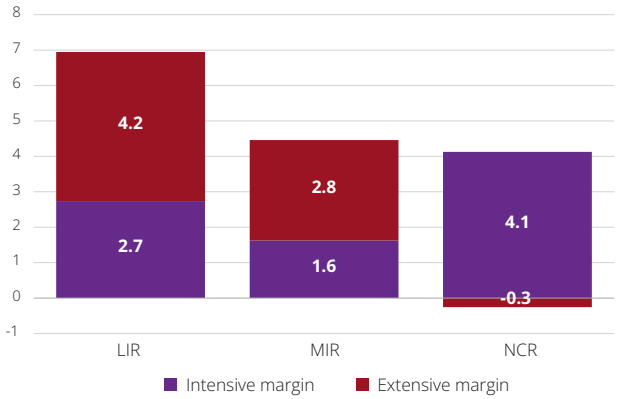
**Wage employment share**
(Index NCR=1)




Source: PSA LFS 2000–2022; World Bank staff calculations.

**Figure 3.10**

**Formal employment growth**
(By region income group, 2012–2019 annualized, in percent)




Source: PSA ASPBI; World Bank staff calculations.

**Much of the LIR and MIR growth stems from non-tradable activities.** Expansions in construction, real estate, commerce, and hospitality services (Figure 3.11) suggest a consumption-driven acceleration, with firms primarily expanding to capitalize on rising local demand. Tradable sectors contributed less. External orientation, whether domestic or international, lagged. Only in Calabarzon and Central Luzon was the role of tradables more significant, in part due to their access to the Philippines' international seaports, and absorption of firms leaving congested areas in the NCR.

**More tradables in the growth mix can help create better jobs.** For sustainable and stable economic progress, a more balanced growth strategy combining a focus on consumption-oriented with robust investment and productivity growth in sectors with higher exposure to competition can help create waged jobs that pay better wages. In LIRs, tourism and agri-business can help. For these sectors to prosper, key factor distortions need to be removed, particularly those preventing land clustering and consolidation, or long-term land leases. Connectivity and resilience-related public infrastructure investment can also help crowd in private investment in these sectors.

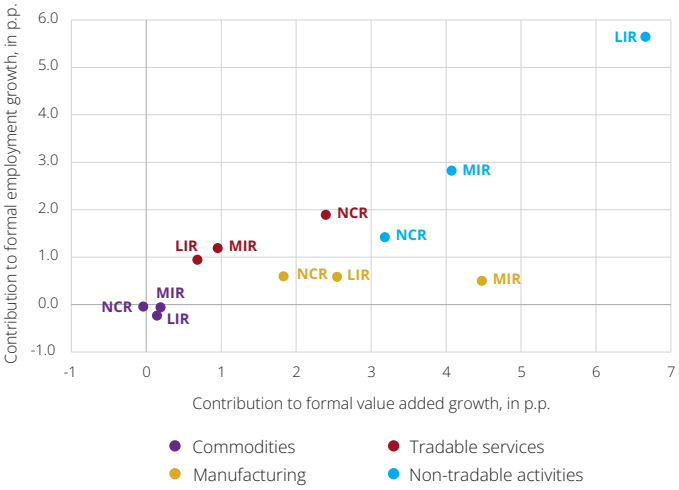
**The decline in TFP and stagnant firms' entry in the NCR highlights tensions between gains from agglomeration and increasing congestions costs.** In the NCR, aggregate TFP growth of formal firms declined to 5 percent per year. This was partly driven by a net decline in the number of firms suggesting low firms' entry. TFP decline and stagnating firms' entry signal a tipping point where the efficiencies generated by dense clusters of businesses, talent, and resources are eroded by congestion-related inefficiencies such as rising rents, limited infrastructure, and long commutes

(Figure 3.11). As urban economies grow wealthier, benefits of proximity and networks are increasingly offset by escalating costs of living, regulatory bottlenecks, and spatial constraints.

**Figure 3.11**

**Sectoral contributions to employment and value-added growth**

(By region income group, 2012–2019 annualized, in percent)




Source: PSA ASPBI; World Bank staff calculations.

**Commuters in Metro Manila lose an average of 113 hours annually to traffic delays, reducing their productivity.** Metro Manila's urgent need for more efficient transport solutions is highlighted by the 2023 TomTom Traffic Index, which ranked it the world's most congested city (out of 387 cities). These dynamics highlight the need for urban planning and policy interventions to address congestion costs, such as investments in infrastructure, affordable housing, and zoning reforms, to sustain agglomeration-driven growth and maintain the NCR's economic vitality and capacity to innovate. They also show the importance of expanding infrastructure in adjacent regions that will continue absorbing relocating firms.

**Infrastructure development investment around the Luzon Economic Corridor, supporting connectivity between Subic Bay, Clark, Manila and Batangas, is an example.**[52] Without this, the NCR risks falling into a cycle of low productivity and job growth, with negative economy-wide effects.

# Mobility, connectivity and decentralization to accelerate subnational development

**To sustain regional growth, it is essential to ease constraints on labor mobility, market connectivity, and decentralized service delivery.** Labor mobility helps address regional disparities by allowing workers to move where jobs are available. Improved transport and digital infrastructure facilitate the flow of goods, services, and ideas, expanding local market access. Decentralizing services such as health, education, and administration can boost responsiveness and equity, provided that local governments have the capacity to deliver.

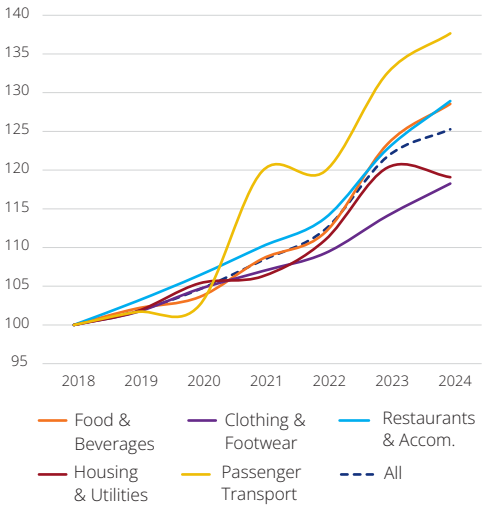
52 Key Luzon Economic Corridor investment projects include the North Luzon and South Luzon expressways connecting Manila to northern and southern Luzon; the Manila-Clark Railway and North-South Commuter Railway; and the port of Manila and Batangas port. The corridor is expected to lead to substantial investment in manufacturing, agri-business, real estate and infrastructure and tourism.

## Mobility of labor

**A major constraint to labor mobility is the difficulty that Filipino workers face in finding jobs that match their educational qualifications when moving between regions.** Barriers such as language and cultural differences, lack of social networks and unfamiliarity with local job markets, limit opportunities and affect interview performance. Employers may prefer candidates with local experience or education credentials, overlooking migrants' qualifications. This can be driven by regional differences in education quality, as well as bias against people from different places, which force migrants to accept roles for which they are, in principle, overqualified, increasing talent misallocation. These vertical skill mismatches can discourage labor mobility between regions as they are more prevalent among migrant workers than local workers, with job-worker matches between the two much larger in the NCR than in MIRs (see Box 5).

### Figure 3.12

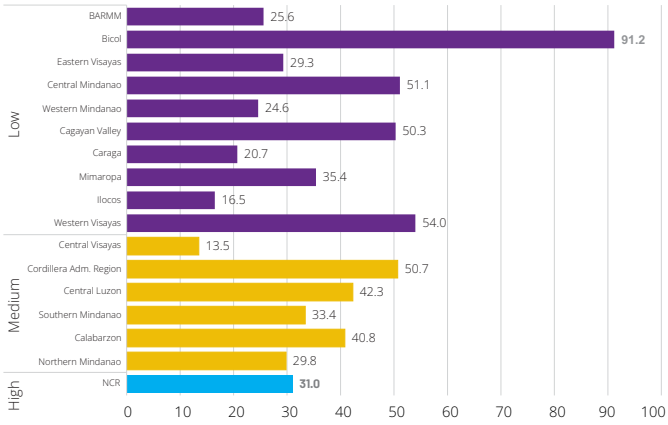
**Consumer prices**
(Index 2018=100)




Source: PSA; World Bank staff calculations.

### Figure 3.13

**Passenger transport services prices**
(2018–2024 growth by region, in percent)




Source: PSA; World Bank staff calculations.

**In addition, passenger transport costs have risen steeply in several Philippine regions, further hindering labor mobility.** Since 2018, the average cost of passenger transport services has increased by nearly 40 percent, with some regions experiencing even sharper rises (Figure 3.12). Regional consumer price data indicate that several LIRs, such as Cagayan Valley, Bicol, and Soccsksargen, have seen increases of over 50 percent (Figure 3.13). These trends highlight the broader challenges facing the passenger transport sector in the Philippines. Outdated public utility vehicles, such as traditional jeepneys, contribute to a lack of scalability, poor air quality, and environmental degradation, ultimately reducing efficiency and reliability in the sector and leading to higher costs for users (ADB, 2012). The Government has pursued modernization, with resistance from operators concerned about financial implications and governance challenges.

**High passenger transport costs and long travel times hinder workers' mobility and inflate commuting expenses.** This is particularly the case in Metro Manila, where daily commutes are often prolonged.[53] The 2010 and 2020 censuses reveal a net outflow of working-age individuals from the NCR to nearby cities within the Calabarzon and Central Luzon regions, such as Angeles, Calamba, San Pablo City, and Lipa. Despite relocating, many workers continue to commute to jobs within the NCR. In 2020, 21 percent of the 5.8 million employed individuals in the NCR—over 1 million people—commuted daily between Metro Manila and these regions. This heavy traffic strains the existing transportation infrastructure, as alternatives to road-based passenger transport remain underdeveloped.

## Box 5

### Vertical skills mismatches of domestic migrants

**The prevalence of vertical skills mismatches varies significantly depending on a migrant's origin region, pointing to region-specific barriers to labor migration.** The likelihood of tertiary educated workers who moved out of LIRs holding a high-skill job depends on the migrant's specific region of origin (Figure 3.14). While tertiary educated migrants from Ilocos or Cagayan Valley are almost as likely to hold a high skill job as non-migrants, the gap is particularly large for workers originating from Soccsksargen (17-percentage-point gap to non-migrants) and from the Bangsamoro Autonomous Region (10-percentage-point gap to non-migrants). A similar pattern is observed among secondary educated workers that migrated: the likelihood of getting stuck in a low-skill occupation depends on the region of origin (Figure 3.15).

**Figure 3.14**

**Probability that a tertiary-educated worker residing in the MIRs and NCR is employed in a high-skill occupation**

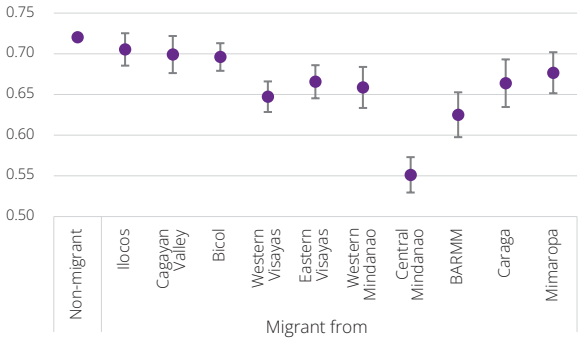
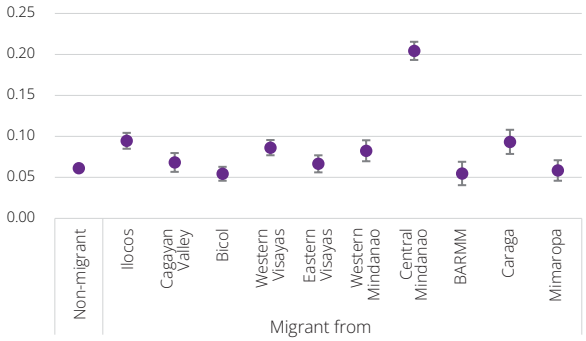



**Figure 3.15**

**Probability that a worker with at least secondary education residing in the MIRs and NCR, is employed in a low-skill occupation**




Source: Sample Household Questionnaire of the Census of Population and Housing 2020; World Bank staff calculations. Note: Error bars indicate 95 percent confidence intervals. See Background Note for a detailed description of methodology.

53 The LFS does not capture information on commuters' travel times, but anecdotal evidence suggests that daily travel times of workers in Metro Manila can often exceed 4 hours.

**About one-third of the difference in job-worker matches between migrants and non-migrants is due to ethnicity effects, suggesting opportunities for enhancing labor intermediation services.** Data from the 2020 population census identify nearly 300 distinct ethnic backgrounds. This finding aligns with extensive literature demonstrating that ethnic networks improve job-matching efficiency(Dustmann et al., 2016), especially in contexts where formal labor market intermediation is weak or inaccessible (Carranza and McKenzie, 2024). Consequently, the results highlight the potential to enhance labor market intermediation services, promoting more effective job searches and relocation across the Philippines. Such improvements are important, as reliance on social networks for job information disadvantage individuals without connections, leading to higher unemployment risks and long job searches (Cingano and Rosolia, 2012).

**The gap in the quality of job-worker matches between migrants and non-migrants is much larger in the NCR than it is in MIRs.** According to both the 2010 and 2020 census, of the workers who move out of LIRs, 70 percent move to MIRs while only 30 percent move to the NCR. While tertiary educated workers who move out of LIRs are, on average, more likely to find a high-skill job in line with their education when they move to the NCR than when they move to MIRs, the gap to workers without a migration background is much larger in the NCR. Focusing on facilitating jobs' search of migrants in the NCR may therefore be helpful.

## Connectivity of markets across regions

**Connectivity infrastructure has been key for spatial convergence.** Despite a large infrastructure gap and high transport costs, the Philippines has seen improvements in its physical transport connectivity, particularly due to enhanced port infrastructure across the archipelago. The development of the Roll-On/Roll-Off (RORO) terminal system is an example. It facilitates seamless intermodal freight transport between land and sea, allowing cargo-bearing vehicles to board and disembark from RORO ships without the need for additional cargo handling (Philippine Development Plan 2023–2028).[54] Prior to 2000, there were only a handful of RORO ports, but these have increased significantly in recent years (Figure 3.16), streamlining inter-island logistics, reducing travel times, and boosting domestic trade between regions.[55]

**The expansion of RORO ports led to increases in firms and jobs in adjacent areas, with effects primarily driven by larger islands with substantial local market access, such as Luzon.** An empirical analysis of the expansion of the RORO port network shows that municipalities that experienced a larger decrease in distance to the nearest RORO port saw larger increases in wage employment and in the number of firms compared with municipalities on the same island that remained further away from RORO ports (see Box 6).

**The employment response to the RORO expansion depends on islands' local market access.** In the island of Luzon, the most populous, which includes Manila, a 100-km reduction in distance to

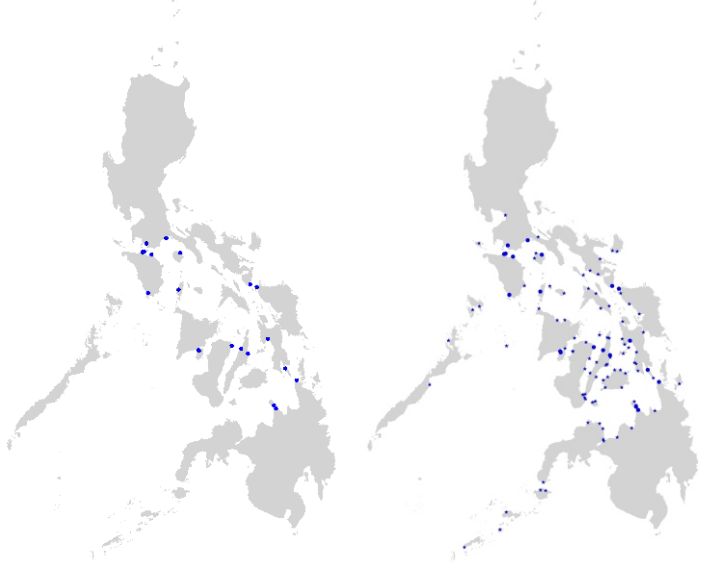
54 National Economic and Development Authority (2023).

55 Go (2022) finds that provinces connected by the RORO port system engage in 39 to 42 percent more trade than comparable provinces without access to this infrastructure.

**Figure 3.16**

**The expansion of the RORO port system over time**

a. RORO ports in 2000 b. RORO ports in 2020



Source: Go (2022); World Bank staff elaboration.

the nearest RORO port was associated with an increase of 18 percent in private sector wage jobs, compared with a 4.9-percent increase on the island of Leyte, which sits at the 25th percentile of the market access distribution. This aligns with previous findings suggesting that trade benefits of RORO expansion were unevenly distributed, with the largest gains concentrated in places close to the largest demand centers (Go, 2022).

**These findings stress that growth in more remote and poorer regions of the country remains dependent on local consumption as inter-island connectivity is still costly.** The robust performance on LIRs observed over the past decade was largely driven by an expansion in non-tradable activities. An expansion of tradable sectors that would integrate these local economies with Manila and surroundings has yet to materialize. A key reason is that even though RORO ports have helped to lower domestic maritime transport costs, they are still high. Transporting a 20-foot container equivalent unit (TEU) from Manila to Cagayan de Oro in the south of the country can be twice as expensive as moving the same cargo via transshipment through Kaohsiung to Taipei, China. Business groups have also confirmed that domestic trade costs can be prohibitive on certain routes.

**Public investment is strategically focused on physical connectivity to drive economic growth.** Of the 207 infrastructure flagship projects (IFPs), 139 are dedicated to physical connectivity, representing US$155.1 billion in planned investment. Four have been completed to date, 50 are under construction, and the rest remain in planning. To expedite IFP implementation, the President issued Executive Order No. 59, streamlining the permitting for IFPs on April 30, 2024. Financing reveals key challenges: 28 percent rely on public-private partnerships (PPPs), 12 percent on general appropriations, and the remainder on official development assistance. This combination of external and private funding underscores the need for efficient project execution and robust national and local governance to ensure that they create maximum opportunities for spatial convergence.

## Box 6

### Spatial employment effects of the RORO port expansion

**In the early 2000s, the Philippines began a significant expansion of Roll-On/Roll-Off (RORO) port facilities across the country.** This policy reduced costs and transit times for trade between islands in the Philippines (Go, 2022). Background analysis prepared for this chapter examines whether the RORO port expansion also contributed to job growth in firms at subnational levels. Since RORO ports can influence employment by enhancing connectivity to other parts of the country, the impact may vary depending on the size of the island. The two main research questions are:

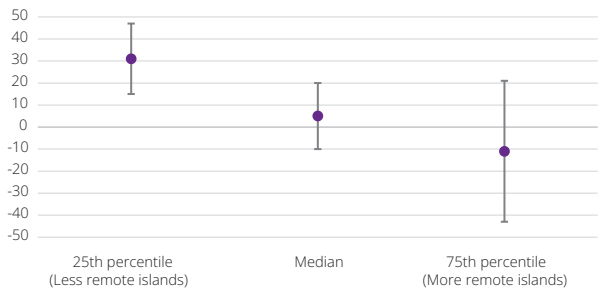
- Does a reduction in distance from RORO ports increase wage and private employment within municipalities?
- Does the effect depend on the size of the island?

**A 100 km reduction in distance from RORO ports increases wage employment by 12 percent and the number of employers by 11.6 percent.**[56] However, these effects vary significantly across islands. Larger islands with greater local market access experience much stronger impacts, whereas smaller, more remote islands see minimal effects. Figure 3.17 illustrates the marginal effects of a 100 km reduction in distance from RORO ports across different islands along the distribution of remoteness. The influence of remoteness is notably stronger for wage employment compared with the number of employers.

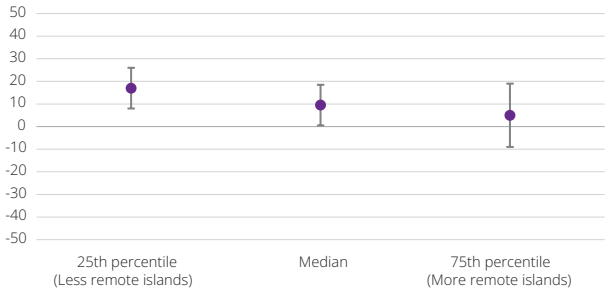
### Figure 3.17

**The spatial employment effects of the RORO port expansion**

a. Wage employment




b. Employers




Source: Based on Eberhard-Ruiz et al. (2024). See Background Note for a methodological description. The figure plots the marginal effects of a reduction in distance from RORO ports by 100km for different islands along the distribution of remoteness from markets, a measure of market access.

56 Using data on the locations of RORO ports, the distance between each municipality and all RORO ports on a given island was calculated. The analysis focused on the minimum distance between a municipality and a RORO port and tracked how this distance changed over time with the establishment of new ports. This information was linked with wage and private employment data from the 2000 and 2020 Census.

## Decentralization of local services delivery

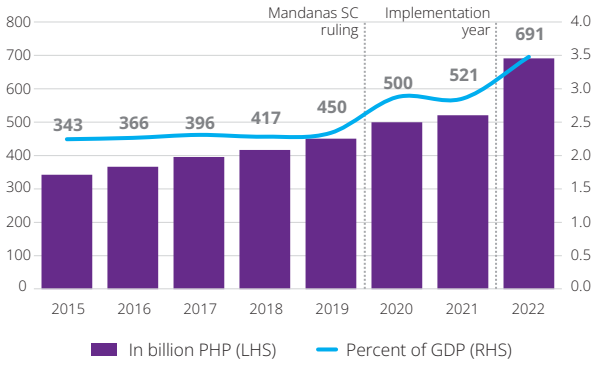
**Sustaining spatially balanced development requires fostering opportunities and capabilities with a targeted focus on the needs of each specific community.** As such, LGUs play a crucial role due to their proximity to these communities. Empowered by the 1991 Local Government Code, LGUs have significant administrative and regulatory powers to address human capital needs, close infrastructure gaps, and implement tailored strategies. LGUs manage over 80 percent of the road network, potentially influencing trade and market access, or investment attraction through Local Economic Development initiatives. However, their effectiveness is limited by capacity, access to funding, and inefficient tax structures. Although many basic services have been devolved to LGUs, the national government has retained control in many sectors, as the LG Code allows national agencies to provide services if LGUs lack resources or capacity.[57]

**The 2019 Mandanas ruling expanded LGU resources by increasing the Internal Revenue Allotment (IRA/NTA) and amplified the need for effective LGU governance.**[58] In 2022, (the first year of implementation of the SC ruling), the IRA/NTA substantially increased to 4.4 percent of GDP, up from around 2.9 between 2015 and 2019 (Figure 3.18). While this pushed subnational spending to 6 percent of GDP (Figure 3.19), nearing levels of subnational spending in upper middle-income countries, resource allocation disparities remain, favoring wealthier LGUs. The challenges highlight the importance of revising intergovernmental fiscal policy to balance equity and effectiveness, alongside capacity-building programs to support LGUs in assuming devolved functions.

### Figure 3.18

**Transfers to LGUs have increased following the Mandanas ruling**

(Constant 2018 PHP billion | Percent of GDP)

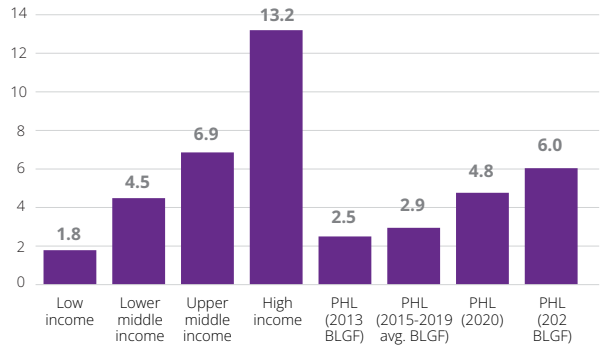



Source: Bureau of Local Government Finance (BLGF); World Bank staff calculations.

### Figure 3.19

**Subnational government spending increased over the decade, nearing the levels of upper-middle income countries**

(In percent of GDP)




Source: Bureau of Local Government Finance (BLGF); 2020 OECD-World Observatory; World Bank staff calculations.

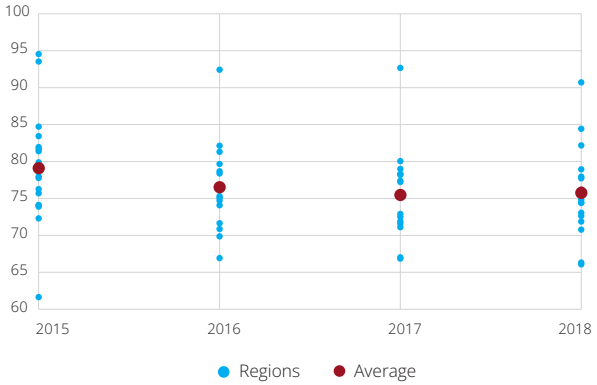
57 Philippines Economic Update, Navigating a Challenging Recovery, June 2021.

58 The Mandanas-Garcia Supreme Court (SC) ruling refers to the final decision on the two separate petitions filed before the SC (consolidated on October 22, 2013) : (i) the petition filed by Congressman Hermilando I. Mandanas and other local officials vs. Executive Secretary Paquito N. Ochoa, Jr., et al. (G.R.No. 199802); and (ii) Congressman Enrique T. Garcia, Jr. vs. Executive Secretary Paquito N. Ochoa, Jr., et al. (G.R. No. 208488). The internal revenue allotment is now denominated the National Tax Allotment (NTA).

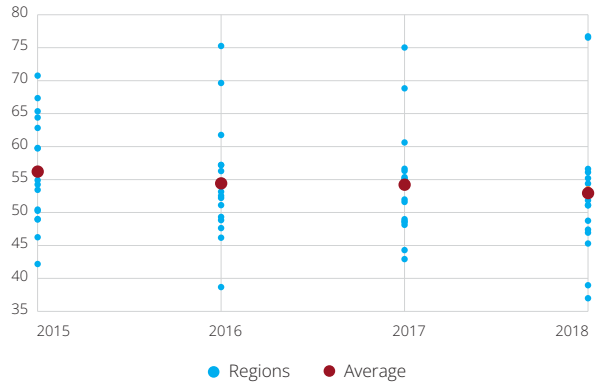
## Figure 3.20

**Budget execution rates vary across regions with low execution of capital outlays, 2015–2018**

a. Total spent as percent of budgeted




b. Capital outlays as percent of budgeted




Source: COA LGU Financial Statements; World Bank staff calculations.

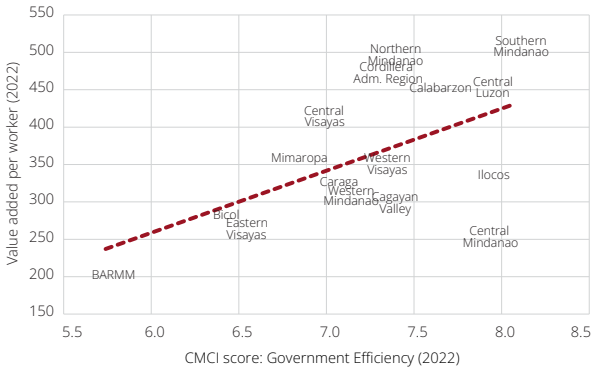
**Despite increased funding, capacity constraints in budget and investment management limit the ability of LGUs to enable quality job creation and growth.** From 2015 to 2018, average budget execution rates across regions declined from 79 to 76 percent, with regional variations (Figure 3.20). Some regions spent as little as 60 percent, while others spent nearly 100 percent of their final budgets. The most severe constraints are in capital outlays, with average execution rates just above 50 percent, indicating inefficiencies in planning, project management, and administrative processes. This led to delayed or incomplete projects and increased reliance on national spending, undermining growth. Addressing bottlenecks in public investment processes and improving regulations on PPPs could enhance absorptive capacity and infrastructure development.[59]

**Regional disparities in LGU capacity mirror economic inequality across the Philippines' administrative regions.** Metrics such as the "Seal of Good Local Governance" and the Cities and Municipalities Competitiveness Index (CMCI) show stronger performance in economically advanced regions (the NCR and Calabarzon). Good governance and business-friendly environments correlate with labor productivity (Figure 3.21) and a larger share of formal jobs (Figure 3.22), but inefficiencies in regulatory processes—such as complex permit procedures, inconsistent national regulation implementation, and poor governance—discourage investment and hinder job creation. Improving LGU regulatory efficiency and transparency is essential for fostering quality jobs and sustained convergence.

59 Republic Act No. 11966, also known as the "PPP Code of the Philippines," and its Implementing Rules and Regulations (IRR) provide a new legal framework with new responsibilities for LGUs and specific processes for solicited and unsolicited PPPs.

## Figure 3.21

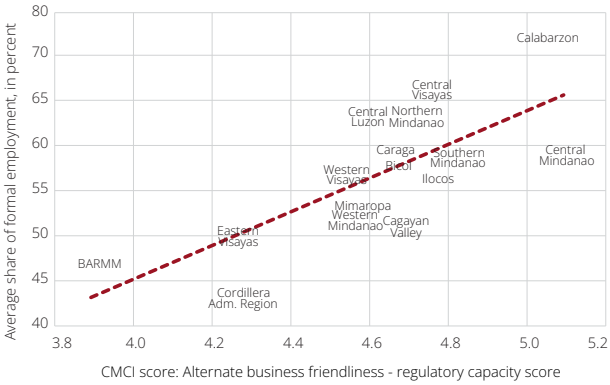
**More efficient LGUs tend to be more productive**




Source: Source: DTI; World Bank staff calculations. Note: Excludes NCR; Governance efficiency pillar of the CMCI is made up of indicators that measure compliance to national directives, presence of an investment promotion unit, compliance with ARTA citizens charter, capacity to generate local resources, capacity of health and school services, getting business permits, peace and order, and social protection.

## Figure 3.22

**Stronger regulatory capacity is associated with higher formal employment**




Source: Source: DTI; World Bank staff calculations. Note: Excludes NCR. The business friendliness score is comprised of the CMCI sub-indices from economic dynamism and government efficiency pillars. It is made up of indicators that measure safety compliance, cost of living and cost of doing business, financial deepening, presence of professional orgs., and compliance with national directives.

# CHAPTER 4
# DIGITAL TECHNOLOGIES, JOBS AND PRODUCTIVITY

*This chapter explores the benefits of digital technologies for better jobs and growth in the Philippines. Adopters grow faster and pay higher wages. Yet, Philippine firms are only partially digitalized, lagging peers. Smaller firms face a wider digital gap than larger ones. Uncertainty about technologies among entrepreneurs and gaps in information, management quality, and human capital are key deterrents of adoption. Limited competition, scarcity of digital skills, and infrastructure gaps are also constraints. The Philippines has significant room for improvement in these areas. Digital platforms and digitally enabled services, particularly through the IT-BPO sector, are crucial in the context of the AI revolution, as escalators for job creation in competition- and innovation-rich activities. This chapter presents recommendations to accelerate the diffusion of digital technology.*

## Digital technologies

**Digitalization is linked with higher productivity in the Philippines.** Investing in IT, software, and data technology in the Philippines is linked to increased TFP, particularly for services sector firms (Figure 4.1, panel a).[60] This relationship holds after controlling for sector-level differences (Figure 4.1, panel b). Integrating advanced digital technologies into general business functions (GBFs, HR management, production planning, marketing, etc.) is associated with higher labor productivity.

**Digital adoption is also linked to higher wages.** Digital technologies are changing the nature of work by providing more job opportunities to both skilled and unskilled workers. In the Philippines, adoption and intensive use of digital technologies positively correlate with wages.

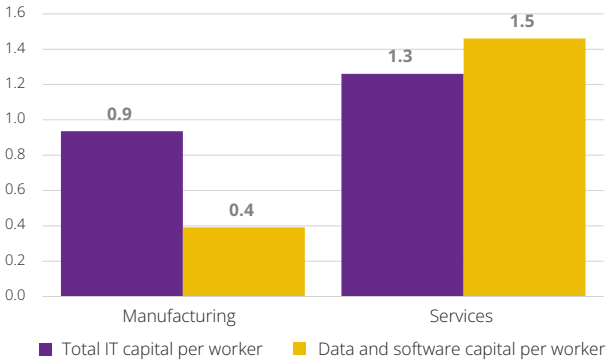
**Despite the productivity premium of digitalization, most firms in the Philippines rely on analog technologies.** The Philippines has low technology sophistication compared with its peers, with the average Philippine firm scoring 1.5, indicating primarily manual processes (Figure 4.2).[61] In manufacturing and services, firms score lower than their regional peers, with notable disparities between top-performing and average firms. For example, frontier firms in the Philippines (the top 10 percent in productivity) are 1.5 points less sophisticated than frontier firms in the Republic of Korea. Bottom-tier firms are 0.7 points behind their peers in the Republic of Korea.

60 In the case of data analytics, for instance, productivity increases are three times higher in services than in manufacturing.
61 Or using a computer using standard software such as Microsoft Excel.

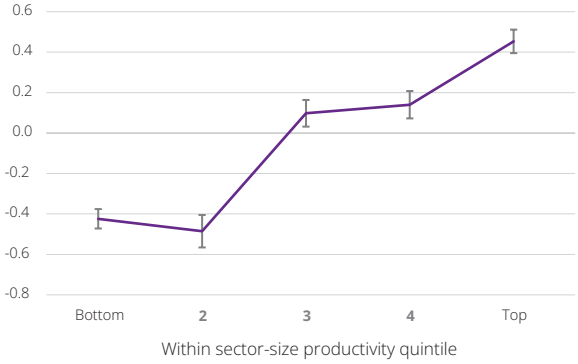
## Figure 4.1

**Within-firm TFP changes from IT capital or data and software capital per worker increases**

a. Within-firm TFP changes from IT capital or data and software capital per worker increases.




b. Ave. labor productivity difference from sector-size mean and within sector-size productivity quantile

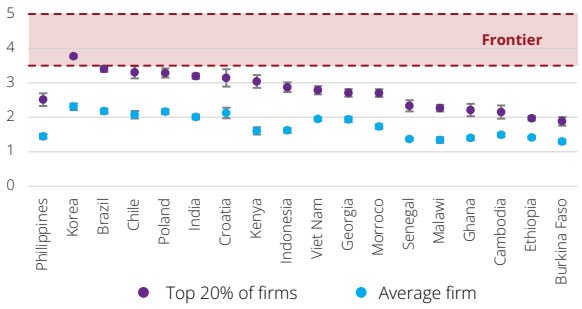



Source: World Bank (2024f) (left); Firm Adoption of Technology Survey 2024; World Bank staff calculations. Note: Panel (b) plots the correlation between labor productivity and technology sophistication within sectors to account for the differences in production functions across sectors.

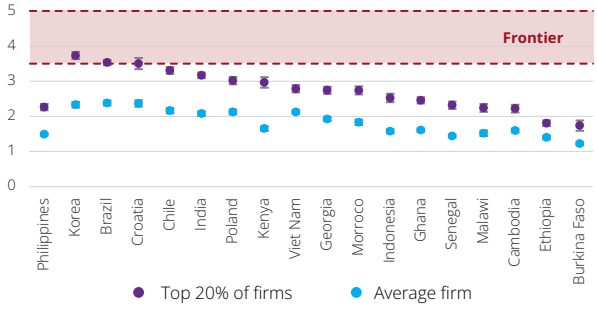
## Figure 4.2

**Adoption of digital software to perform business functions in selected countries**

a. GBF index in the manufacturing sector by country




b. GBF index in the services sector by country




Source: Firm Adoption of Technology Survey 2024; World Bank staff calculations. Note: The figure plots each country's average level of technology sophistication for general business functions (GBFs) at the intensive margin. Results are based on ordinary least squares (OLS) estimation using sampling weights and controlling for sector, country, and size. GBF = general business function.

**Technology diffusion in the Philippines is slower than in peer countries such as India, Viet Nam, and Brazil.** Off-the-shelf technologies, such as those for business administration and quality control, are underused. For example, while 86 percent of firms have internet access, 62 percent lack full adoption of tools such as mobile apps, enterprise resource planning (ERP) systems, and specialized business software. Among firms adopting digital tools for business administration (23.7 percent), 73.8 percent use them minimally.

**A pronounced digital divide exists between large and small firms.** Larger firms are more digitally advanced, while smaller firms struggle with basic digitalization. Sector-specific data show that many firms cluster below their sector's average sophistication, and smaller firms rarely adopt Industry 4.0 technologies such as industrial robots. Cloud computing is a notable exception, though penetration rates remain low compared with peers.

**These limitations in digital adoption hinder firms' ability to compete locally and globally, and to scale up and create jobs.** Firms are yet to bridge the gap in fully integrating digital technologies into daily operations, further slowing the transition to advanced systems seen in global peers.

# Determinants of technology adoption

**Understanding the factors influencing technology adoption is essential for formulating effective policy solutions.** Some of these barriers and drivers are related to the capabilities of management and staff in firms, requiring appropriate management quality and human capital for upgrades. Other factors are related to the incentives that firms face to adopt technology, the credit ecosystem, and the availability of adequate human capital.

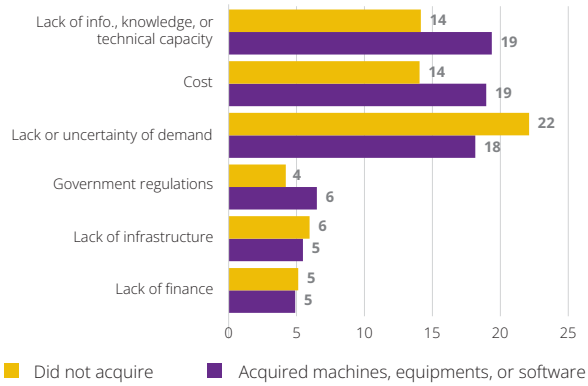
## Internal determinants

**Lack of information about technologies hinders technology adoption.** In the Philippines, lack of information is reported as the main barrier to technology adoption (Figure 4.3). Potential adopters of new technologies need to recognize the value of acquiring them, for example by understanding their economic benefits. From a strategic perspective, managers need to understand how acquiring technological capacity will lead them to compete and thrive in the marketplace. Filipino managers perceive that the returns to digitalization may not be high and may also be uncertain.

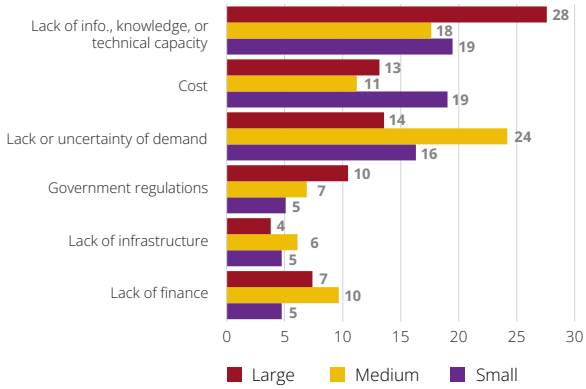
**Figure 4.3**

**Perceived barriers to adopt technology by firms in the Philippines**

a. Barriers to adopting by type of firm, in percent of firms




b. Barriers to adopting by firm size, in percent of firms




Source: Firm Adoption of Technology Survey 2024; World Bank staff calculations

**Firms with higher levels of digitalization exhibit greater human capital and better managerial qualities.** Firm-level evidence shows that managers who have studied abroad and have previous experience with large companies are more likely to adopt and use more sophisticated technologies. Meanwhile, college-educated managers are more likely to adopt digital technologies, but the impact does not extend to intensive use.

**Behavioral biases also hamper the willingness to adopt digital technologies.** Philippine firms overestimate their technology sophistication, reducing the incentive to upgrade. A firm's decision to adopt and upgrade technologies is influenced by how it perceives its position compared with its competitors. This can lead to complacency and a reluctance to adopt technologies as firms' self-assessments are inconsistent with their actual economic position as determined by the GBF index.[62]

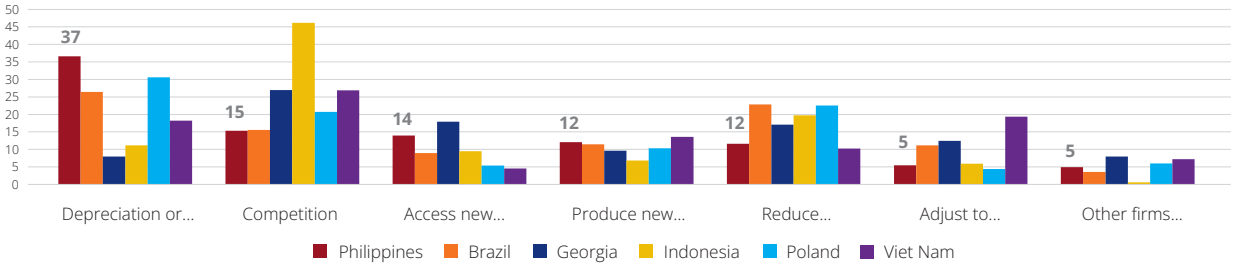
## External determinants

**Lack of competitive pressure may be delaying technology adoption in the Philippines.** Unlike other countries, where survey results point to competitive pressures being the main driver of technology adoption, most managers in the Philippines point to the replacement of technology when this becomes necessary as a primary driver. Competitive pressure seems insufficient to motivate Philippine firms to adopt new technologies (Figure 4.4). Removing distortions in capital markets that impede de jure entry of foreign firms in key sectors and streamlining complex business permitting that impedes the entry of foreign firms can help add competition, leading to faster technology upgrading.

**In the Philippines, advanced digital technologies diffuse faster in industries that show limited competition (or that are at risk of competition).** The speed of diffusion of technologies such as ERP for business administration and automated systems for quality control is slightly faster for sectors that operate under a healthier competition environment.

### Figure 4.4

**Drivers of technology adoption**
(In percent of firms)




Source: Firm Adoption of Technology Survey 2024; World Bank staff calculations.

**Competition and knowledge transfer significantly influence technology adoption among companies in international markets.** Firms trading with MNCs are more likely to use advanced digital technologies. International market access boosts productivity through competition and learning, promoting technology adoption. Trading with MNCs facilitates knowledge diffusion, enhancing local firms' capabilities.

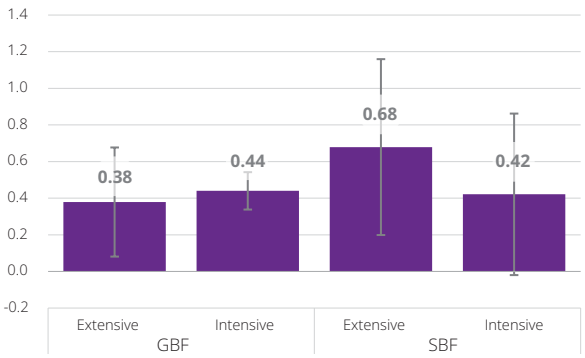
62 The survey analyses included re-scaling values from 1 to 5 and taking quintiles of the distribution). The FAT survey asks firms to assess their technological level relative to similar firms within the country on a scale of 1 to 10.

**Exporting firms in the Philippines are more technologically sophisticated than domestic-oriented ones** (Figure 4.5, panel a), **and more likely to use digital technologies.** Firms linked to MNCs adopt digital technologies more frequently than those without such connections (Figure 4.5, panel b). Utilizing digital assessment tools, connecting suppliers with international buyers, and updating certifications for local providers can improve performance. Global digital platforms provide opportunities for freelancers. Integration into global markets fosters technology sophistication, highlighting the need to reduce trade and investment costs.

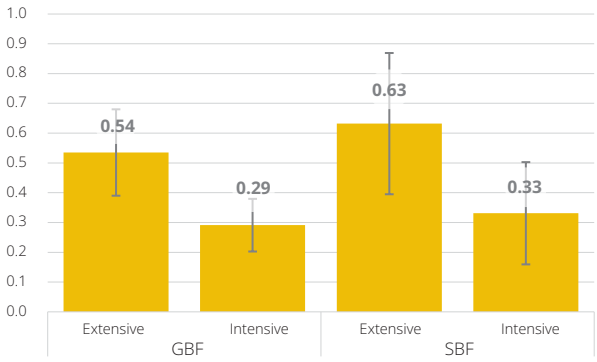
## Figure 4.5

**Correlation between trade and technology adoption**

a. Exporting status




b. Link with MNEs




Source: Firm Adoption of Technology, Survey, Philippines 2024; World Bank staff calculations. Note: The figure plots each country's average level of technology sophistication for general business functions (GBFs) at the intensive margin. Results are based on ordinary least squares (OLS) estimation using sampling weights and controlling for sector, country, and size. GBF = general business function.

## Digital and managerial skills

**Digital occupations are increasingly common in the workplace in the Philippines, especially in sophisticated segments of the services sector.** Digital workers (those primarily using digital technologies and the internet), comprise a growing percentage of the labor force. In 2023, digital workers accounted for 12 percent of the workforce, with 9 percent in the formal sector.

**Digital occupations are often filled with the most educated workforce.** In the Philippines, individuals who have completed secondary education, and especially those with tertiary education, are more likely to secure formal digital job positions than those with a primary education or less. Regular and digital jobs now require more digital skills from the workforce. Digital skills require individuals to have a good grasp of basic literacy and numeracy skills, which are typically acquired through formal education and training. Maximizing gains from digitalization will require the build-up of digital capabilities in the labor force.

**Compared with its peers, the Philippines suffers from an important deficit in ICT skills** (Figure 4.6). Foundational digital skills encompass an individual's ability to use technology and digital devices to carry out basic tasks: using social media for product promotion, employing basic arithmetic formulas in a spreadsheet, sending, and receiving emails, using search engines to locate

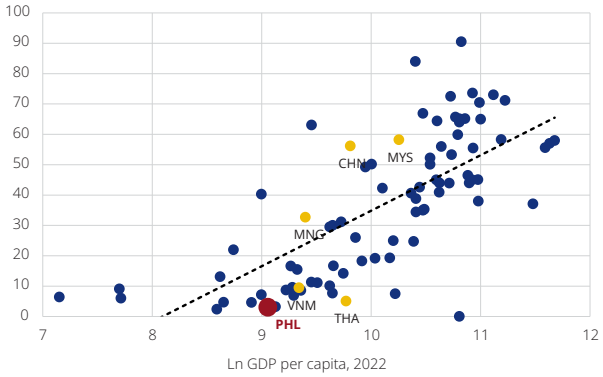
relevant information, creating presentations, and transferring and storing files. At the intermediate level, workers should be capable of independently utilizing more advanced software and technologies for analysis, creation, management, and design. In the Philippines, the percentage of individuals using the copy-and-paste function to duplicate or move information within a document was lower than most regional peers. Only a small portion of the population possesses basic ICT skills to send emails with attached files. Using basic arithmetic formulas in a spreadsheet or creating electronic presentations with presentation software is limited.

## Figure 4.6

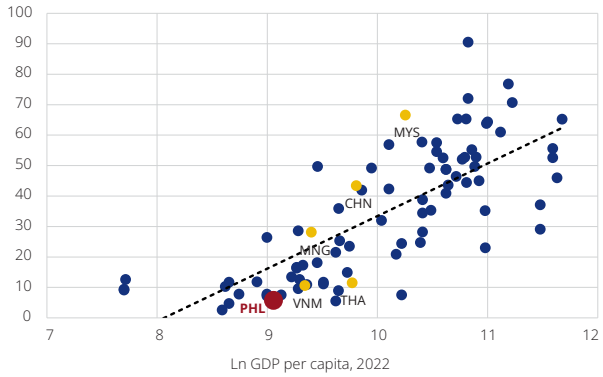
**Prevalence of foundational digital skills in selected countries**

(ICT and digital skills prevalence, in percent, 2022)

a. Installing and configuring software




b. Transferring files




Source: ITU; World Bank staff calculations.

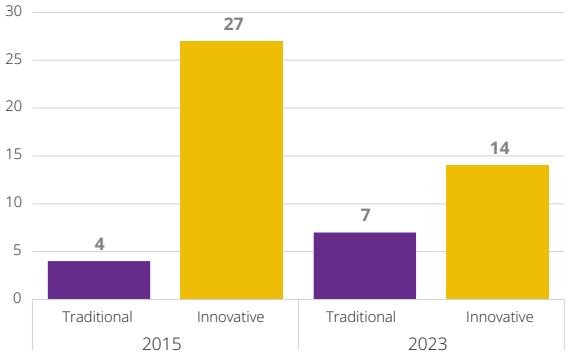
**Philippine firms' management quality and human capital lag peers, creating further barriers to digitalization.** Limited capabilities reduce the incentive to adopt more advanced technologies, as firms cannot effectively use them to improve their production processes. This gap is evident across multiple indicators: the Philippines ranks below MICs such as Georgia and regional peers such as Viet Nam and Indonesia regarding human capital. It has a lower supply of researchers than expected for its development level. TESDA and DOLE support on-the-job training and vocational education tailored to industry needs (for example, the EBET initiative) are crucial in building the necessary capabilities, and upskilling and reskilling workers.

**Innovative firms are affected the most by limited skills in the labor force.** Having skilled workers and appropriate incentives are crucial for successfully adopting modern technologies. In the Philippines, firms have indicated that there can be a shortage of necessary skills, with innovative firms twice as likely to be dissatisfied with workforce education compared with traditional firms (Figure 4.7, panel a). A comparison of educational levels shows that the Philippines ranks below many regional peers, only surpassing Lao PDR and Cambodia, signaling the negative impact of the lack of advanced education on the workforce (Figure 4.7, panel b).

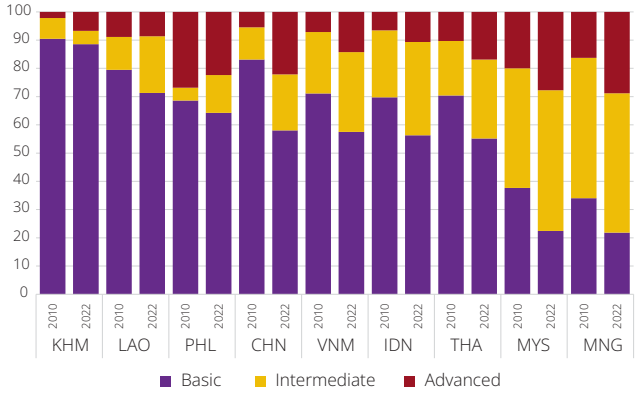
## Figure 4.7

**The extent to which the workforce education attainment constrains firms in the Philippines**

a. Share of firms identifying an inadequately educated workforce as a major constraint




b. The educational level of the working-age population in East Asian countries



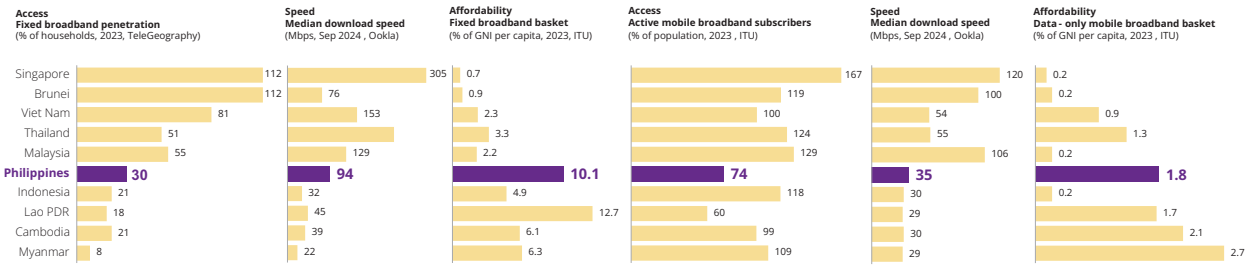

Source: WBES; World Bank staff calculations.

# Access to digital connectivity

**The opportunities that digital technology adoption offer depend on reliable and competitive digital connectivity.** Infrastructure, particularly in the telecoms sector, is crucial in facilitating businesses' adoption of digital technology. Higher internet download speed correlates with higher TFP and sales, showing evidence of the importance of connectivity infrastructure.

## Figure 4.8

**Digital connectivity access, speed, and cost for selected East Asian countries**




Source: Kanehira et al. (2024).

**Access to fixed broadband in the Philippines is lower than that of its peers, with small firms particularly underserved. About 87.5 percent of Philippine firms access internet.** This is lower than in other MICs such as Georgia (95 percent). Most medium and large firms in the Philippines have access to basic internet, but not small ones. Implementing productivity-enhancing technologies, such as ERP systems and Industry 4.0, relies on access to fixed broadband rather than mobile broadband. The Philippines' access to fixed broadband (proxied by households) is 30 percent, much lower than Viet Nam (81 percent), Malaysia (55 percent), and Thailand (44 percent)

(Figure 4.8). Catching up with regional peers in terms of access to, speed and cost of digital connectivity requires a combination of more competition and more investment in infrastructure.

## An enabling institutional environment for digitalization

**To create opportunities for firms to participate in the global digital economy, cybersecurity, online protection, and internationally compatible cross-border data regulations are key.** While personal data protection features a robust regulatory framework, trust and data utilization improvements are needed. The World Bank's B-Ready 2024 ranks the Philippines lowest among four ASEAN peers in cybersecurity and online safety indicators. The Philippines has no specific legislation or regulatory controls for non-personal or industry data, limiting the gains from supporting secure, fair, and trusted data exchange in business ecosystems to enable aftermarket or other data-driven innovative services. Also, cross-border data transfers are vital for digital market expansion and international investment but require global coordination and compatible data-protection regimes. After ratifying the Regional Comprehensive Economic Partnership (RCEP) in early 2023, the Philippines must adhere to cross-border data flow commitments. Uncertainty around data localization and restrictive AI legislative proposals further affect cross-border data flows.

# Digital platforms

**Digital platforms offer growth and job creation opportunities.** Digital platforms are online systems or networks that facilitate transactions between users and businesses. Typical examples are e-commerce, streaming, and the freelance and gig economy, together with financial platforms or educational ones.

**Philippine firms have adopted digital platforms but often underuse them in their operations.** About one in four firms in the Philippines have a website or use social media for business purposes, which is higher than in peers. For sales, firms are employing more advanced technologies such as social media (45 percent), digital platforms (10 percent), websites (13 percent), and electronic orders (3 percent), as seen by the share of firms adopting and using digital technologies for this specific purpose. On-premises sales, however, remain the primary sales method (82 percent). These platforms have yet to integrate into business operations fully.

**More firms are using payment platforms, but few use them extensively.** Since the pandemic, digital transactions have surged in the Philippines, with monthly digital payments growing from 1 percent of total payments in 2013 to 20.1 percent in 2020.[63] However, intensive use of these platforms remains limited, and cash remains the primary payment method in firms. Gaps in last-mile payment facilities and low consumer trust hamper intensive use of these technologies.

63 World Bank (2022).

**Digital labor platforms are creating new opportunities for Filipino freelancers, but gaps in digital capabilities limit workers' ability to maximize gains.** Platforms have enabled gig work, which has gained global momentum. Demand for online contractors has surged in the Philippines and across the region, with job postings nearly tripling between 2017 and 2021. By 2022, the Philippines ranked among the top providers of online labor globally, following South Asian countries (India, Bangladesh, and Pakistan), the United States, and the United Kingdom. However, the Philippines experienced an overall decrease in the freelancing market share from 5.7 percent in 2018 to 3.5 percent in 2022, with Ukraine, the Russian Federation, and others closely following. Filipino online workers primarily engage in lower-paying clerical and data-entry tasks, commanding 13.3 percent of the market share in these areas. In contrast, their presence in higher-skilled, better-paying roles such as software development and technology, remains limited at just 1.4 percent, suggesting a critical gap in digital skills.

## Box 7

### Revolutionizing healthcare in the Philippines using digital technology

**The Philippine healthcare management information systems sector, also known as the healthcare services industry in the Philippines, has been growing strongly.** These services include care management, claims processing, billings, and collections. In 2022, industry revenues reached US$3.6 billion, which is 11 percent of the overall IT-BPO sector, up from US$2.4 billion in 2016. Employment in the sector was at 173,000 in 2022, representing 11 percent of the overall IT-BPM sector workforce. The United States contributes between 75 and 80 percent of the overall revenue for this sector. Recent advances in digital technologies, combined with insights gained from the pandemic, present opportunities for the Philippines. For instance, telemedicine can offer healthcare services to Filipinos residing in remote locations. Telemedicine platforms allow healthcare professionals to consult with patients remotely, thereby enhancing access to healthcare in underserved regions of the Philippines.

**Telemedicine, care management, and claims processing can enhance the export of high-value services within the IT-BPO sector.** In addition, it increases job opportunities for Filipino doctors, nurses, and other healthcare professionals, allowing them to provide services internationally, particularly to countries with aging populations experiencing shortages in healthcare professionals, without leaving the Philippines. Remote rendering of services can also provide significant advantages to Filipino healthcare professionals who might otherwise choose to migrate abroad. Many Filipino healthcare workers seek employment overseas due to opportunities in the global healthcare sector. According to data from the Philippine Overseas Employment Administration (POEA) and the Commission on Filipinos Overseas (CFO), a total of 350,361 doctors, nurses, and midwives left the country for overseas work from 1990 to 2017.

# Disruptive technologies and jobs: Artificial intelligence

**Some jobs in the Philippines are at risk of technology displacement.** AI exposure and AI's potential complementarity can affect employment. The Philippines has slightly fewer jobs comprising routine tasks than its peers. However, the Philippines is more exposed to AI's displacement effect than other EAP countries due to its higher engagement in cognitive services sectors, such as contact centers in the IT-BPO sector.

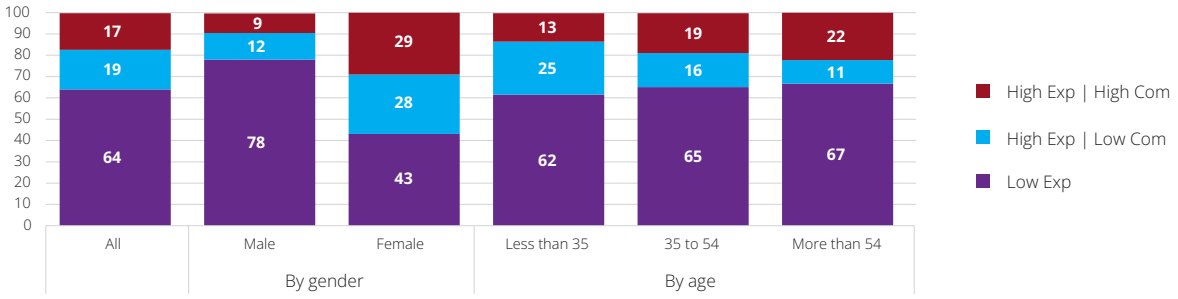
**The Philippines' workforce has relatively low skills, but the prevalence of nonroutine jobs reduces risks** (Figure 4.9). The high share of low-skill occupations employed and the relative prevalence of routine task content in professions drive susceptibility to automation. Employment concentration on physical non-routine tasks, such as construction, transport, and agriculture, has reduced the exposure to AI; however, front-facing professions such as sales, telemarketing, and customer services remain at risk.

**AI also presents new opportunities for new jobs and more efficient digital technologies, but reaping the benefits requires investments in technology adoption and workers' skills.** The development of AI will complement and improve existing digital technologies in tasks as diverse as production planning, supply chain management, or quality control. As discussed, however, leapfrogging is rare, and Philippine firms need to first invest in accumulating digital capabilities and getting ready to adopt AI-augmented technologies.

## Figure 4.9

**Most employment in the Philippines has low exposure to AI, but about one-fifth is highly exposed with low complementarity, suggesting potential risks**

(Exposure and complementarity to AI by 4-digit ISOC occupation, in percent of group employment, 2023)




Source: LFS 2023; Felten, et al. (2021); Pizzinelli et al. (2023); World Bank Staff calculations. Note: We adopt the Cucio and Hennig (2025) occupational classification at 4-digit ISCO 08, based on Felten et al. (2021) and Cazzaniga et al. (2024), which defines occupations as task lists. Felten et al. (2021) developed an AI exposure index measuring task overlap with AI capabilities, while Cazzaniga et al. (2024) introduced a complementarity score considering skill level and societal context. Combining these, occupations are classified as "high exposure/high complementarity," "high exposure/low complementarity," or "low exposure," using sector median exposure and complementarity scores as cutoffs.

# CHAPTER 5
# CLIMATE RESILIENCE AND FIRMS

*Climate-related events affect the private sector through two channels. First, the direct effects of climate events or natural disasters on firms' decisions and performance. This matters because of the high exposure of firms to weather events in the Philippines. Second, the effects of changes in consumer preferences or national or international regulations on firms' energy efficiency decisions. This chapter recommends specific policies that enable firms to adapt to climate events, comply with international emission standards, and improve efficiency by reducing energy intensity.*

## Firms' resilience to climate change and extreme weather events

**Climate change is shaping economic outcomes, modes and structures of production.** The increased prevalence and unpredictability of both extreme infrequent and less extreme recurrent weather events impact the performance of firms exposed to these conditions. It also has effects on firms', households' and governments' behavior in terms of where to locate, and where to allocate the extra dollar for investment purposes to adapt to it. The devastating prospects of climate change have also led to public and private efforts to mitigate it to a certain extent, which typically implies decoupling economic growth from emissions.[64] This can happen through increasing energy efficiency, or through transitioning to renewable energy sources with lower emissions.

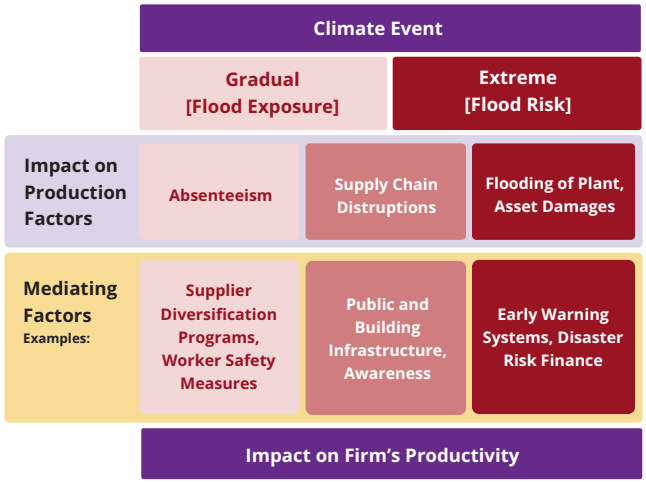
**This chapter looks at the impact of climate change on firms in the Philippines, focusing on increased rainfall and flood risks.** It examines how these climate-related events affect manufacturing and services firms' performance and decisions.[65]

**Changing weather patterns matter for firms' economic performance** (Figure 5.1). As climate-related events become more frequent, unpredictable, and severe, businesses face disruptions in supply chains, production processes, and damage to assets. Firms that invest in climate-resilient infrastructure, adjust business models, and are prepared to manage risk arising from weather conditions are more likely to succeed. National and local governments that invest in quality infrastructure to cope with weather events, and that create an ecosystem for firms to follow suit will attract more business.

64 Various sources of emissions contribute to climate change. The measure used here is $CO_2$ equivalent.
65 A discussion on the impact on agriculture can be found in the latest Philippine Climate Change Development Report (Balota et al., 2023).

**Figure 5.1**

**From climate events to firms' performance**




Source: World Bank staff elaboration.

**Increasing exposure to small-scale floods affects firms' performance and decisions.** Small-scale floods and extreme weather events (EWEs) represent distinct types of risks with different implications for businesses. The former, triggered by moderate to heavy rainfall or regional storms, have a gradual but steady impact on firms. They increase costs, delay supply chains, and cause higher absenteeism. In contrast, EWEs, such as catastrophic floods or storms, are high-impact, low-probability events that cause significant damage to firms' infrastructure, often leading to temporary but severe setbacks in operations. The immediate effects of EWEs may be more destructive, while their impact on long-term firms' performance is shorter in duration.[66] Instead, the cumulative effects of repeated small-scale floods are persistent on firms' performance. Policy responses differ. Resilient infrastructure and long-term adaptation are more critical for recurrent smaller floods, disaster risk finance or emergency response systems more for mitigating effects of extreme events.

## Exposure to climate risks

**This section focuses on one source of climate risk: increased precipitation and its effect on flood risk.** Based on the analysis, it is the most relevant for the private sector in the Philippines.

**Increased precipitation raises the exposure of Philippine firms to floods.** In economic hubs such as Metro Manila, Laguna, and Cebu, daily precipitation exceeding 50 mm (among the highest levels globally) is not uncommon. High precipitation is often concentrated within a few days when it reaches levels that typically cause local flooding. In Cebu, for example, the average daily precipitation exceeded 50 mm in several geocoded locations on 10 days in 2023, and on 18 days in Quezon, affecting firms' operations, workers, and suppliers. In contrast, areas such as Bulacan and Tarlac face lower, though still elevated flood exposure. Insufficient local infrastructure exacerbates risks, which have increased as rainfall levels in Metro Manila alone are 40 percent higher now than they were in the 1960s.

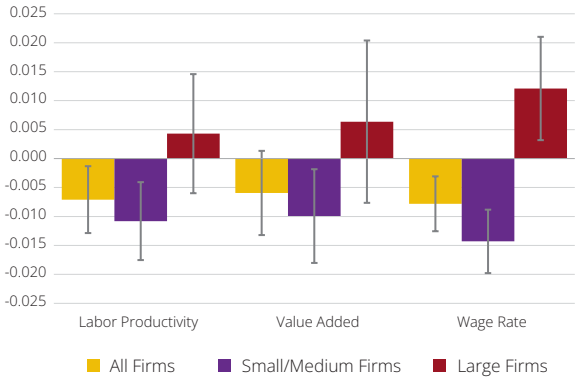
66 Balboni et al. (2024)

## Performance

**Increased flood exposure negatively affects wages, firms' value added and productivity.**[67] Firms located in neighborhoods in which heavy-rain days were above average, had reduced value addition, lower labor productivity, and therefore paid lower wages. An additional day of heavy rain (defined as more than 100 mm of precipitation) is associated with an average 1-percent reduction in the firm's real value added and its labor productivity (Figure 5.2), while wage rates decline by 1.4 percent.[68] This is related to disruptions that rain and floods cause on the mobility of people, and inputs, and higher maintenance and repair costs due to partial flooding of premises.

### Figure 5.2

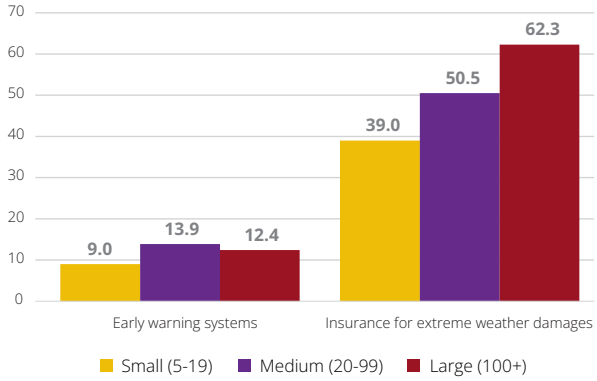
**Flood exposure and firm outcomes, by size**
(Proportional change)




Source: PSA ASPBI/CPBI 2012-2021; weather satellite data at barangay level using the World Bank's Firms and Climate Adaptation Policy Tool; World Bank staff calculations. Note: Estimates from pooled cross-section OLS. Firm, industries, provinces, and year controls were included.

### Figure 5.3

**Adaptation practices by size**
(Percentage of firms implementing them)




Source: Bureau of Local Government Finance (BLGF); 2020 OECD-World Observatory; World Bank staff calculations.

**The effects of climate shocks on firms depend on firms' size, sectors in which they operate, and capacity to adapt.** The negative effects primarily impact services sector SMEs rather than manufacturing or large firms. Larger companies, especially exporters and MNCs, show greater resilience due to better management and credit access.[69] Small firms are particularly vulnerable due to limited adaptation capabilities. Only 9 percent of small firms use early warning systems (12/14 percent for medium/large firms), and 39 percent have weather damage insurance (51/62 percent for medium/large firms) (Figure 5.3 ).

67 To understand the effects of extreme climate conditions on firms' performance, barangay-level data on weather conditions over the past decades is combined with geo-referenced firm-level data.

68 These effects control for differences across provinces, such as infrastructure quality, economic conditions and general provincial-specific climate patterns, as well as sector-specific characteristics, such as sensitivities to weather conditions, and for year-effects, such as macro, policy, or national level climate variations.

69 Grover and Kahn (2024).

## Box 8

### Adapting to climate change: firms and public policy

#### Effects of climate change on firms' performance and decisions

- *Labor Market and Productivity:* Extreme heat reduces labor productivity and disrupts job stability, particularly in labor-intensive sectors. Lyu et al. (2024) find that extreme heat is associated with a decline in the labor share within Chinese manufacturing firms, driven by the substitution of capital for labor as firms attempt to maintain output. Each additional day of extreme heat above 90°F leads to a 0.13-percentage-point decrease in labor share. Stronger effect in financially constrained and low-skill intensive firms.
- *Supply Chain and Operational Resilience:* Firms with diversified supply chains are more resilient during extreme weather events. Production network disruptions, as seen during natural disasters such as the Japanese 2011 tsunami, highlight the importance of diversifying suppliers and establishing financial links to mitigate the risks from localized shocks. This aligns with the findings for the Philippines, where firms have shown a tendency to leave high-risk areas, indicating their response to climate risks in their long-term planning.
- *Resilience and Employment:* Firms with diversified supply chains and adaptive measures recover faster, preserving jobs. Key strategies include resilient infrastructure, cooling systems, flexible work options, and robust health and safety measures.

#### Job Protection

- *Infrastructure:* Flood control systems (e.g., Singapore's basins, Bangkok's flood gates) reduce job losses.
- *Financial Support:* Grants and climate insurance help firms, especially SMEs, invest in resilience.
- *Training and Flexibility:* Managerial training, public awareness, and flexible labor markets protect jobs by enabling faster recovery and worker reallocation.
- *Business Continuity Plans:* In Japan, business continuity plans (BCPs) are a key part of the strategy to help firms adapt to climate change. The Japanese government rates firms on the quality of their BCPs, which include detailed strategies for maintaining operations during and after extreme weather events. Firms with high ratings access concessional finance and incentives. Plans cover establishing backup sites, diversifying supply chains, training employees for emergency response, and early warning systems.
- *Streamlined Construction Permits:* Simplifying permitting in climate-vulnerable municipalities to facilitate investment in flood walls and resilient buildings, particularly benefiting SMEs.
- *Risk-Based Health and Safety Regulations:* Standards for resilient buildings and worker safety through health and safety licenses and inspections (e.g., temperature limits and ensuring safe commuting arrangements).

## Firms' decisions

**Extreme climate-related events shape the long-term strategies of Philippines firms, influencing where firms choose to set up shop.**[70] Around 6 percent of Philippine firms relocate each year, with the highest movement out of Metro Manila (8.3 percent), Central Visayas (5.9 percent), and Calabarzon (4.7 percent). Services sector firms (6.2 percent) are more mobile than manufacturers (4.8 percent), and large firms (9.2 percent) relocate more than small ones (4.8 percent). Notably, firms tend to exit areas with high precipitation and favor drier regions, reflecting an internalization of weather risks in their planning. This trend underscores the importance of resilient infrastructure for LGUs aiming to attract and retain investment.

## Box 9

### Protecting tourism jobs from climate change

**Tourism contributes to 6.2 million jobs for Filipinos and US$9.1 billion in annual exports.** Exports have grown at 9.1 percent per year since 2010. The sector contributed 8.6 percent to the country's GDP, and it is a powerful job creator. By some estimates, it employs 12.9 percent of total employment.

**The tourism sector is vulnerable to natural disasters and climate change.** Many popular tourist destinations are coastal areas susceptible to coastal erosion, coral bleaching, and storm surges. These events can damage infrastructure, degrade natural attractions, and disrupt local livelihoods dependent on tourism. The increased frequency and intensity of natural disasters threaten the stability and appeal of these destinations, making it important to address these vulnerabilities for the sustainability of the tourism industry. In 2023, around 45 percent of all tourism arrivals were in these highly vulnerable coastal and island destinations.[71] SMEs in the hotel and hospitality segment reported a higher average damage due to extreme weather as a share of revenue compared with larger firms and firms operating in segments other than tourism (World Bank 2024b). In December 2021, when Typhoon Odette hit Central Philippines, 90 percent of tourism facilities in Southern Leyte in the poor region of Eastern Visayas, suffered losses, more than 260 tourism establishments in Cebu were damaged. In Siargao, the cost of damage reached PHP 1.6 billion.[72]

70 Investment and hiring decisions remain largely unaffected, as firms view such events as fleeting disruptions rather than permanent shifts, avoiding adjustments to capital and workforce that are hard to reverse.

71 Department of Tourism.

72 See https://www.rappler.com/philippines/dot-typhoon-odette-damaged-tourism-facilities-southern-leyte/, https://www.rappler.com/philippines/tourism-firms-cebu-damage-typhoon-odette-january-7-2022/ and https://www.pna.gov.ph/articles/1178624#:~:text=Based%20on%20the%20February%208,2%20billion.

**To enhance the resilience of the Philippine tourism sector the following should be considered:**

- Tourism destination management plans incorporating climate and disaster risk assessments will help to mitigate climate change effects. Upgrading public and private infrastructure to withstand extreme climate-related events and integrating climate-smart technologies in construction and operations, will also be useful steps.
- Strengthening disaster risk management and crisis response mechanisms at the local level, including early warning systems and health and safety protocols, will improve preparedness and recovery from natural disasters.
- Promoting sustainable tourism practices, such as reducing greenhouse gas (GHG) emissions and effectively managing waste, is crucial for preserving the natural environment.

# CHAPTER 6
# POLICY ROADMAP

## Introduction

**This chapter outlines a reform roadmap to place the Philippines on a higher trajectory of GDP, employment, and wage growth, aligned with its national development plans.** The roadmap is structured around three pillars introduced in the Preface: foundational investments in infrastructure and human capital; improved business regulations and governance; and targeted interventions to mobilize private capital. These aim to achieve four intermediate outcomes: fostering spatial convergence; boosting firms' productivity and innovation; improving resource allocation; and deepening regional and global integration.

**If implemented, the proposed reforms are estimated to boost GDP growth, employment outcomes, and wages.** The implementation of the reforms is estimated to increase the annual average real GDP growth rate from 5.4 percent under a business-as-usual (baseline) scenario, to 6.8 percent, over the next 15 years, which results in GDP being 23.6 percent greater by 2040 than in the absence of reforms. Labor outcomes are expected to improve, with over 5.1 million jobs created for an increasing labor force of 4.9 million people by 2040, in addition to the 65 million jobs that would prevail in the baseline. Interventions that lead to increased labor force participation (mainly of women, where the Philippines lags the region), human capital accumulation and skills upgrading, and increased productivity drive the results. Between 2025 and 2040, average real wages are estimated to increase by 12.9 percent above the baseline.

**The focus on job creation for the Philippines matters.** Unemployment rates have remained low and stable. However, the country's relatively high fertility rate will continue to expand the labor supply. Labor force participation remains below potential, in aggregate, and particularly for women, where the Philippines lags Viet Nam by 10 and 17 percentage points, respectively. With higher educational attainment for females than males, this gap takes a toll on efficient talent allocation. In addition, technological disruptions such as GenAI pose an important challenge for jobs in IT-BPO, one of the country's most dynamic sectors.

**Quality job creation also matters.** Convergence to higher income levels also requires that these jobs created are linked to higher productivity. In the Philippines, agriculture accounts for 22 percent of jobs, but for less than 9 percent of value added. As a result, the median agricultural income is only 2.1 times the national poverty line, compared with about 4 and 3.6 times in industry and services, respectively. Thus, we measure quality of job creation through the proposed reforms by tracking the impact on wages.

**The impact assessment framework in the report combines macro, micro, and spatial economic modeling with sector-specific analysis in a dynamic general equilibrium setup.** It accounts for direct and indirect impacts, reform costs, and their dynamic interconnections and externalities.

# Policy recommendations

**Policy recommendations are organized by the four intermediate outcomes that they support and grouped under three pillars, (I) investments, (II) regulations, and (III) private capital mobilization, to which they correspond** (Table 2). Specific channels are identified to link policy tools with the four outcomes based on thematic intervention areas such as governance, investment, technology, and labor market reforms. Within each channel, actionable policy measures are outlined, addressing sectoral and cross-sectoral challenges. The reform roadmap serves as a comprehensive framework for policy actions, linking broad strategic objectives to targeted, actionable recommendations designed to address key developmental constraints.

**Table 2**

**Policy Recommendations**

| **Outcome 1: Sustain Spatial Convergence** | |
|---|---|
| *Improved governance of Local Government Units (Pillar II)* | • Revise the LG code to clarify functional mandates and ensure allocation of skilled HR matching devolved areas.<br>• Consider equity in the formula for LG transfers.<br>• Introduce performance-based and conditional transfers.<br>• Increase fiscal autonomy for revenue generation. |
| *Foster regional integration (II & III)* | • Expand transport infrastructure (e.g., RORO ports, airports).<br>• Lift cabotage restrictions.<br>• Unify permitting for cargo using multi-modal transport, to amplify impact of RORO ports. |
| *Investing in skill formation (I)* | • Invest in early childhood education and stunting agendas to boost human capital, reduce stunting and enable female work.<br>• Implement Enterprise Based Education and Training and Digital Skills Act.<br>• Invest in STEM and emotional skills. |
| *Adapt to climate events to reduce volatility (I & II)* | • Invest in public climate adaptation infrastructure (flood control stations) prioritizing high risk locations with economic agglomeration.<br>• Streamline permitting for resilient construction to facilitate private investment in adaptation.<br>• Support firms' awareness of the impact of climate events and support to diversify suppliers.<br>• Add resiliency of materials as a 'value-for-many' consideration in public procurement rules. |

| | |
|---|---|
| *Accelerate investment (II & III)* | • Pass Comprehensive Tax Reform and upgrade tax administration sustain public infrastructure investment with fiscal prudency.<br>• Advance procurement reforms to create additional fiscal space. Publicly disclose information on public procurement price dispersion.<br>• Publicly disclose cost-benefit analyses of publicly funded projects. |
| **Outcome 2: Boost Firms' Productivity and Innovation** | |
| *Foster technology adoption (II & III)* | • Unlock financing for SMEs by revising single borrower's limits.<br>• Improve natural disaster risk insurance for SMEs.<br>• Expand the guaranteed coverage of the PhilGuarantee to SME investments in innovation.<br>• Improve the credit infrastructure ecosystem and collateral registry to unlock SME lending.<br>• Diffuse technologies anchored in market demand through ecosystem solutions and university-industry collaboration.<br>• Introduce suppliers' development programs to connect SMEs to large, MNCs.<br>• Expand mandates for energy audits to a broader set of firms to encourage energy efficiency technology adoption.<br>• Align tax incentives in CREATE MORE with investments in energy efficiency. |
| *Foster Innovation (II & III)* | • Introduce M&E and convergence budgeting systems to reduce the fragmentation of innovation programs and increase public expenditure efficiency.<br>• Crowd in risk capital by allowing government-owned pension funds to invest in PEVC funds subject to risk appetite, following careful analysis of risk impact.<br>• Clarify the legal structure and the registration process for the establishment of PEVC funds. |
| **Outcome 3: Allocate Resources Better** | |
| *Increase competition (II)* | • Strengthen sector regulators (NTC, MARINA, CAB) to enhance competition in public and network services.<br>• Decouple the roles of developer and regulator of regulatory agencies (CAAP, PPA) and remove private sector representation in regulatory agencies to prevent capture.<br>• Strengthen the energy regulator (ERC) in promoting competition, protecting customers, and investigating abuse of power.<br>• Implement the Beneficiary Ownership Registry for increased transparency in public procurement. |
| *Improve business regulation (II)* | • Harmonize national and local government regulations for business permits to enable data exchange, system interoperability, ID, and authentication systems (Implement Philippine Business Portal).<br>• Simplify business entry and location permitting through automation availability and electronic signature systems.<br>• Strengthen insolvency procedures to enable digital services (e-courts) and reduce time to resolve an in-court liquidation proceeding. |

| | |
|---|---|
| *Reduce factor and input market distortions (II & III)* | • Carefully consider introducing a part-time employment framework to boost FLFP.<br>• Simplify employment permitting, to reduce hiring costs.<br>• Revise lay-off processes, to align time-it-takes to dismiss with regional good practices.<br>• Allow long-term land leases to foreign companies.<br>• Align wages with labor productivity to boost employment and revise minimum wage mechanism, considering regional disparities in labor productivity and in standards of living.<br>• Streamline procedures for land clustering and joint ventures in agriculture.<br>• Re-consider double deductions of energy costs from profit taxes. |
| **Outcome 4: Better Integrate Regionally and Globally** | |
| *Reduce trade costs (II & III)* | • Negotiate and implement EU-PH FTA, Korea-PH FTA, Canada-PH FTA.<br>• Streamline non-tariff measures to reduce trade costs.<br>• Promote exports to reduce discovery costs of exporters and potential exporters. Leverage trade attaches.<br>• Modernize customs to lower costs of clearing customs on export and import sides.<br>• Streamline VAT refunds to exporters to reduce exporters' operative costs. |
| *Reduce investment costs (II & III)* | • Implement the PSA amendment to reduce costs of key enabling services inputs and attract more investment.<br>• Evaluate impact of investment incentives and deploy well-coordinated investment promotion and facilitation. |
| *Reduce costs of moving people (II)* | • Introduce mutual recognition in key professions to narrow the skills gap in the short term.<br>• Relax barriers to services trade in Mode 4. |

Source: World Bank staff elaboration. Note: Main affected pillars are indicated in parenthesis.

# From recommendations to results

**Estimating the impact of these policies on growth and jobs is complex.** This section presents a step-by-step approach to estimate the impact of reforms on key economic outcomes (e.g., growth, jobs, wages), focusing on the macroeconomic outcomes and specific impacts on jobs and welfare.[73]

**Step 1: Baseline Projection.** A baseline projection is established using the MFMOD-GJ model to represent the economy before the implementation of the reform roadmap.[74] The framework generates a long-term growth trajectory, providing a comprehensive analysis of the contributions of demographics, labor market dynamics, structural change, and human capital investment to aggregate output. This step is essential for estimating the business-as-usual (baseline) scenario, serving as a benchmark for impact evaluation.

73 There is no consensus approach for this type of impact assessment. The best methodology depends on the policy type, area of reform, available data and evidence. While tools for measuring growth impact are more readily available, this is not the case for labor outcomes.

74 The MFMod (Macro-Fiscal Model) is a framework used by the World Bank for macro analysis and forecasting Burns et al. (2019).

**Step 2: Micro-level Evidence:** Next, we evaluate the partial equilibrium impact of our recommended policies on specific economic outcomes. This is done using sectoral models or micro-data on firms, households, or the labor force. By estimating changes in prices, supply-side factors or demand-side factors, micro-level evidence provides a first-round assessment of the direct impact of recommended policies on the economy, including the mechanisms through which they operate. It has limitations, however, to account for the general equilibrium effects.

**Step 3: Linking Micro-level Evidence to the Macroeconomic Model.** This step integrates the policy impact into a multi-sectoral general equilibrium economic model, CGE/MANAGE, to estimate economy-wide effects, accounting for economy-wide interactions and feedback effects across markets and sectors. Initially, the business-as-usual scenario is calibrated to match the baseline projection of MFMOD-GJ model, representing the growth trajectory of the current economy. Evidence-based mechanisms are mapped to corresponding policy instruments within the model, with extensions made as necessary. Policy changes are introduced incrementally to assess the economic impact of individual policies and their potential complementarities. Box 10 provides detailed examples of the methodologies used to estimate the effects of the recommended policies.

**Step 4: Impacts on Growth, Jobs and Welfare.** In the final stage, the CGE/MANAGE model compares policy outcomes with the baseline scenario to assess the macroeconomic effects on GDP, consumption, as well as sectoral impacts on employment and wages. This enables policymakers to formulate targeted strategies that foster job creation while aligning with sector-specific economic objectives. In addition, it facilitates a comprehensive analysis of welfare and public finance by evaluating the effects of policies on household consumption and government expenditures. This step provides a thorough assessment of the policy's overall economic impact, offering insights into both its benefits and potential challenges.

## Box 10

### Examples of policy recommendations and their impact evaluations

**Recommendation: Remove foreign investment limits in transport and telecom (PSA amendment).**

The Public Service Act (PSA) amendment in the Philippines, enacted in April 2023 (yet to be implemented), removed foreign ownership restrictions in key infrastructure sectors such as railways, subways, airports, airlines, domestic shipping, expressways, tollways, and telecommunications. Previously, foreign ownership was limited to a maximum of 40 percent.

The reform is expected to have significant economic impacts by increasing competition, attracting FDI, and improving efficiency in affected, and downstream sectors. Partial equilibrium estimates based on elasticities estimated using econometric analysis on firm-level data suggest that the PSA could lead to a 3.2 percent increase in productivity and GDP.

In a general equilibrium setting, this higher productivity would raise wages by 2.8 percent annually, stimulating households to increase their participation in the labor market and ultimately leading to 200,000 additional jobs in 2040.

**Recommendation: Invest in early years for improved education and health outcomes, and boost FLFP.**

The Philippines lags Southeast Asian peers in female labor force participation (FLFP). As of 2022, the country's FLFP rate stood at 50.8 percent, significantly lower than the regional maximum of 75.6 percent.[75]

Investing in early-year workers to strengthen childhood education and health, particularly initiatives aimed at reducing stunting, is expected to alleviate household caregiving burdens on women, thereby facilitating greater labor market participation. This reform is projected to narrow the FLFP gap by 15 percent relative to the regional maximum over a five-year period, corresponding to an annual increase of 1.1 percent in FLFP.

Within a general equilibrium model, higher FLFP is estimated to drive a 1.7 percent annual increase in employment by 2040, with the services sector experiencing the most significant growth at 2 percent per year. This expansion is expected to enhance household income and stimulate demand for goods and services. However, the resulting increase in labor supply may exert downward pressure on wages, requiring complementary actions to boost labor productivity, to create better jobs for the increasing labor force.

# Results

## Impact on economic growth

**Proposed reforms are expected to increase annual real GDP growth by 1.4 percentage points, reaching an average of 6.8 percent annual growth rate over 2025–2040** (Table 3). This results from the accumulation of human capital, physical capital, and increased productivity. The average annual GDP growth rate could reach up to 6.8 percent over a 15-year period (2025–2040), compared with 5.4 percent in the reference business-as-usual scenario (baseline). The higher GDP growth rate is mainly driven by higher TFP growth (0.8 pp contribution), higher capital accumulation (0.5 pp contribution and an investment-to-GDP ratio converging to 25.8 percent), and to a lesser extent (0.13 pp) by an increasing labor force participation rate (from 61 to 66 percent).

75 Labor force participation rate, female (as a percent of female population ages 15–64). Source for Philippines data: World Bank Staff calculations based on LFS. Source for regional comparators: WDI-ILO. Note: regional maximum corresponds to Viet Nam.

## Table 3

**Impact of the proposed reforms on real GDP**

| | Real GDP (Deviation from baseline, in percent) | | Real GDP growth (Avg. annual growth rate, in percent) |
|---|---|---|---|
| | 2030 | 2040 | 2025-2040 |
| **All policies** | **13.0** | **23.6** | **6.8** |
| Outcome 1 | 2.2 | 5.2 | 5.8 |
| Outcome 2 | 4.6 | 7.0 | 5.9 |
| Outcome 3 | 1.4 | 3.1 | 5.6 |
| Outcome 4 | 4.1 | 6.1 | 5.8 |

Source: World Bank staff calculations based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024). Note: The sum of the impacts of specific policies do not add up to the total impact of all policies combined due to non-linearities in the model.

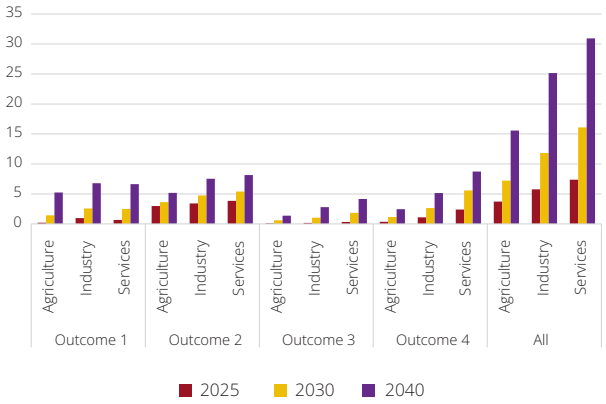
**A sensitivity analysis was conducted to assess the impact of potential low sensitivity of outcome variables to the proposed reforms.** In a low-case scenario of reforms having weak effects on labor participation rate, investment and productivity, real GDP growth is projected to increase by 11 percent with respect to the baseline, compared with the estimated 23.6 percent by 2040. The corresponding average annual growth rate lowers to 6.1 percent, compared with 6.8 percent during 2025–2040.

**On the supply side, growth is expected to be driven by an expansion of the services and industry sectors** (Figure 6.1). Policies under the first outcome area tend to enhance labor market participation the most, benefiting labor-intensive sectors such as services. Policies under the second outcome boost productivity, with large effects in relatively more capital-intensive sectors such as extractives and transport. Outcome 3 policies focusing on allocation show larger effects on productivity in manufacturing and services. Outcome 4 policies favor tradables leading to higher value-added in manufacturing, renewable energy, and transport.

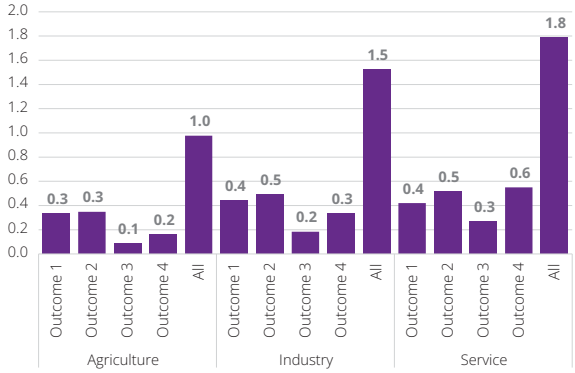
## Figure 6.1

**Impact of the proposed reforms on value added by sector**

a. Deviation from baseline, in percent




b. Average annual growth rate over baseline, p.p. difference




Source: World Bank staff calculations based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024).

**On the demand side, growth is driven by exports, investment (public and private), and government spending** (Table 4). This is because, as part of policy reforms, the Government is increasing spending (investment and consumption) to achieve its objectives. The increase in exports and imports is mainly due to the reduction in trade costs (tariffs and non-tariff equivalent), improved border efficiency under Outcome 4, and increased productivity (under Outcome 2).

## Table 4

**Impact of the proposed reforms on GDP by demand component**

| | GDP components (Deviation from baseline, in percent) | |
|---|---|---|
| | 2030 | 2040 |
| Household consumption | 8.3 | 14.3 |
| Government consumption | 13.3 | 24.0 |
| Investment | 16.5 | 29.7 |
| Exports | 15.0 | 30.0 |
| Imports | 8.1 | 13.8 |

Source: World Bank staff calculations based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024). Note: the sum of the impacts of specific policies do not add up to the total impact of all policies combined due to non-linearities in the model.

## Impact on employment

**Policy reforms are expected to lead to better labor market outcomes: increased jobs, with higher labor market participation rate.** Employment under the policy reform scenario is 6.5 and 7.8 percent higher than the baseline in 2030 and 2040, respectively (Table 5). Policies under each of the four outcomes lead to higher employment. The most significant gains come from policies that encourage more people, especially females, to participate in the workforce, particularly under Outcome 1. The implementation of the policy package is estimated to raise labor force participation by 4.8 percentage points, adding 4.9 million workers, with nearly 75 percent being female. Together with policies that boost labor demand, employment is projected to rise by 7.8 percent relative to the baseline, or almost 5.1 million additional jobs by 2040.[76]

### Table 5

**Impact of the proposed reforms on employment and real wages**

| | Employment (Deviation from baseline, in percent) | | Real wages (Deviation from baseline, in percent) | |
|---|---|---|---|---|
| | 2030 | 2040 | 2030 | 2040 |
| **All policies** | **6.5** | **7.8** | **5.1** | **12.9** |
| Outcome 1 | 2.6 | 2.4 | -1.0 | 0.6 |
| Outcome 2 | 1.5 | 2.2 | 2.6 | 4.3 |
| Outcome 3 | 1.4 | 1.7 | 0.1 | 1.7 |
| Outcome 4 | 0.8 | 1.2 | 3.3 | 5.4 |

Source: World Bank staff calculations based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024).

**Because the reform package leads to increases in labor productivity, it is also expected to lead to better paying jobs.** Overall, it is estimated to lead to increases in real wages of 5.1 and 12.9 percent higher relative to the baseline for 2030 and 2040, respectively (Table 5). Higher labor productivity is the key to higher wages. Policies in the whole reform package are estimated to increase labor productivity by 17.8 percent relative to the baseline by 2040. These gains are especially influenced by policies in Outcomes 2 and 4, which focus on improving worker skills and efficiency, while policies under Outcome 1 that boost labor supply have a slightly negative wage impact in the medium term, while being positive in the long term.

76 A back of the envelope calculation shows internal consistency of the results on employment and GDP growth. The estimated 5.1 million additional jobs by 2040 and 1.4 percent point of additional GDP growth, is consistent with about US$14,000 per additional job (at 2024 prices). Considering the labor share in GDP (estimated at 0.4 percent), this would imply a wage-equivalent (including benefits and other labor surcharges) of about US$5,600 per job yearly (about US$467 a month), consistent with slightly increasing labor costs.

**A balanced set of reforms that combines increased participation with increased productivity is essential to create more jobs with higher wages.** Policies that encourage labor market participation (mainly Outcomes 1 and 3) account for about 81 percent of total employment growth by 2040 (Figure 6.2, panel a), but without labor productivity improvements (mainly Outcomes 2 and 4), wages may face downward pressure. Meanwhile, productivity-enhancing policies stimulate labor demand, ensuring that wage growth keeps pace with employment expansion (Figure 6.2, panel b). Together, these policies facilitate a new equilibrium of higher employment with higher wages.

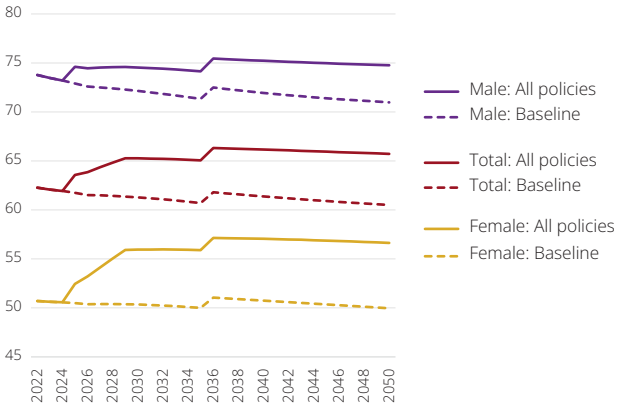
**A sensitivity analysis was conducted to assess the impact of potentially low pass-through of the labor force participation rate to the proposed reforms.** Under a scenario where reforms have a weaker-than-expected effect on the labor participation rate, investment and productivity, employment is projected to increase by 2.6 percent with respect to the baseline, corresponding to an additional 1.7 million jobs, compared with the estimated 7.8 percent and 5.1 million jobs by 2040. The increase in real wage lowers to 9.3 percent from baseline, compared with the estimated 12.9 percent by 2040.

**The services and industry sectors are the main contributors to job creation** (Figure 6.3). Services are responsible for over 70 percent of job creation under the reform scenario, with 3.7 million additional jobs. Within services, the subsectors of transport, real estate and hotels and restaurants are the main drivers of job creation. Industry is the second-largest contributor, with over 1.1 million additional jobs (with non-food manufacturing and construction being the main drivers).

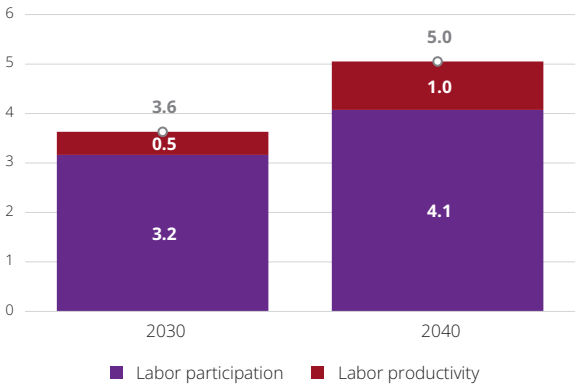
## Figure 6.2

**Impact of the proposed reforms on job creation**

a. Labor force participation rate, in percent




b. Additional jobs, in million

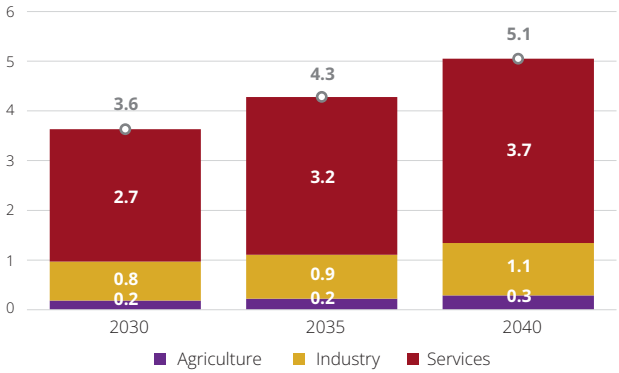



Source: World Bank staff calculations based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024).

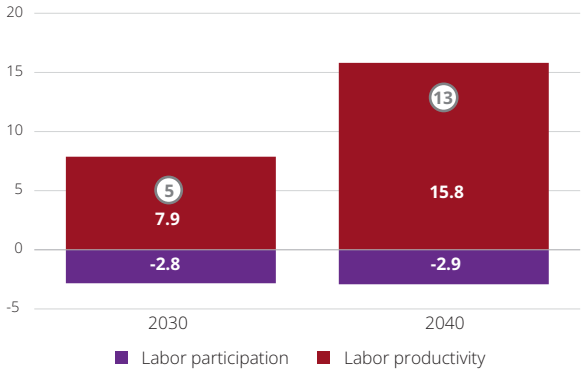
**Figure 6.3**

**Impact of the proposed reforms on structural change and wages**

a. Additional jobs by sector, in million jobs




b.Real wages, deviation from baseline, in percent




Source: World Bank staff calculations based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024).

## Impact on welfare

**Real household consumption per capita is projected to increase steadily over the forecast period as policy reforms drive real GDP growth, job creation, and productivity.** By 2040, the reform scenario results in consumption levels being 14 percent higher than the baseline, with each policy objective similarly contributing to this increase.[77] The growth trajectory reflects sustained improvements in household incomes and public finance, bolstered by policy measures targeting investments, global market integration, and job creation. The trend suggests that reforms not only enhance immediate welfare but also set the stage for long-term, inclusive economic benefits.

# Conclusion

**The structured approach outlined in this chapter helps policymakers link policy actions with outcomes, contributing to evidence-based decisions.** The projected increase in employment and wages highlights the positive impact of policy reforms on economic welfare. This growth trajectory reflects sustained improvements in job opportunities and income levels, driven by investments, global market integration, and targeted policy measures.

**The importance of an integrated approach to policy reforms matters.** For example, policies that support labor market participation and those that support productivity boosts should be conducted together to ensure job creation is not at the expense of productivity and, therefore, wages. Altogether, this reform package provides an actionable and feasible roadmap to bring the Philippines to a higher development path, with an increase in potential output growth, and with more and better paying jobs. Thus, it brings the Philippines closer to achieving its national ambition of creating a middle-class society where nobody is poor, by 2040.

77 The impact of reform package on public finance are also positive. It is estimated to increase tax revenue as a share of GDP by 16 percent and lower public debt as a share of GDP by 9.7 percent, compared with the baseline scenario in 2040.

# Standardized Tables

| | Philippines | | | | | | | Comparators (2014-2023 avg.) | | Source |
|---|---|---|---|---|---|---|---|---|---|---|
| | Pre-AFC | Between Crisis | Post-GFC | Annual Data | | | | | | |
| | 1990-1998 | 1999-2009 | 2010-2023 | 2014 | 2017 | 2020 | 2023 | Regional | Structural | |
| **Real/Income** | | | | | | | | | | |
| Real GDP growth, mkt. prices | 2.8 | 4.4 | 5.2 | 6.3 | 6.9 | -9.5 | 5.5 | 5.2 | 4.7 | MPO |
| Investment, % change | 3.4 | 4.9 | 7.9 | 8 | 10.6 | -27.3 | 8.2 | 6.3 | 5.7 | MPO |
| Public | | 5.0 | 9.6 | -0.7 | 13.6 | -16.7 | -5.9 | 4.7 | 6.6 | MPO |
| Private | | 4.4 | 7.8 | 9.4 | 10 | -29.9 | 12.8 | 6 | 6.4 | MPO |
| Consumption, % change | 3.5 | 4.2 | 5.2 | 5.5 | 6 | -5.3 | 4.7 | 5.3 | 4.2 | MPO |
| Public | 2.9 | 2.8 | 7.1 | 3.6 | 6.5 | 10.5 | 0.6 | 5.1 | 5 | MPO |
| Private | 3.6 | 4.5 | 4.9 | 5.8 | 6 | -8 | 5.6 | 5.4 | 4.2 | MPO |
| Net exports, % change | -18.1 | -23.2 | 15.8 | 0.5 | 9.2 | -35.6 | -0.3 | -16.4 | 19.5 | MPO |
| Exports | 6.8 | 5.3 | 6.7 | 12.1 | 17.4 | -16.1 | 1.4 | 8.3 | 8.3 | MPO |
| Imports | 8.1 | 4.0 | 8.3 | 9.9 | 15.1 | -21.6 | 1 | 8.3 | 7 | MPO |
| Investment, % of GDP | 19.2 | 17.1 | 22.5 | 20.6 | 25.7 | 21.4 | 23.3 | 28.8 | 22.2 | MPO |
| Public | | 2.9 | 4.1 | 2.7 | 4.4 | 4.8 | 5 | 7 | 6.7 | MPO |
| Private | | 14.3 | 18.3 | 17.9 | 21.3 | 16.6 | 18.3 | 22.3 | 13 | MPO |
| Consumption, % of GDP | 87.2 | 87.8 | 85.7 | 84.5 | 84.2 | 88.7 | 87.3 | 63.3 | 83 | MPO |
| Public | 13.6 | 10.8 | 12.3 | 10.9 | 11.3 | 15.1 | 14.3 | 10.7 | 10.2 | MPO |
| Private | 73.6 | 77.0 | 73.4 | 73.5 | 72.9 | 73.6 | 73.1 | 54.5 | 72.7 | MPO |
| Net Exports, % of GDP | -4.7 | -4.9 | -8.1 | -5.3 | -10.2 | -8 | -10.5 | 3.2 | -6.6 | MPO |
| Exports | 25.5 | 30.2 | 26.8 | 24.8 | 28.7 | 27.1 | 27.4 | 53.2 | 17.7 | MPO |
| Imports | 20.8 | 25.4 | 34.9 | 30.1 | 38.9 | 35.1 | 37.9 | 50 | 24.2 | MPO |
| Potential GDP growth | 3.8 | 4.4 | 5 | 6 | 7.3 | -8.9 | 5.3 | 5.0 | 4.7 | MPO |
| Distance to frontier, % of USA's GDP pc | 9.9 | 9.4 | 12.5 | 12.1 | 13.4 | 12.9 | 13.3* | 21.2 | 12.5 | CT |
| Real GDP growth | 2.4 | 4.3 | 5.2 | 6.3 | 6.9 | -9.5 | 5.5 | 5.2 | 4.7 | MPO |
| TFP, contrib. (ppt) | -0.1 | 1.6 | 0.7 | 2.1 | 1.2 | -15.5 | 1.7 | 0.9 | 0.4 | MPO |
| Capital accum., contrib. (ppt) | 1.6 | 1.7 | 3.4 | 3.3 | 4.7 | 2.3 | 3 | 2.8 | 3.3 | MPO |
| Labor, contrib. (ppt) | 0.9 | 1.1 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.8 | 0.8 | MPO |
| **Inclusion** | | | | | | | | | | |
| Poverty rate, $2.15 poverty line | | | 4.6 | 6.5* | 3.0* | 3.0* | 1.9 | 3.1 | 3.9 | MPO |
| Gini Inequality Index | | 47.0 | 43.5 | 44.6* | 42.3* | 40.7* | 40.7* | 37.7 | 34.4 | WDI |
| **Demographics and Jobs** | | | | | | | | | | |
| Dependency ratio, WAP/Pop | 75.1 | 68.4 | 58.8 | 60.2 | 58.1 | 56.6 | 55.3 | 44.7 | 53.7 | WDI |
| Employment to WAP ratio | | | | | | | | | | |
| Male | 73.2 | 72.9 | 71.1 | 72.7 | 71.5 | 65.2 | 71.6 | 77.7 | 71.4 | ILO |
| Female | 45.5 | 45.2 | 45.3 | 47.1 | 43.7 | 41.4 | 46.0 | 56.2 | 24.4 | ILO |
| Labor force participation rate | 63.0 | 62.9 | 62.0 | 64.1 | 61.3 | 56.9 | 61.7* | 72.6 | 54.5 | ILO |

| | Philippines | | | | | | | Comparators (2014-2023 avg.) | | Source |
|---|---|---|---|---|---|---|---|---|---|---|
| | Pre-AFC | Between Crisis | Post-GFC | Annual Data | | | | | | |
| | 1990-1998 | 1999-2009 | 2010-2023 | 2014 | 2017 | 2020 | 2023 | Regional | Structural | |
| Male | 77.2 | 77.0 | 75.0 | 77.1 | 75.3 | 68.9 | 74.6* | 83.2 | 79.3 | ILO |
| Female | 48.7 | 48.6 | 48.6 | 50.8 | 46.8 | 44.5 | 48.4* | 62 | 29.6 | ILO |
| Unemployment rate | 3.8 | 3.7 | 3.0 | 3.6 | 2.6 | 2.5 | 2.2 | 2.9 | 8.1 | ILO |
| Male | 3.5 | 3.5 | 2.8 | 3.5 | 2.5 | 2.4 | 2.0 | 2.8 | 6.6 | ILO |
| Female | 4.1 | 4.0 | 3.2 | 3.7 | 2.7 | 2.7 | 2.5 | 3.1 | 12.5 | ILO |
| **Structural Change** | | | | | | | | | | |
| GDP growth, factor prices, % change | 2.8 | 4.4 | 5.2 | 6.3 | 6.9 | -9.5 | 5.5 | 5.2 | 4.8 | MPO |
| Agriculture | 0.8 | 4.3 | 1.4 | 1.9 | 4.2 | -0.2 | 1.2 | 2.5 | 3.6 | MPO |
| Manufacturing | 2.6 | 3.4 | 5.2 | 7.5 | 7.0 | -13.1 | 3.6 | 3.7 | 5.5 | MPO |
| Services | 3.5 | 5.1 | 6.0 | 6.7 | 7.4 | -9.1 | 7.1 | 4.8 | 5.1 | MPO |
| Agriculture, % of GDP | 18.1 | 13.9 | 11.1 | 12.3 | 10.2 | 10.2 | 9.4 | 13.9 | 15 | WDI |
| Manufacturing, % of GDP | 36.3 | 33.7 | 30.2 | 31.0 | 30.1 | 28.4 | 28.2 | 39 | 28 | WDI |
| Services, % of GDP | 45.6 | 52.4 | 58.7 | 56.7 | 59.7 | 61.4 | 62.4 | 45.9 | 50.5 | WDI |
| Sectoral composition of employment, % | | | | | | | | | | |
| Agriculture | 43.3 | 35.7 | 27.8 | 30.4 | 25.4 | 24.8 | 23.7* | 37.7 | 38.3 | WDI |
| Manufacturing | 16.1 | 15.9 | 17.2 | 15.9 | 18.3 | 18.3 | 18.9* | 22.2 | 21.5 | WDI |
| Services | 40.6 | 48.4 | 55.0 | 53.6 | 56.3 | 56.9 | 57.4* | 40.1 | 40.2 | WDI |
| Sectoral composition of employment (LFS), millions | | | | | | | | | | |
| Agriculture | | | | 11.2 | 10.3 | 9.8 | 10.1* | | | PSA |
| Industry | | | | 6.0 | 7.4 | 7.3 | 8.6* | | | PSA |
| Services | | | | 20.1 | 22.7 | 22.8 | 29.0* | | | PSA |
| Total | | | | 37.3 | 40.3 | 39.8 | 47.7* | | | PSA |
| **Employment Structure** | | | | | | | | | | |
| Waged employment in Employment, % | 50.2 | 50.6 | 60.1 | 57.5 | 62.4 | 62.6 | 63.9* | 53.4 | 54.1 | ILO |
| Average Years of Schooling | 7.0 | 7.8 | 8.5 | 8.5 | 8.7 | 8.9* | | 7.3 | 5.9 | PWT |
| **External** | | | | | | | | | | |
| Current account balance, % of GDP | -3.0 | 1.0 | 0.6 | 3.6 | -0.7 | 3.2 | -2.6 | 2 | -1.8 | MPO |
| FDI, % of GDP | 1.6 | 1.4 | 2.0 | 1.9 | 3.1 | 1.9 | 2.0 | 3.3 | 1.5 | WDI |
| Exchange rate, LCU/USD | 28.1 | 49.2 | 48.2 | 44.4 | 50.4 | 49.6 | 55.6 | | | MPO |
| REER, index 2015=100 | | 90.2 | 96.0 | 95.4 | 92.8 | 99.7 | 97.5 | | | MPO |
| Reserves, months of imports | 2.8 | 5.6 | 9.5 | 9.9 | 7.8 | 12.3 | 7.7 | 5 | 5.1 | WDI |
| **Memo items** | | | | | | | | | | |
| Credit to private sector, % of GDP | 29.6 | 31.3 | 41.8 | 37.6 | 45.6 | 52.0 | 48.3 | 92.9 | 38.6 | WDI |
| Interest rate, annual average | 15.0 | 7.4 | 3.7 | 3.1 | 3.0 | 2.7 | 6.3 | 6.3 | 9.6 | MPO |

| | Philippines | | | | | | | Comparators (2014-2023 avg.) | | Source |
|---|---|---|---|---|---|---|---|---|---|---|
| | Pre-AFC | Between Crisis | Post-GFC | Annual Data | | | | | | |
| | 1990-1998 | 1999-2009 | 2010-2023 | 2014 | 2017 | 2020 | 2023 | Regional | Structural | |
| Inflation, CPI | 10.1 | 5.0 | 3.5 | 3.6 | 2.9 | 2.4 | 6.0 | 6 | 6.8 | MPO |
| Fiscal deficit, % of GDP | -0.8 | -2.8 | -3.6 | -0.6 | -2.1 | -7.6 | -6.2 | -2.3 | -4.6 | MPO |
| Public debt, % of GDP | 52.6 | 59.9 | 48.4 | 43.4 | 40.2 | 54.6 | 60.1 | 41.6 | 60.9 | MPO |
| -of which external, % of GDP | 22.2 | 27.6 | 16.7 | 14.5 | 13.4 | 17.3 | 18.9 | 11.5 | 17.7 | MPO |

Source: World Bank staff calculations. Note: Regional peers include Indonesia, Malaysia, Thailand, and Viet Nam. Structural peers include Bangladesh, Morocco, Egypt, and Sri Lanka. WAP: Working-age population. CT: Convergence Tool. *Indicates that no data was available for that particular year, instead data corresponding to the closest (in general, previous) year was used. PSA labor data and ILO modeled estimates may differ due to methodological variations in data collection, definitions, and estimation techniques.

# Standardized Employment Table

| | Annualized growth | | | | | | | Composition | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Philippines | | | | | Comparators (2009-2022 avg.) | | | | |
| Sector | 2001-2009 | 2009-2022 | 2020 | 2021 | 2022 | Regional | Structural | Philippines | Regional | Structural |
| Total Employment | 2.5 | 2.5 | -5.7 | 10.9 | 6.6 | 1.2 | 1.1 | 100.0 | 100.0 | 100.0 |
| Agriculture | 1.8 | -0.7 | 2.1 | 7.7 | 1.5 | -1.2 | -1.4 | 21.9 | 25.8 | 27.6 |
| Mining | 4.0 | 2.6 | 1.3 | -8.1 | 31.7 | 1.4 | 1.9 | 0.5 | 0.6 | 0.5 |
| Manufacturing | -0.4 | 2.3 | -10.2 | 8.0 | 8.5 | 2.5 | 1.2 | 8.1 | 17.0 | 13.6 |
| Utilities | 2.2 | 1.5 | -9.9 | 7.1 | 12.5 | 5.7 | 4.6 | 0.4 | 0.7 | 0.8 |
| Construction | 2.0 | 7.0 | -9.5 | 16.3 | 1.4 | 2.1 | 3.3 | 9.7 | 7.4 | 10.0 |
| Wholesale & Retail Trade | 3.6 | 3.6 | -2.6 | 19.9 | 6.4 | 2.2 | 2.6 | 22.0 | 17.3 | 16.6 |
| Transport & ICT | 2.8 | 2.8 | -13.7 | 2.9 | 10.1 | 1.9 | 3.1 | 8.2 | 4.9 | 7.5 |
| Hospitality | 4.9 | 4.9 | -22.1 | -4.5 | 28.6 | 2.8 | 4.6 | 4.0 | 6.8 | 3.0 |
| Finance | 3.0 | 4.7 | -2.3 | 10.9 | 4.4 | 4.9 | 1.7 | 1.4 | 1.6 | 1.1 |
| Business Services | 9.9 | 8.0 | -4.3 | 12.7 | 18.7 | 6.4 | 5.7 | 6.1 | 3.9 | 2.3 |
| Public Administration | 2.6 | 4.0 | -6.7 | 6.0 | 4.1 | 1.1 | 0.3 | 6.1 | 3.9 | 4.3 |
| Education | 2.7 | 2.3 | 2.1 | 10.5 | 3.2 | 1.5 | 3.1 | 3.3 | 4.3 | 5.3 |
| Health | 4.3 | 4.0 | 1.4 | 21.0 | 1.8 | 4.1 | 2.6 | 1.5 | 2.2 | 1.9 |
| Other Services | 3.1 | 1.3 | -12.7 | 13.2 | 8.5 | 0.5 | 2.9 | 6.9 | 3.7 | 5.4 |

Source: PHL LFS; ILO; World Bank staff calculations.

# Standardized Projections Table

| | BAU | | All policies | |
|---|---|---|---|---|
| | MT | LT | MT | LT |
| | 2025-2040 | 2041-2050 | 2025-2040 | 2041-2050 |
| | | | | |
| Real GDP (value added at factor cost), compound average annualized growth rate (%) | 5.6 | 5.0 | 7.2 | 6.5 |
| TFP, contribution (ppt) | 1.7 | 1.7 | 2.6 | 2.4 |
| Capital accumulation, contribution (ppt) | 3.4 | 3.0 | 4.0 | 3.7 |
| Labor, contribution (ppt) | 0.5 | 0.4 | 0.7 | 0.4 |
| Human capital, contribution (ppt) | | | | |
| Demand Composition of Real GDP, (share, %) | | | | |
| Private consumption | 69.5 | 70.2 | 69.5 | 70.1 |
| Government consumption | 12.0 | 12.0 | 12.0 | 12.0 |
| Private investment | 21.4 | 19.9 | 21.4 | 19.3 |
| Government investment | 3.5 | 3.5 | 3.5 | 5.4 |
| Net exports | -6.4 | -5.5 | -6.4 | -6.7 |
| Sectoral composition of Real GDP, (share, %) | | | | |
| Agriculture | 6.6 | 5.3 | 6.4 | 4.8 |
| Manufacturing | 23.1 | 21.5 | 23.2 | 22.1 |
| Services | 70.2 | 73.2 | 70.4 | 73.0 |
| Working age population (WAP), end of period (000's) | 108 105 | 121 474 | 108 105 | 121 474 |
| Labor force participation rate (% of WAP), end of period | 61.4 | 60.5 | 66.2 | 65.7 |
| Male | 71.9 | 71.0 | 75.2 | 74.8 |
| Female | 50.7 | 50.0 | 57.0 | 56.6 |
| Employment ('000), end of period | 64 865 | 71 846 | 69 915 | 78 031 |
| Unemployment rate, end of period, (%) | 2.2 | 2.2 | 2.2 | 2.2 |
| Labor productivity, end of period, (Index 1 =2025) | 1.8 | 2.6 | 2.0 | 3.4 |
| Average Hourly Earnings per worker, end of period, ($) | 2.8 | 3.4 | 3.2 | 4.2 |
| Sectoral composition of employment, end of period, (%) | | | | |
| Agriculture | 12.8 | 15.1 | 12.8 | 14.4 |
| Manufacturing | 22.7 | 21.9 | 22.7 | 21.9 |
| Services | 64.5 | 63.1 | 64.5 | 63.7 |

Source: World Bank staff calculations. based on World Bank country specific MANAGE, a dynamic CGE model, calibrated using the recent Philippines SAM for the base year 2018 (Beyene et al., 2024).

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